Feasibility Analysis

Version 1.1

For Release

VuduVations
Sean Halverson

Document Change Control

Approved Document Owners	Contact	Role
Sean Halverson	[redacted]	Author/ Editor

Date	Version	Changes Summary	Changes By
Oct. 06 2021	1	Established Version 1	Sean Halverson
Dec. 30 2021	1.1	Modifications and additions to Algorithmic Trading	Sean Halverson

Table of Contents

1. Executive Summary

"Business usage alone can transform a commodity into a common medium of exchange. It is not the state, but the common practice of all those who have dealings in the market, that creates money." - Mises, Theory of Money and Credit

The GSHI Feasibility Analysis's primary objective was to conduct a 10-week study on the cryptocurrency ecosystem, organize, classify, understand it, then identify a path forward with a model that could look and operate like a bank and build trust in a trustless setting.

This GSHI Feasibility Analysis (FA) culminated in a series of research methods that used consulting data analysis and techniques to create a logical framework for identifying and organizing these emerging markets in cryptocurrency. This FA will present a logical taxonomy that identifies current classifications as well as clusters of protocols. This will help reveal how many collaborators are involved in the cryptocurrency ecosystem and where they fall into place.

This F&A achieves market alignment where we start by explaining the core concepts of decentralization. We use the Decentralized Autonomous Organization (DAO) to identify the algocratic style methods for self-rule. You will quickly see why this market of over $40 billion is fast emerging as a legitimate crypto economy.

The next sections will examine where Decentralized Finance and Centralized Finance (CeFI) fit in the cryptocurrency sphere and why the hybrid CeDeFI has emerged as an alternative model that provides new fundamentals for investing. We believe that the four core components of cryptocurrency foundations (DAO, DeFI, CeFI, and CeDeFI) can be understood based on the collective data and knowledge we are presenting in this FA. The additional sections provide detailed information on several aspects: algorithmic trading, loans, non-fungible tokens, Peer2Peer lending, Liquidity Pools, and an innovative and newer concept for Tokenizing future yields.

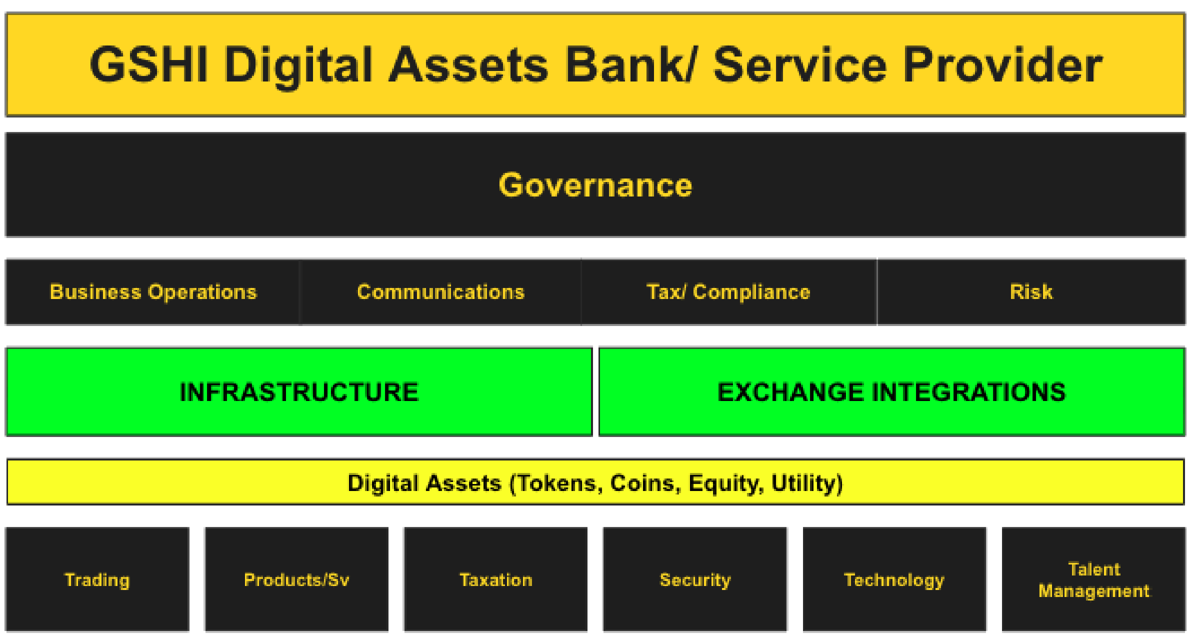
In the end, we identify a customer type that requires a set of certain functions from a crypto platform. These include the types of services one would expect in investing and using cryptocurrency as their new medium of value exchange.

1.1. Our Evolution of the Feasibility Analysis

How our perspective of the cryptocurrency and blockchain market has changed during this engagement.

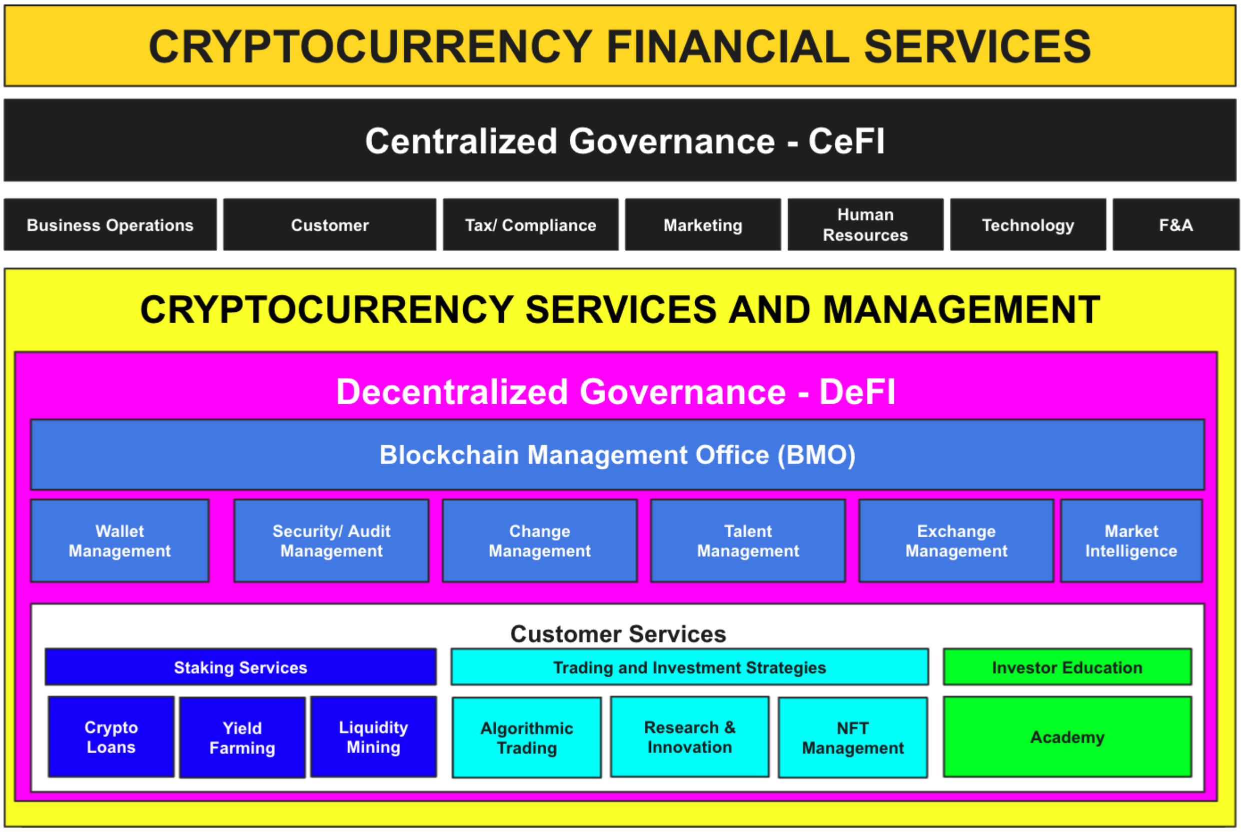
Our Initial View

Mid-Project View



Current View

The demarcation between decentralized and centralized functions is through the Blockchain Management Office (BMO.)



2. The Cryptocurrency Market

The cryptocurrency market provides a new economic model which enables customers to trade intrinsic value for virtual currencies, including fiat currency. The market may be global in which buyers and sellers make trades worldwide, or local transactions are completed within a particular country or amongst a group sharing a common interest.

The cryptocurrency market features high volatility, nascent trading methods, ever-increasing capitalization, and the high availability of market data. It has been proven that the cryptocurrency market is financially viable in comparison to other markets. The algorithms on which the cryptocurrencies work have also been tested in other areas. Although the cryptocurrency market appears to be independent of other financial markets, there is still a strong influence by most major economies. The technology used to mine cryptocurrency is a viable alternative to traditional markets like gold. This market has a lot of appeal as a result.

These attributes have attracted significant capital. However, there have been very few studies to attempt to create profitable trading strategies for the cryptocurrency market.

In the early days of cryptocurrency (estimated at 2010), if you wanted to trade Bitcoin, you had to find an individual willing to trade with you. The first official Bitcoin transaction occurred on May 22, 2010, when Florida-based Laszlo Hanyecz traded 10,000 Bitcoin for two pizzas from a local pizza store. Today's value of 10,000 Bitcoin exceeds $450,000,000. Since cryptocurrencies gained traction, the infrastructure demand required to keep the liquidity of the cryptocurrency market fluid enabled several new exchanges to spring up over the coming years.

Globalization’s positive and negative effects have significantly impacted various state, geopolitical, and global financial crises. This led to representatives from different social classes looking for alternatives to store or augment funds. Moreover, entrepreneurs and individuals in various countries are becoming more aware of the benefits of cryptocurrency. Cryptocurrencies are highly sought-after in many countries. The spread of cryptocurrency is primarily due to financial instability, currency fluctuations, and capital flight restrictions.

As the size of the market grew, more new clients were continuously being attracted to the market. This growth has enabled the reach of cryptocurrency to expand into other businesses, and depending on legislation; governments may find it useful as well. Nowadays, because of the rapid expansion of the cryptocurrency market, the number of retail and institutional investors has grown significantly, and competition has become fiercer between those offering cryptocurrency liquidity and trading services. There are now over 9000 cryptocurrencies available worldwide in the market today[1] (Austin 2021).

---

[1] Austin, Ryan. “Altcoins Are the Alternative Digital Currencies to Bitcoin - Here's What They Are and How They Work.” Business Insider, Business Insider, 1 Sept. 2021, https://www.businessinsider.com/what-is-altcoin#:~:text=Since%20the%20emergence%20of%20Bitcoin,of%20more%20than%209%2C000%20altcoins.

The Cryptocurrency market size today is growing substantially. Already, several well-established companies have entered the marketplace, which poses a substantial risk to traditional finance models. The increase in the market size is the recognition by companies of the importance of investing in the cryptocurrency market and the increases in liquidity spawning new market opportunities.

Because of this significant rise in the number of currencies in the marketplace, there has been a corresponding increase in the number of places you can effectively trade. As previously stated, the increase in liquidity has increased the number of choices available to buyers or traders. There are now several online sites that provide the ability to trade on the cryptocurrency market.

3. The Customer Services

Based on our research using Data Mining and Natural Language Processing Techniques through Extraction and Topic Modeling, this Feasibility Analysis identified these core areas of the cryptocurrency industry and have been categorized into the following clusters for market segmentation:

- **Staking Services** offers current cryptocurrency owners the opportunity to invest their idle assets over some time in the hopes of obtaining rewards by letting GSHI use these proceeds through other services to borrowers.

- **Trading and Investment Strategies** offer trading and investment opportunities for retail and institutional investors. They focus on services related to Algorithmic Trading, Research and Innovation, and Non-Fungible Token Management.

- **Investor Education** is designed to alleviate novice and amateur investors' concerns about the cryptocurrency market. It is also a valuable contribution to the community, which builds the consensus and functional trust for further operationalization.

3.1. Staking Services

Staking means you lock your cryptocurrency through a smart contract to get rewards[2](Binance 2021).

Crypto staking works in the same way as depositing money at a bank. An investor locks up his assets and receives "interest" in return, while the bank lends the money out to others. The difference is that, unlike bank rates that are often below 1%, the cryptocurrency staking market often generates APY above 10%, and over 100% is not uncommon for investor returns.

Participants can lock tokens into a smart contract (their stake), and at specific intervals, the protocol randomly assigns one of them the right to validate the next block. The probability of being selected is usually proportional to how many coins you have. Thus, the more tokens you have, the greater the chance of receiving the reward[3](Becker 2021). This can increase the token's value by restricting the supply. If the network uses a Proof-of-Stake (PoS), the tokens are often used to govern the future direction of the blockchain platform.

In contrast to Bitcoin, where miners must prove their work (Proof of Work, or PoW) to receive the reward from solving the mathematical puzzle and securing the right to add the next block to claim their reward.

PoS systems allow coins to be staked (investors to put skin in the game) in order to forge new blocks on the blockchain. Participants are then rewarded. Randomization ensures that no one entity has a monopoly on forging.

3.1.1. Crypto Loans

Many crypto enthusiasts are encouraged to "HODL" (Hold On for Dear Life) their assets, which is keeping them safe in their wallets until the value of their chosen currency increases. Just as you wouldn't feel comfortable leaving cash in a bank at low-interest rates, the common question is: How can you make your digital currency grow?

The core concept of crypto lending is simple: Borrowers can use crypto assets as collateral for a fiat loan or stablecoin loan. Lenders provide the assets needed to repay the loan at an agreed-upon rate. The reverse can be true: Borrowers may use their crypto assets as collateral to borrow against other crypto assets[4](Crowell 2020.)

---

[2] Binance Academy (2021) What is staking? Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/what-is-staking (accessed 1 October 2021).
[3] Becker S (2021) What to know about staking - the process of locking up crypto holdings to earn rewards and interest. Business Insider, Business Insider. Available from: https://www.businessinsider.com/staking-crypto (accessed 1 October 2021).
[4] Crowell B (2020) Crypto lending, explained. Cointelegraph, Cointelegraph. Available from: https://cointelegraph.com/explained/crypto-lending-explained (accessed 27 September 2021).

The lender will earn interest from the borrower in return for the loan. The borrower will deposit crypto-assets above the principle to protect the investor's investments. This way, the lender can be certain that, if anything goes wrong, that collateral can be used as compensation[5] (CoinMarketCap 2021).

A crypto loan has several benefits. Unlike traditional banking, your credit score won't be assessed, which makes it easier to borrow money from people with poor credit histories, those who aren't financially well off, or self-employed workers whose incomes don't match the bank's lending criteria. Flexible repayments are also possible.

3.1.2. Yield Farming

Although Yield farming is a newer concept than Staking, the two concepts share many similarities. Yield farming is a way to provide liquidity to a pool with other 'Farmers'" that uses various protocols to increase yield based on different market variables[6](Vermaak 2021). Farmers are those who actively look for the highest yielding assets and rotate between pools to maximize their returns (often referred to as rotating crops.)

Yield farmer's are very secretive about the best yield farming strategies. Why? The more people know about a strategy, the less effective it may become[7].

Yield farming refers to the practice of lending or staking crypto assets in order to earn high returns and/or rewards in additional cryptocurrency. This risky, volatile, and innovative application of Decentralized Finance (DeFi) has seen a rise in popularity thanks to innovations in liquidity mining. The still-nascent decentralized finance sector is seeing Yield farming as the largest growth driver. It has helped it balloon to a market cap of $500m to $10billion in 2020.

Yield farming protocols encourage liquidity providers to lock up or stake their crypto assets in a smart contract-based liquidity pool. These incentives could be a percentage and interest from lenders measured in Annual Percentage Yields (APY) that are used to express the returns. The value of issued returns drops as more investors add liquidity to the pool requiring the rotation of the crops[8].

---

[5] CoinMarketCap (2021) What is crypto lending?: CoinMarketCap. CoinMarketCap Alexandria, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/article/what-is-crypto-lending (accessed 15 October 2021).

[6] Vermaak W (2021) What's the difference between staking, yield farming, and liquidity mining?: Certik Foundation blog. CertiK Foundation Building Secure Blockchain Infrastructure. Available from: https://www.certik.org/blog/whats-the-difference-between-staking-yield-farming-and-liquidity-mining (accessed 17 September 2021).

[7] Binance Academy (2021) What is yield farming in Decentralized Finance (DEFI)? Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/what-is-yield-farming-in-decentralized-finance-defi (accessed 14 September 2021).

[8] CoinMarketCap (2020) What is yield farming?: CoinMarketCap. CoinMarketCap Alexandria, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/article/what-is-yield-farming (accessed 16 September 2021).

3.1.3. Liquidity Mining

Liquidity Mining can be described as the same thing as having a savings account that earns interest. You agree to let bankers use your money for a certain interest rate when you deposit your money at the bank[9]. You agree to pool liquidity on an exchange in order to support the project's growth, in return for a reward for a voice in the future of the protocol.

The main difference between Yield Farming and Liquidity Mining is that later liquidity providers are compensated not just with fee revenue but also the platform's own utility token[10].

Like Yield Farming, Liquidity Mining provides liquidity through cryptocurrencies to Decentralized Exchanges (DEXs). DEXs reward those who bring capital to an exchange, as liquidity is the main goal of any exchange[11]. Many decentralized exchanges provide a governance model that allows users to vote, which makes them decentralized and is a major driver for the cryptocurrency community.

To ensure decentralization, developers usually distribute the cryptocurrency fairly. As a second source of rewards, governance tokens become an important part of Liquidity Mining. Many protocols allow Liquidity Providers to be rewarded with traditional yield rates, but also include governance tokens[12]. This further incentivizes liquidity mining, since LPs receive an additional stream of income. They also have the opportunity to take part in governance and get a share in the protocol. They can also change the protocol and its workings, or add liquidity pools through different methods that vary between DEX's that also provide a governance token for voting.

You deposit trading pairs in Liquidity Mining. This means that the pool will always have two different cryptocurrencies for borrowers to pull from. Each trading pair is therefore defined in its own smart contracts or pool. Token A can be sent to the address for trading. In return, the token B will be received at the indicated exchange rate.

3.2. Trading and Investment Strategies

[9] Eliot Couvat E (2021) Liquidity mining (staking) for social tokens. GCR - Home -. Available from: https://globalcoinresearch.com/2021/09/14/liquidity-mining-staking-for-social-tokens/ (accessed 1 October 2021).
[10] Vermaak W (2021) What's the difference between staking, yield farming, and liquidity mining?: Certik Foundation blog. CertiK Foundation Building Secure Blockchain Infrastructure. Available from: https://www.certik.org/blog/whats-the-difference-between-staking-yield-farming-and-liquidity-mining (accessed 17 September 2021).
[11] Wiesflecker L (2021) What is liquidity mining, and how does it work? Medium, Coinmonks. Available from: https://medium.com/coinmonks/what-is-liquidity-mining-and-how-does-it-work-d0ab491e607 (accessed 2021).
[12] What is liquidity mining? (2020) Bit2Me Academy. Available from: https://academy.bit2me.com/en/what-is-liquidity-mining/ (accessed 27 September 2021).

The Trading and Investment Strategies include the categories that allow the investor to use tools and processes for investment research and trading. These investments can be based on individual strategies or a combination of individual and community strategies. This term has not been defined by the industry. We will instead call it Collective Trading Intelligence, which produces a culmination of community investor strategies.

3.2.1. Algorithmic Trading

Algorithmic trading, which is programmatically executed trading strategies instead of manually placing orders, is a type that trades. Algorithmic trading is when traders use algorithms to program computers to buy, sell and hold assets.

Algorithmic trading has many benefits, including reducing costs and saving time. Multiple lines of code can simultaneously execute trades and check market conditions across multiple exchanges. All this while being available 24/7. The popularity of cryptocurrency algorithmic trading is evident among traders of all levels, novices, and professionals alike. This is because the software can quickly analyze the market and respond to it efficiently and speed that no human could match[13]. This software can have a significant impact on cryptocurrency markets by increasing liquidity and volatility and affecting their market rates.

Today, 58% of online traders are millennials and 75% of cryptocurrency traders are millennials. This includes Bitcoin, Ethereum, Polkadot, or any of the nearly 9000 cryptocurrencies that currently exist[14].

3.2.2. Research and Innovation

In the last few years, there has been an increased interest in what is technology innovation research and what type of research methods are being used for cryptocurrency. Some types of research methods are commonly known such as open-ended grants, commercialization, basic and applied research grants, and philanthropy. Other types of research methods are less well-known, including sociological surveys, political polls, technological proposals, and innovations policy[15]. While many seem to focus on what is technology innovation research and

---

[13] What is cryptocurrency algorithmic trading and how is it changing the way we trade? (2021) Arbismart. Available from: https://arbismart.com/blog/what-is-cryptocurrency-algorithmic-trading-and-how-is-it-changing-the-way-we-trade/ (accessed 17 September 2021).

[14] Using algorithms to trade in crypto currencies (2020) Small Business Insider. Available from: https://smallbusinessinsider.co.uk/using-algorithms-to-trade-in-crypto-currencies/#_edn2 (accessed 16 September 2021).

[15] Ashoor, A. and Sandhu, K. (2019) 'Framework and Model for Cryptocurrency Innovation and its Impact on Economic Transformation', International Journal of Innovation in the Digital Economy, 10(4), p. N.PAG. Available at: https://search.ebscohost.com/login.aspx?direct=true&AuthType=shib&db=edb&AN=137587866&site=eds-live&scope=site (Accessed: 3 October 2021).

what types of research methods are being utilized, it appears that some have overlooked the most important part of the innovations process - the public engagement.

When innovation occurs, it is generally not in response to a private sector company or governmental agency. Typically, technology is developed by an individual, a group of people, or a community. Often, individuals with an interest in advancing technology will seek other individuals with an interest in advancing technology as well, but without necessarily consulting a governmental agency. For example, when someone develops a new product or technology, they will often consult with groups and individuals interested in using the new product or technology before developing it into a business application. However, without this form of public consultation and engagement, technology innovation would not be possible.

Historically, most businesses and governments do not take public innovations seriously. This means that when innovation happens, such as the case with cryptocurrency, it is often not well received by the status quo, leaving them feeling threatened and unable to devise alternative strategies to maintain the structure. Governments and businesses will require significant degrees of tangible research and the results of this research to declare the cryptocurrency market as a new economy.

The expansion of digital finance continues to penetrate developing countries that lack basic financial services (e.g., El Salvador in 2021). This has led to the future adoption of cryptocurrency services to the nonfinancial sector and new developments in digitally transforming standard bank services into cryptocurrency capabilities utilized by smartphones. In 2023, 53.8 percent, more than half the population, will have a smartphone capable of accessing and using cryptocurrencies[16]. With the increasing acceptance of digital finance innovations and the need to keep up with the competition, GSHI must be competent and focused on the needs of underdeveloped entities.

3.2.3. Non-Fungible Token (NFT) Management

The entire Non-Fungible Token (NFT) market cap on the Ethereum blockchain is worth more than $14.27 billion, and therefore outperforms for example cryptocurrencies like Bitcoin Cash and Litecoin. At the current market rate, NFTs would be in the 12th spot just below Uniswap[17].

A NFT is actually a physical unit of information stored on a computer database, which certifies that a particular digital item is actually unique and cannot be copied or cloned. Basically, a NFT is used to represent things like photos, movies, audio, and any other kind of digital data. Any

[16] Gaubys J (n.d.) How many people have smartphones? [Jul 2021 update]. Oberlo, Oberlo. Available from: https://www.oberlo.com/statistics/how-many-people-have-smartphones (accessed 21 September 2021).
[17] Hoogendoorn R (2021) NFT market cap bigger than bitcoin cash or litecoin. DappRadar Blog RSS. Available from: https://dappradar.com/blog/nft-market-cap-bigger-than-bitcoin-cash-or-litecoin (accessed 24 September 2021).

information which can be created in the future by using a computer program can be placed on a non-fungible token.

Non-fungible tokens, unlike cryptocurrencies where every token is the same, are exclusive and limited in number. The item cannot be duplicated or copied. For example, bitcoin is fungible. It can be traded for another bitcoin to get the exact same thing. A one-of-a kind trading card cannot be resold. If you trade it for another card, you would get something completely different. These tokens could be digital assets, or tokenized versions of real-world assets. They can be used to prove ownership and authenticity in the digital realm, but they cannot be interchangeable.

3.2.4. Investor Education

The goal of any good Cryptocurrency investor education program is to teach both groups how to recognize value in Cryptocurrencies and how to use their inherent advantages to maximize their profits.

Many investors are not fully acquainted with the concept of what cryptocurrency is. Some people think that it is a fad, a passing trend that will soon be forgotten, but the truth is that the future value of cryptocurrencies is increasing exponentially. As more people embrace the idea of being their own futures trader, the increases in both retail and institutional investments will result in numerous financial and investment mediums quickly taking shape. Many of these investors will use their own cryptocurrency investments to indirectly help fund the furtherance of financial crisis efforts, while some may contribute to different elements such as Non-Fungible Tokens or Liquidity Mining.

Cryptocurrency is becoming omnipresent as a digital asset that is bought and sold over the Internet and can be anything from artwork to computer programs to E-Books to applications. In the past, many of the largest online platforms were the world's first currency trading platforms and were used by millions around the world seemingly overnight. By providing a better understanding of how cryptocurrency works, customers will be more privy to adoption at the same pace.

Aiding others in understanding cryptocurrency will help realize that there are two schools of thought when it comes to investing in cryptocurrencies. There are those who hold cryptocurrency investment in the hopes that it will become a mainstream investment vehicle. Others who are trying to make a profit by taking a long position on the future of cryptocurrencies - both methods are demonstrating significant opportunities for different reasons.

4. Cryptocurrency Regulations, Legislation, Compliance, and Audits

4.1. Difference Between Cryptocurrency Legislation and Regulations

It is not an easy task to distinguish between "legislation" and "regulation, 'or as Levi-Faur puts it, 'regulation is hard to describe, not least because it means different things to different people."

Does the meaning of legislative terms, such as statutes, acts, and laws, differ in countries with different characteristics? For example, in the United States, Congress adopts Acts of Congress. In the United Kingdom, Parliament adopts the Act of Parliament. Leyes (laws in Mexico) refers to the different types of legislation that the Mexican Congress makes. Lois (laws in France) refers to the legislation enacted by the French Parliament. **Although this question is open to debate, it is much easier to identify and define legislation than to identify and define regulation.** The United States has federal agencies that issue regulations. However, in the UK, regulations are often referred to as one type of regulation and various ordinances, orders, and rules. Similar to Australia, legislation instruments include codes, notices and ordinances, proclamations, and regulations. For example, in the EU, regulations, directives, and decisions are binding laws[18].

The difference between legislation and regulation is that legislation is the act or process of making certain laws while regulation is maintaining the law or set of rules that govern the people.' - askanydifference.com

For this document, we recommend that you view legislation in the form of something being introduced into government, that If enacted, will become a regulation that is enforced by the laws of the same jurisdiction.

4.2. Cryptocurrency Legislation

In a broader context, the impact of cryptocurrency legislation in the financial sector cannot be ignored. Cryptocurrency is nothing more than a new technology that has been brought to the public, much the same way the Internet was, in its most nascent form, introduced to the world before everyone had an email address. Cryptocurrency has an enormous potential to create significant applications in a number of industries and could even overcome the traditional financial instruments like stocks and bonds where crypto wallets are as common as email addresses are today.

There are several cryptocurrency legislation solutions being debated in both the US and EU. Some countries such as India, China, and South Korea have taken up the issue because there is significant political and social pressure being applied to the various governments of these regions to introduce some form of legislation to regulate the use of cryptocurrency.

---

[18] authors, All, et al. “Legislation and Regulation: Three Analytical Distinctions.” Taylor & Francis, 12 Mar. 2020, https://www.tandfonline.com/doi/full/10.1080/20508840.2019.1736369.

In the US, there have been proposals to introduce legislation that would impose limitations on the activities of virtual currencies in place of imposing heavy penalties on individuals or businesses involved in activities involving cryptocurrency, simply due to not understanding the technology where it is easier to say no to the innovation then it is to find a path into it.

Similarly, in Europe, several pieces of legislation have been introduced to curb cryptocurrency activities. However, none of these measures has had the desired effect. In some cases, the efforts have even led to the further restriction of cryptocurrency. In others, it appears the only way to curb cryptocurrency activity in the European Union is for the European governments to classify it as a financial instrument that may be misused and therefore require a licensing system for its use.

The European Commission's Regulation of Markets in Cryptoassets (MiCA), as part of the EU's Digital Finance Strategy, is a proposal that was developed in 2018 to regulate crypto-assets defined as a new class of digital asset designed for other cryptocurrencies that do not qualify as financial instruments, including utility tokens and payment tokens, all fall within the scope of the regulation. MiCA is intended to provide a single licensing system across all member countries for cryptocurrencies by 2024.

MiCA will only prove a success; however if the barriers to entry are not limiting to the efforts of smaller start-ups. The EU commission's impact assessment of the proposal currently estimates 35000-75000 EUR one-off costs for the whitepaper, 2.8-16.5 EUR million one-off compliance costs for unregulated entities, plus 2.2-24 million recurrent compliance costs (capital reserves, reporting, IT-security, governance, etc.). These financial and administrative burdens could prove impossible for some of the younger market participants[19].

The changes MiCA attempts to address will help protect consumer funds from cyber-attacks and theft, as well as malfunctions that are under the control of cryptocurrency exchanges[20]. However, some crypto-specific risks are not addressed in the regulation. Despite the fact that the regulation holds cryptocurrency exchanges responsible if consumer assets are lost due to fraud, cyber-attack, negligence, or other causes, it doesn't extend compulsory insurance requirements to these types of cases or accidental losses.

One of the primary concerns of regulators around the world is the impact on the country's macro-economy. The worry is that if cryptocurrency becomes popularized and therefore

---

[19] Hansen P (2021) New crypto rules in the European Union – Gateway for mass adoption, or excessive regulation? New Crypto Rules in the European Union – Gateway for Mass Adoption, or Excessive Regulation? Available from: https://law.stanford.edu/2021/01/12/new-crypto-rules-in-the-eu-gateway-for-mass-adoption-or-excessive-regulation/ (accessed 4 September 2021).

[20] Cengiz, Firat, et al. "What the EU's New Mica Regulation Could Mean for Cryptocurrencies." EUROPP, 5 July 2021, https://blogs.lse.ac.uk/europpblog/2021/07/05/what-the-eus-new-mica-regulation-could-mean-for-cryptocurrencies/.

becomes the de facto standard in countries, where demand for the local currency falls. In addition, if this occurs, the government is unable to change the current laws which may prevent the government from maintaining aggressive taxation of its citizens' wealth. The end result of this would be a significant decrease in the country's revenue and consequently affect public spending and growth, a move common with decentralized finance. China, until recently was home to some of the largest crypto exchanges in the world. However, Chinese authorities view cryptocurrencies as speculative instruments that lack intrinsic value and are susceptible to price swings. They also see them as a way to circumvent capital control. Instead, the Chinese authorities have backed the creation of a digital currency[21].

Another concern is the impact that cryptocurrency legislation may have on the country's stability. Many have speculated that because of this new technology there will be an increase in hacking where tens of millions of dollars can be lost in a single moment[22]. It is believed that if this is the case that the legislation may actually reduce the effectiveness of numerous government agencies involved in national security as they are unable to identify hackers quickly. In addition, this would mean that the cybercrime landscape will be more volatile. The best way to address this is by regulating the sale of cryptocurrency to combat any unnecessary risk, while still ensuring that its users have a safe and secure environment to operate from[23].

[21] Person, and Andrew Galbraith Samuel Shen. “Cryptocurrency Exchanges Scramble to Drop Chinese Users after Beijing's Ban.” Reuters, Thomson Reuters, 27 Sept. 2021, https://www.reuters.com/technology/cryptocurrency-exchanges-rush-cut-ties-with-chinese-users-after-fresh-crackdown-2021-09-27/.
[22] Reiff, Nathan. “The Largest Cryptocurrency Hacks so Far.” Investopedia, Investopedia, 12 Sept. 2020, https://www.investopedia.com/news/largest-cryptocurrency-hacks-so-far-year/.
[23] Anwar Sheluchin PhD Student. “A National Digital Currency Has Serious Privacy Implications.” The Conversation, 14 June 2021, https://theconversation.com/a-national-digital-currency-has-serious-privacy-implications-130520.

EU Policymakers Propose Tighter Regulation of Crypto Transfers

The European Commission made the proposal to help crack down on flows of illicit money.

European Union (EU) policymakers have proposed tightening regulations on the transfer of crypto assets by requiring companies to collect details of senders and recipients.

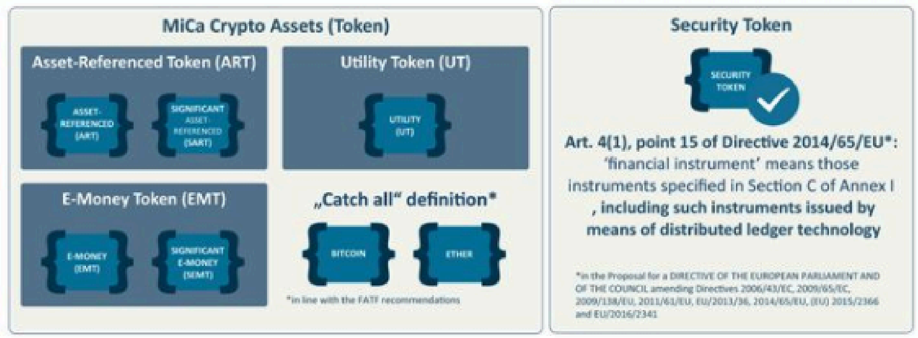
- The European Commission, the EU's executive arm, made the proposal to help crack down on the flow of illicit money.
- Such a law would extend to crypto the Financial Action Task Force's "travel rule" that already applies to wire transfers.
- The aim of the legislation would be to "ensure full traceability of crypto-asset transfers" and "allow for prevention and detection of their possible use for money laundering and terrorism financing," the Commission said.
- The travel rule requires service providers to exchange identifying information such as customer's name, address and account details.
- Extending these requirements to crypto transfers would prohibit the use of anonymous crypto wallets.

The people want decentralization and transparency- transparent economics.

Survey finds Europeans want home countries to regulate crypto, not EU

A recent poll regarding European cryptocurrency policy saw the participation of 31,000 respondents across 12 member states of the European Union.

Redfield & Wilton Strategies carried out a survey for Euronews, polling 31,000 respondents from Estonia, France, Germany, Greece, Hungary, Italy, Latvia, Lithuania, the Netherlands, Poland, Portugal and Spain.



GSHI watchlist.

MiCa EU

The European Commission's Regulation of Market in Crypto-assets (MiCA) proposal is a regulatory framework developed since 2018 to help regulate currently out-of-scope crypto-assets and their service providers in the EU and provide a single licensing regime across all member states by 2024.

4.3. Cryptocurrency Regulation

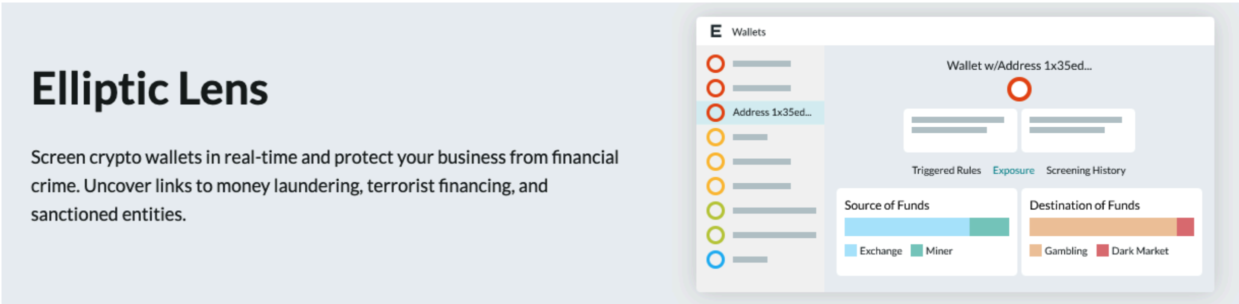
There is much confusion among merchants and consumers as to the Cryptocurrency Regulations EU and the Cryptocurrency Regulations UK. Both of these terms are often used interchangeably by most people. They both refer to a form of virtual payment, which can be used on the Internet and across mobile devices worldwide. In essence, cryptocurrency protocols provide platforms for end-users to pay for goods and services in a digital currency rather than fiat or cash. New legislation, put forth by the European Union, covers all financial institutions and brokers who offer any type of virtual payment service to collect additional AML and KYC information for those that are conducting cryptocurrency transfers[24]. Some of the things that the Cryptocurrency Regulations EU and Cryptocurrency Regulations UK both cover are how virtual currencies are issued, how and how they may trade, how their prices are set, what types of activities may be conducted with virtual currencies.

One of the biggest problems with this is that not every country or region is adequately compliant with the cryptocurrency regulations. This is because some people, and governments in the EU do not view the world of digital currencies as being something that they should interfere with[25]. In other regions and countries, the cryptocurrency regulations are much more strict. For instance, in the European Union, there are many legal battles ongoing between various European countries and some are fighting over whether or not private digital currencies are allowed to be traded freely within their jurisdictions. Even within those regions, some regions are only partially covered, leaving traders and investors in an area without significant legal protection.
The emergence of crypto-asset risk management firms such as Elliptic provide regulatory compliance services that banks can navigate the rapidly-evolving crypto ecosystem with enough confidence to[26]:

- Understand their overall exposure to crypto-assets, through their customers' activities.
- Identify customers whose crypto-asset activities pose heightened money laundering or sanctions risks.
- Assess the risk profiles and KYC procedures of different crypto-asset exchanges, and enable them to work more closely with these businesses.

---

[24] McBride, Landon. “New EU Proposal Looks to Tighten Regulations for Sending Cryptocurrency.” Cointelegraph, Cointelegraph, 20 July 2021, https://cointelegraph.com/news/new-eu-proposal-looks-to-tighten-regulations-for-sending-cryptocurrency.
[25]Cengiz F (2021) What the EU's new mica regulation could mean for cryptocurrencies. EUROPP. Available from: https://blogs.lse.ac.uk/europpblog/2021/07/05/what-the-eus-new-mica-regulation-could-mean-for-cryptocurrencies/ (accessed 3 September 2021).
[26] Elliptic (2019) Elliptic discovery enables banks to assess their exposure to cryptoassets. Blockchain Analytics & Crypto Compliance Solutions. Available from: https://www.elliptic.co/media-center/elliptic-discovery-enabling-banks-identify-crypto-exposure (accessed 5 September 2021).



Cryptowallet screening for KYC/AML Source: elliptic.co

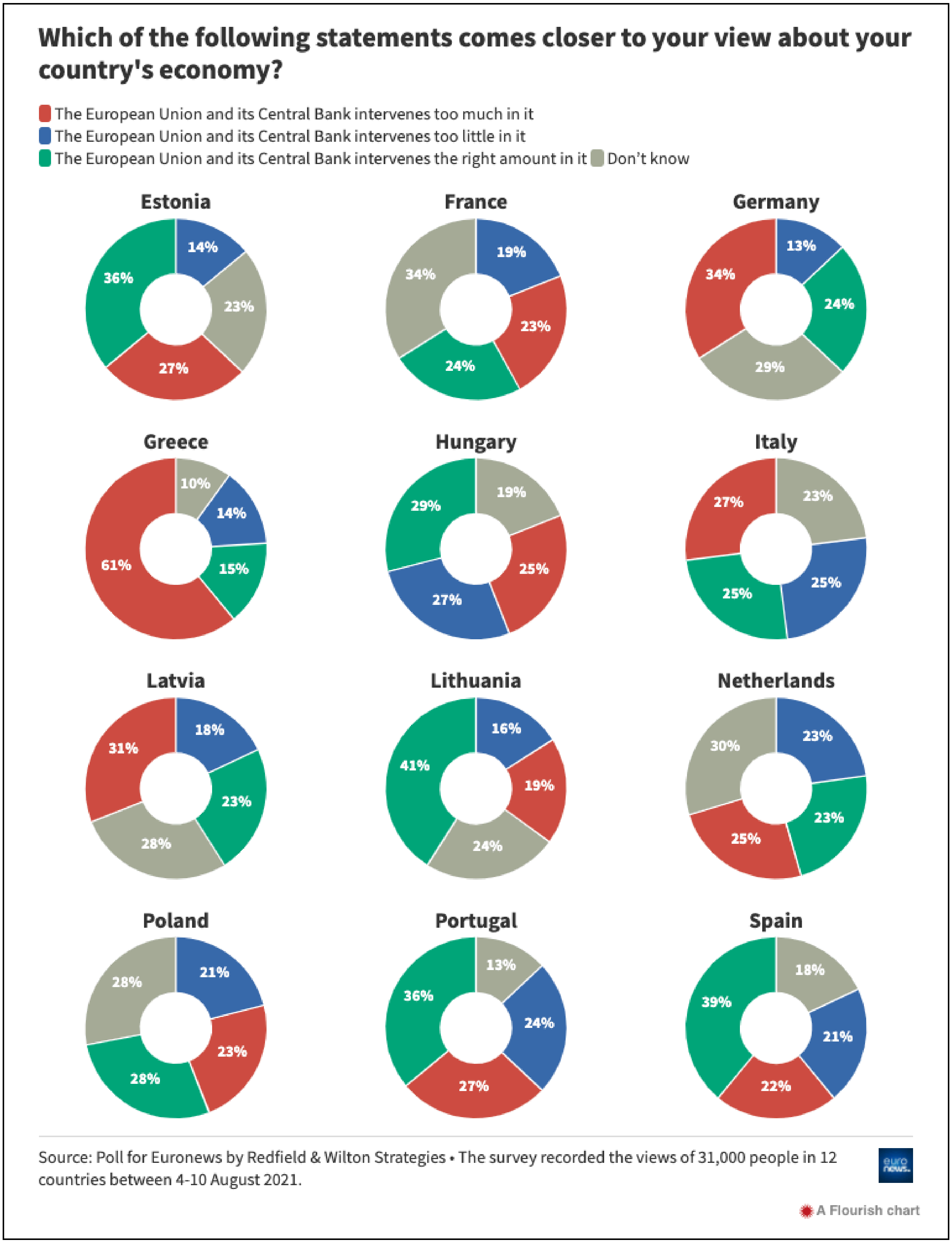


Figure X: euronews.next

The problem with regulating the entire cryptocurrency marketplace is that it would be nearly impossible to police and regulate over 9000 altcoins[27]. There are too many players in the market and the landscape is constantly changing. This causes a problem, especially when new technologies or laws come out that make certain methods of conducting business with cryptocurrency users illegal or harmful to the general public[28]. Because of this, the responsibilities of the regulatory agencies assigned to regulate the currencies themselves are limited[29].

Luckily, some cryptocurrency regulators have recognized the importance of regulating the cryptocurrency themselves, in an effort to avoid issues that could be harmful to both users of cryptocurrency and the general public who interact with it during a transaction. In fact, some cryptocurrency regulators are working on an algorithm that determines which cryptocurrencies are good to trade in and which ones are bad. This allows for maximum growth in the amount of the total value of all cryptocurrency on the market, but prevents the general public from being ripped off by unscrupulous publishers of "get rich quick" schemes. Unfortunately, it still leaves the investor in a position where he must exercise due diligence and caution when selecting which cryptocurrency they wish to invest in.

GSHI will greatly benefit from the development of a standardized interface that allows investors to easily access the information that they need in order to make informed decisions regarding which cryptocurrency to purchase. By creating standardized interfaces for the different cryptocurrencies, investors will be able to get the same information and applications that they would if they were visiting a physical store. This will allow for easier comparisons between the various currencies that are being traded on online exchanges such as those found on major financial institutions such as NYSE, NASDAQ, and AMEX.

In order to be approved by the regulators, a diverse collection of metrics must be calculated in order for a particular cryptocurrency to be listed on a cryptocurrency exchange. These measurements can be anything from supply and demand indicators, to profit margins, to metrics that compare different currencies against one another. The use of these metrics, coupled with the mathematical algorithm that is used to identify which cryptocurrency should be included in the marketplace, gives rise to a marketplace that will be free of scammers, rip-offs, and frauds.

A key goal of these regulations is to maintain the safety and liquidity of the cryptocurrency.

---

[27] Austin, Ryan. “Altcoins Are the Alternative Digital Currencies to Bitcoin - Here's What They Are and How They Work.” Business Insider, Business Insider, 1 Sept. 2021, https://www.businessinsider.com/what-is-altcoin#:~:text=Since%20the%20emergence%20of%20Bitcoin,of%20more%20than%209%2C000%20altcoins.

[28]Alper, Tim. “Pos Coins, Lightning, Defi & Dexes in Danger as US Bill Chaos IntensifiesTim.” Crypto News, https://cryptonews.com/news/proof-of-stake-coins-in-danger-as-us-infrastructure-bill-cha-11337.htm.

[29] Ilkogretim Online - Elementary Education Online, 2021; 20 (3): pp. 1541-1550 http://ilkogretim-online.org, doi: 10.17051/ilkonline.2021.03.172

As technology continues to improve and grow, new forms of regulation will surface as well. One such development will be created for a global community of cryptocurrency users in the form of a marketplace e.g., W3 (World Wide Web 3)[30]. A future version of the aforementioned regulations will be a list of "exchanges" that can list digital currencies, all of which must meet the same minimum standards set forth by the respective jurisdictions for both the exchange and customers; all who are participating global users who have the ability to participate in this emerging market without having to worry about the same issues that apply to businesses who deal in traditional money. A future regulation may even define a new virtual currency category altogether, or may attempt to apply regulations to certain segments of the entire market.

In the end, it is very important for everyone, including business owners and consumers, to understand the different facets of these regulations. By doing so, we will be able to reap the full benefits of the free market that goes into cryptocurrency trading. This will further help to keep our financial services free of frauds and scams, allowing us to enjoy fair prices and convenient accessibility.

4.4. Cryptocurrency Compliance

One of the first considerations that any company should take is whether they will need to obtain individual licenses to service and trade cryptocurrency customers' accounts. Depending on the company's size and location, there may be a need to designate one individual to manage cryptocurrency compliance while the company completes and maintains its necessary licensing processes. The responsibility of this individual is to determine what is needed to maintain appropriate compliance and then fill out all the required forms. Once the papers have been completed and submitted to proper departments and authorities, the company will be given a license to operate.

Once the license has been granted, the next step may be determining what type of monitoring will be performed. Two primary methods of monitoring may be used by a company: manual monitoring and automation. Manual monitoring involves receiving reports on a weekly or monthly basis and analyzing the information. Automation is more hands-on; however, the software must generate an adequate report daily. Both methods are necessary for cryptocurrency compliance, as losses may not be properly investigated if a company does not monitor activity closely enough[31].

GSHI will benefit greatly from including the correct monitoring tools e.g., keyword triggers for compliance notifications events.

[30] https://www.w3.org/
[31] McBride L (2021) New EU proposal looks to tighten regulations for sending cryptocurrency, Cointelegraph. Available from: https://cointelegraph.com/news/new-eu-proposal-looks-to-tighten-regulations-for-sending-cryptocurrency (accessed 4 September 2021).

When a business becomes aware that one or more of its employees may be using an inappropriate method of cryptocurrency communications, it may want to further examine what is cryptocurrency compliance. In addition to the above-mentioned obligations, companies may also want to determine what types of outside sources it is providing information to. Some external sources may need to be verified to ensure they are not leaking information to the public.[32]

When a business chooses to implement what is Cryptocurrency Compliance, it may find itself on the cutting edge of the ever-changing digital landscape. By having a solid foundation in place, a company can be sure to thrive in the online world. Companies that choose to maintain what is Cryptocurrency Compliance will likely find their company to grow and flourish. The best way to do this is to maintain the information that is Cryptocurrency Compliance. By doing so, any and all security breaches or other issues may be avoided.

4.5. The Controversy About Audits

The Financial Conduct Authority, UK, the regulator in charge of overseeing the role of banks and other financial institutions in facilitating trading and investing, is now considering making it mandatory for digital currencies like BitUSD, BitUX, and Monero to undergo a cryptocurrency audit. This move comes on the heels of two major news events. The first was the announcement by the FSA that it will be investigating whether or not these types of coins and tokens have been created in a manner that they can be considered a “real” currency. The second came in the form of news that the MiCA, the financial regulator in the island nation of Jersey, is looking into the matter.

There are at least two main arguments against making this type of audit mandatory. The first comes from those who feel that cryptocurrency is simply a new name for the same thing. ‘Those who base their arguments on this feeling that it is impossible to separate the worth of currencies based on their design.’ It is argued that a cryptocurrency audit would open up a can of worms that will damage all future cryptosystems.

Those who support a cryptocurrency audit feel that those against doing so are acting in bad faith. They argue that those against an audit would do whatever they can to muddy the waters to convince investors to invest in a particular currency rather than another. If investors follow the law, then they will not be committing fraud or another crime. Investors would lose confidence in the system, they argue, and the system’s value would fall. If this happened, there would be an immediate and negative impact on economic growth. This is why many investors need to get in touch with those responsible if something does go wrong. They would also need access to records if and when they are needed, and support for an appeals process.

[32] European Parliament P and Council P (2020) Proposal for a Regulation Of The European Parliament And Of The Council on Markets in Crypto-assets, and amending Directive (EU) 2019/1937. Brussels: European Commission.

4.6. Smart Contract Audits

Smart code audits are crucial to validating any cryptocurrency mechanism supported by the market. This is evident because millions of dollars worth of funds were at risk due to numerous code vulnerabilities that have exploited many blockchain protocols such as Ethereum, resulting in the chain splitting in 2016. An attacker used a “recursive call bug,” which allowed him to steal millions of dollars of ETH from a DAO. Following disagreements, the community decided to return the funds by force. This led to a hard fork.[33]

Smart contract audits are conducted by professionals who specialize in smart contract technology. These experts work together with the programmers and other specialists to verify that a certain application is compliant with the terms and conditions of the decentralized Internet work.

Smart contract audits are conducted by scrutinizing the functions of the smart contracts themselves. Basically, this kind of an audit involves the verification of every line and detail of the smart contract. Such auditing will help the developers improve the operation of their applications. The main reason why there is a need for such auditing is the fact that some applications may be vulnerable to certain kinds of security flaws. Through the help of these tools, we can expect to protect ourselves from the damages that these flaws may bring.

Smart contract audits that are done is the review which aims to make sure that the functionality of the decentralized application is not limited or compromised. When the developers use smart contract based software, they make sure that every transaction that goes through the said platform is protected and monitored perfectly. As a result of this, the transactions become secure and safe. With this being said, the developers as well as users are assured that they will get what they pay for - a secure and fully functional governance platform.

4.7. Smart Audit Companies and Summary

Company	Company Summary
Audithor	Audithor offers smart contract and blockchain technology security and performance audits, smart contract and dapp development, and digital forensics services. It is a diverse and dynamic group of professionals forming a coherent whole on a project-per-project basis.
Authio	Authio is a full service smart contract security firm focused on mitigating technical debt and security risk for projects building decentralized applications on Ethereum networks.

[33] CoinMarketCap (2021) Smart contract audit: CoinMarketCap. *CoinMarketCap Alexandria*, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/glossary/smart-contract-audit (accessed 4 September 2021).

Blockchain Consilium	Blockchain Consilium is a Smart Contract Security Auditor that specializes in DeFi (Decentralized Finance), NFT (Non-Fungible Tokens), ERC20 Tokens and Crowdsale smart contract auditing. With 96 audits completed and $2.55Bn+ secured, Blockchain Consilium offers an unmatched experience of professional smart contract auditing services.
Certik	Certik describes themselves as the premier blockchain and smart contract cybersecurity startup.
ChainSecurity	ChainSecurity are the winners of multiple Ethereum Foundation grants and the creators of Securify, a security scanner for Ethereum smart contracts, as well as Chaincode Scanner, a security analyzing tool for hyperledger fabric smart contracts. They also offer custom tailored smart contract audits to their clients.
Chaitin	Chaitin Future Technology Co.,Ltd providing professional security auditing and penetration testing services for exchanges, blockchains, contracts, wallets, mining pools and other blockchain projects.
Cheetah Mobile Security	Cheetah Mobile Security is security company from China. Their research covers business security, security audits, and public opinion monitoring. They are committed to building a healthier, more secure blockchain ecosystem.
CoinFabrik	CoinFabrik audits smart contracts and apps looking for exploitable security flaws, errors, inefficiencies and any unexpected behavior.

5. Decentralized Autonomous Organizations

5.1. How Does a Decentralized Autonomous Organization Benefit the Bottom Line?

A Decentralized Autonomous Organization (DAO), sometimes also known as a Decentralized Autonomous Corporation (DAC), is a company governed by smart rules encoded into its computer code that are entirely self-enforcing and independent of any central government. DAOs are internet-native organizations collectively owned and managed by their members. They have built-in treasuries that are only accessible with the approval of their members. Decisions are made via proposals the group votes on during a specified period. The concept is very much similar to that of a Decentralized Autonomous Government. However, i**n a**

Decentralized Autonomous Organization, each member possesses a computer terminal with a stored copy of the smart contract that governs the behavior of all members[34].

Everyone who belongs to the DAO has a vote when it comes to making decisions for the organization; however, no single individual or group can change the code that governs the program[35]. This means that a DAO is entirely controlled and governed by its computer code. In fact, one could argue that the entire point and purpose of owning a DAO is the fact that no single entity controls it.

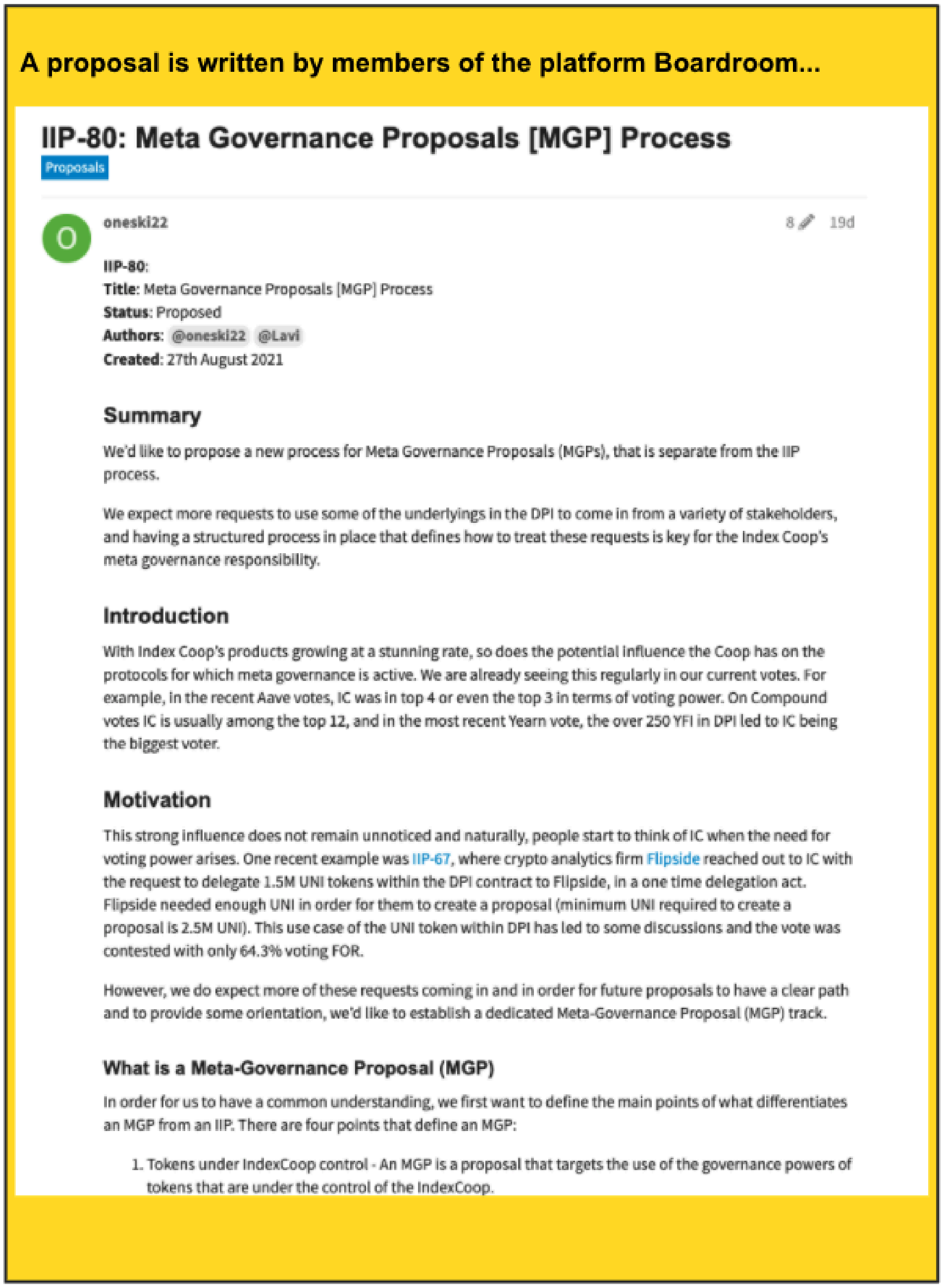
5.2. Decentralized Autonomous Organization (DAO)

There are many different theories as to what is the purpose of a DAO. Some people theorize that it is a way for wealthy individuals to control major functions of society through their investment capital. In other theories, it is a way for workers to band together to form a massive movement. No matter what the original motivation behind owning a Decentralized Autonomous Organization (DAO), one thing is clear: a DAO will be a highly efficient method of pooling collective intelligence and action in order to solve collective goals[36].

One of the most unique features of a Decentralized Autonomous Organization is its ability to adjust its internal rules at any given time without having to wait for a popular consensus[37]. Therefore, a Decentralized Autonomous Workforce (DAW) has the ability to coordinate its actions in a multitude of ways. For example, a decentralized workforce might decide that they want to go on strike. With the help of a Decentralized Autonomous Manager, the workforce might decide to refuse to work until their demands are met. The following illustrations are an example of how a member of a DAO can create and submit a proposal for voting. In this example, we are using Boardroom token governance.

[34] Matt Hussey AH (2021) What is a Decentralized Autonomous Organization (DAO)? Decrypt, Decrypt. Available from: https://decrypt.co/resources/decentralized-autonomous-organization-dao (accessed 3 September 2021).
[35] Government by algorithm (2021) Wikipedia, Wikimedia Foundation. Available from: https://en.wikipedia.org/wiki/Government_by_algorithm (accessed 3 September 2021).
[36]
[37] Dao Overview - a list of Ethereum's top Daos and dao structures (2021) DeFi Rate. Available from: https://defirate.com/daos/ (accessed 4 September 2021).

The proposal is submitted with the related information.[38]



A proposal is written by members of the platform Boardroom...

IIP-80: Meta Governance Proposals [MGP] Process

Proposals

oneski22 8 19d

IIP-80:
Title: Meta Governance Proposals [MGP] Process
Status: Proposed
Authors: @oneski22 @Lavi
Created: 27th August 2021

Summary

We'd like to propose a new process for Meta Governance Proposals (MGPs), that is separate from the IIP process.

We expect more requests to use some of the underlyings in the DPI to come in from a variety of stakeholders, and having a structured process in place that defines how to treat these requests is key for the Index Coop's meta governance responsibility.

Introduction

With Index Coop's products growing at a stunning rate, so does the potential influence the Coop has on the protocols for which meta governance is active. We are already seeing this regularly in our current votes. For example, in the recent Aave votes, IC was in top 4 or even the top 3 in terms of voting power. On Compound votes IC is usually among the top 12, and in the most recent Yearn vote, the over 250 YFI in DPI led to IC being the biggest voter.

Motivation

This strong influence does not remain unnoticed and naturally, people start to think of IC when the need for voting power arises. One recent example was IIP-67, where crypto analytics firm Flipside reached out to IC with the request to delegate 1.5M UNI tokens within the DPI contract to Flipside, in a one time delegation act. Flipside needed enough UNI in order for them to create a proposal (minimum UNI required to create a proposal is 2.5M UNI). This use case of the UNI token within DPI has led to some discussions and the vote was contested with only 64.3% voting FOR.

However, we do expect more of these requests coming in and in order for future proposals to have a clear path and to provide some orientation, we'd like to establish a dedicated Meta-Governance Proposal (MGP) track.

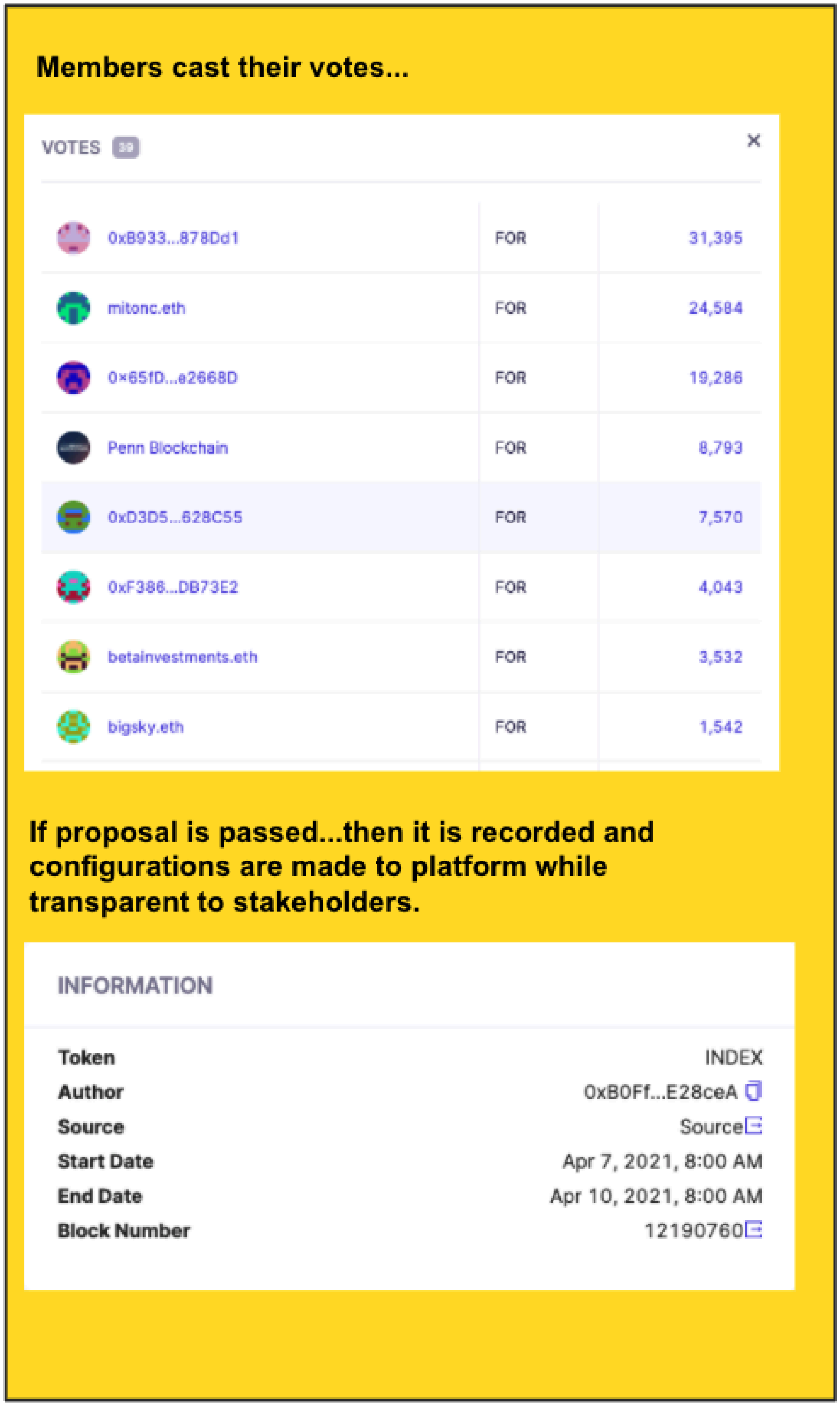
What is a Meta-Governance Proposal (MGP)

In order for us to have a common understanding, we first want to define the main points of what differentiates an MGP from an IIP. There are four points that define an MGP:

1. Tokens under IndexCoop control - An MGP is a proposal that targets the use of the governance powers of tokens that are under the control of the IndexCoop.

[38] For full text: https://gov.indexcoop.com/t/iip-80-meta-governance-proposals-mgp-process/2492

Members then cast their votes, and if the proposal is passed, it is recorded as so.



Transparency is created and can be viewed by anyone, including regulators.

The core of this proposal...

Current IndexCoop Token Holdings & Proposal Thresholds

Protocol	Tokens required (/delegated) to create a proposal	# of tokens IC has	% of threshold requirements	Governance Tool	Comment
Aave	80k (0.5%)	102k	100%	Aave Governance V2 (short time lock)	Controls Aave V1 and V2, risk parameters, token distribution, fees, etc.
	320k (2%)	102k	31.8%	Aave Governance V2 (long time lock)	Controls changes to the AAVE and stkAAVE token and updates to governance parameters
Compound	65k	42.8k	65.8%	Governor Bravo	DPI holdings only
Uniswap	2.5M	1.76M	70.4%	Governor Bravo	
Yearn	1	283	100%	Snapshot	
Balancer	0	85.8k	100%	Snapshot	

Decentralized Autonomous Organizations provide the transparency into governance that most regulators seek; but seem to fail grasp the concept of a Algocracy in place of traditional Centralized Governance Models.

As mentioned above, a decentralized system allows for a much larger degree of collaboration and leadership. If top management and employees feel that they have the power to make decisions, they will be much more likely to make those decisions. Because everyone is working together in a decentralized environment, there is less resentment towards an executive's decisions.

The bottom line is that a company's performance can only go so far as its employees. By decentralizing the control of a company, it gives employees more confidence in their jobs and their capacity as employees. Because the bottom line is that no one person can direct a company without input from everyone, the more employees there are, the better the company performance will be.

Why should top managers or employees participate in a decentralized system? Because they are given the freedom and the power to make decisions themselves rather than following an already established and rigid process. If top managers or employees don't like the current rules, they can simply create a new set of rules that will benefit them and everyone else.

A Decentralized Autonomous Organization increases openness and accountability, two very important qualities in a company. This means that the company has a much better chance of maintaining a good quality of service or product than a company where only a handful of people actually controls everything[39].

How does this improve bottom-line performance? Top employees and managers can make informed decisions about how to optimize the company's resources, its production, and its overall output.

Let's look into the near future. The Internet-native organizations may decide to run their DAO's without a governing board. Since the Internet has everything, they will just look at their decentralized autonomous organization charter as the reference point, because they have put all their eggs into this basket where anyone can change anything at any time, creating a charter with an unstable framework. So as you can see from above, even if the Internet-native organizations call the shots, the governance process cannot be left to anyone. This was the case in 2016 when MIT published an article on why spending $130M on a DAO was not in the best interest of the investors. Or as quoted by Tom Simonite *"like so many ideas buzzing around the cryptocurrency world, DAO probably won't live up to its own hype. Indeed, the DAO's success at attracting funds can be seen as an indicator of a bigger problem facing cryptocurrencies."*[40] Fast forward six years from the time of this quote to today and the DAO

[39] Liebkind J (2021) Daos, blockchain, and the potential of ownerless business. Investopedia, Investopedia. Available from: https://www.investopedia.com/news/daos-and-potential-ownerless-business/ (accessed 4 September 2021).

[40] Simonite T (2020) The "Autonomous Corporation" called the DAO is not a good way to spend $130 million. MIT Technology Review, MIT Technology Review. Available from: https://www.technologyreview.com/2016/05/17/160160/the-autonomous-corporation-called-the-dao-is-not-a-good-way-to-spend-130-million/ (accessed 5 September 2021).

market cap is over $40 billion with over $5 billion in 24-hour trading volume and continues to thrive under self-rule.

A DAO is where the future of corporate governance and smart contract technology intersect. Since no one will be in charge of the governance process, it becomes imperative that the corporate governance experts (the people who study, analyze, design, develop, deploy and oversee) become members of the digital community. By being a part of the digital community, these experts will be able to influence the future development of cryptocurrency technologies. This way, the future of corporate governance and smart contract technologies will be based on open source technology, which has the flexibility to integrate into any system, and it also has the ability to extend its functionality beyond the current system.

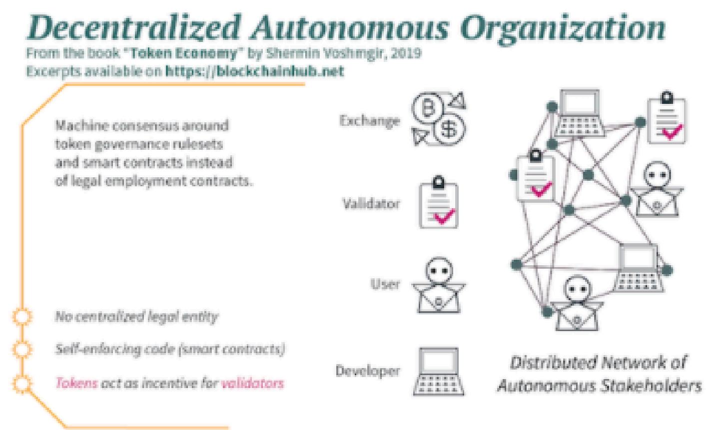
Figure (x) The Principal-Agent Dilemma and DAO Compared

Trusted
(Principal-Agent Model)

A DAO is a solution to the principal-agent dilemma.

This dilemma is a conflict in priorities between a person or group (the principal) and those making decisions and acting on their behalf (the agent).

Trustless
(Algocracy)



Smart contracts can be extremely helpful for automating transactional processes and reducing inputs that humans have to provide for simple tasks[41]. A Decentralized Autonomous Organisation's goal is not to reduce human inputs, but to eliminate them completely. Although it's still an idea on paper, the DAO is a business that utilizes an interconnected network of smart contracts to automate all its necessary and non-essential processes.

5.3. Algocracy in the DAO

To give a more comprehensive definition of Algocracy, it is important first to define what Algarchy is. For clarity, the dictionary definition of Algarchy is "organized society in which political decision-making is carried out through a decentralized autonomous system," which is a deviation from the traditional models of a single political system, state, or nation-less civilization. Government by the algorithm isn't an option, which is why "Decentralized Autonomous Organizations" are becoming more common. DAO is also a deviation from the traditional models of government, but the overall aim is still to reestablish the rule of a single political system. Therefore, for an entity to become part of the Decentralized Autonomous Organizations, it must fulfill several specific conditions laid down by the creator or the governing board of the DAO.

Creating such a DAO is to ensure the protection of property rights, guarantee access to knowledge and resources, guarantee the development of economic and technological competence, and, lastly, promote economic efficiency and growth. A DAO can only come about when a certain number of participants participate and agree to these conditions[42]. However, this system doesn't mean that the entire society of the world will have to get involved and contribute to such a venture because a DAO is merely a collection of individuals and businesses within a highly customized and specialized business unit that may also include start-ups or existing businesses looking to utilize the benefits of a more organized management process. So, how does an Algocratic phenomenon actually come about?

The core concept of Algocracy is actually the application of algorithms and other forms of computer code to solve certain business problems. It's clear that in this day and age, there are a lot of companies who would want to use Algocratic tools as a way of solving problems and increasing competitiveness. The problem is not that there are not many companies out there who can provide these kinds of solutions; the problem lies in the fact that most organizations today seem to be stuck with the old and traditional management styles. These styles are often called "the ivory tower syndrome". Traditional managers tend to treat their employees as intellectual equals with great potential for the organization, whereas the newer generation of managers tends to view employees as simple mechanisms to boost sales and quarterly profits.

---

[41] Liebkind J (2021) Daos, blockchain, and the potential of ownerless business. Investopedia, Investopedia. Available from: https://www.investopedia.com/news/daos-and-potential-ownerless-business/ (accessed 4 September 2021).

[42] Aneesh A (n.d.) Technologically Coded Authority: The Post-Industrial Decline in Bureaucratic Hierarchies.

Algocracy

Algocratic Governance is the foundation of Decentralized Autonomous Organizations

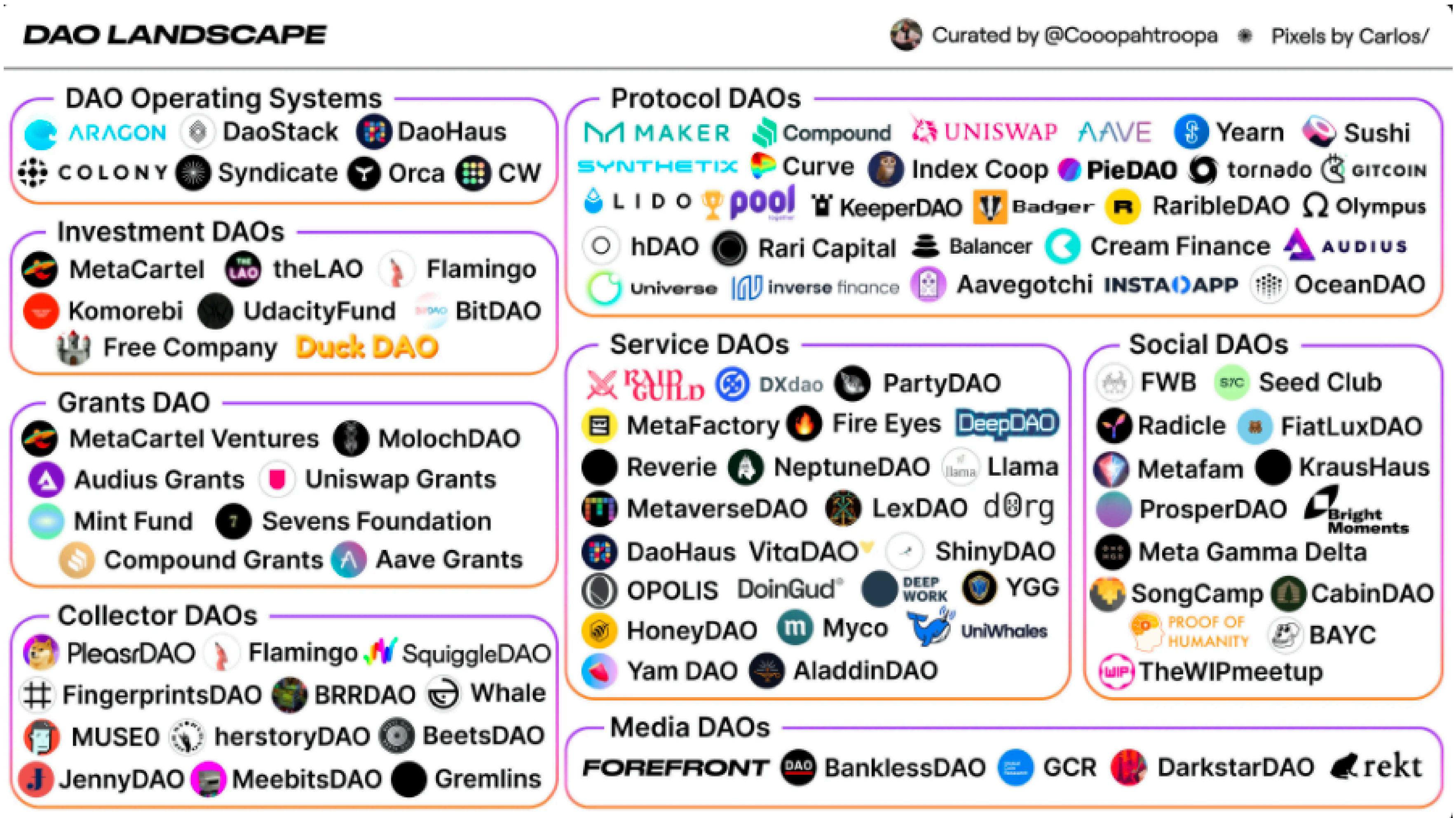
> Government by algorithm is an alternative form of government or social ordering, where the usage of computer algorithms, especially of artificial intelligence and blockchain, is applied to regulations, law enforcement, and generally any aspect of everyday life such as transportation or land registration.
>
> Source: Wikipedia

This image of a stiff upper-class control over an underdeveloped economy is exactly what many people have against the concept of an Algocracy. However, it is a misconception to think that large and established organizations can only use algorithm-driven or algorithmic technologies. Smaller organizations may use these methods and even use them successfully, given the right circumstances and the right impetus.

Smaller organizations still have a lot of room for innovation. They can choose to develop their algorithmic technologies or buy scalable off-the-shelf applications from existing providers. They can then combine these technologies with other forms of governance to build new systems. Alternatively, they can develop their internal governance structures to ensure fair and equitable allocation of resources. Most importantly, they can gain a competitive advantage through collaboration with existing and relatively larger organizations. Such a governance system allows smaller organizations to adopt and utilize algorithmic technologies without relying on outside institutions that can potentially be hostile to their interests.

A potential competitor for the Algarchy is not the corporation itself but the organizations and businesses that use and implement certain algorithmic systems. Algocracies are most effective when they are adopted within a context that is not explicitly political, yet at the same time highly technical and complex. This kind of context is most appropriate for the development of algorithmic technologies and other forms of algocracy. Given that future competition for space and time on the Internet remains fierce, the time is ripe for organizations to develop their own opaque, algorithmic system for governance. The implications of this are profound; as more organizations rely on complex algorithms to serve their customers and serve their mission objectives, the business of making such systems becomes increasingly important.

Top Decentralized Autonomous Organization Landscape[43]



[43] Peaster WM (2021) How to join a dao. Bankless, Bankless. Available from: https://newsletter.banklesshq.com/p/how-to-join-a-dao (accessed 5 September 2021).

6. Decentralized Finance (DeFI)

6.1. The DeFI Market Snapshot

The Decentralized Finance Market Capitalization is $43B with a $5B Trading Capitalization.

6.2. What is Decentralized Finance

Decentralized finance refers to an innovative new form of online financing that avoids the traditional reliance on banks, financial intermediaries, or other financial institutions to provide financial services. Instead, this type of financing relies on self-services ledgers generated based on the real-time collection and processing of data from participants in the network through smart contracts[44].

Decentralized Finance (or DeFi) allows traders to access services such as trading, borrowing, and lending without the need for financial institutions or middlemen (such as a central exchange like Coinbase, Binance, or Gemini)[45]. DeFi exchanges use automated market-making, rather than relying on exchange liquidity. They do this by using liquidity provided to them by ordinary traders who pool their holdings with other traders to create supply.

Self-service financing is rapidly becoming the most popular means of acquiring small business loans, venture capital, high-risk private equity, and other types of personal financing. This form of financing is also becoming increasingly popular among start-ups looking for available funding sources for their growing operations. The key attraction to the concept of decentralized finance lies in its ability to avoid the problems that are common with conventional bank financing[46]. This form of financing is also highly adaptable and efficient.

Decentralized digital currency is another attraction to the concept of self-service ledgers used as the backbone for decentralized finance. Fiat currency is an easy way to complete transactions but is rarely used in everyday financial services because it is expensive, wasteful, and prone to errors. By contrast, crypto-based systems are based on the protocol of a decentralized ledger concept to allow users to transact using any major Internet-ready computer regardless of its operating system.

---

[44] van der Merwe, A. (2021) ‘A Taxonomy of Cryptocurrencies and Other Digital Assets’, Review of Business, 41(1), pp. 30–43. Available at: https://search.ebscohost.com/login.aspx?direct=true&AuthType=shib&db=ent&AN=148423022&site=eds-live&scope=site (Accessed: 29 September 2021).
[45] What is defi in crypto? The Complete Defi & Yield Farming Crypto Tax Guide (2021) Cryptocurrency Tax Software, ZenLedger. Available from: https://www.zenledger.io/guides/the-complete-defi-crypto-tax-guide (accessed 14 September 2021).
[46] Decentralized finance (2021) Wikipedia, Wikimedia Foundation. Available from: https://en.wikipedia.org/wiki/Decentralized_finance (accessed 4 September 2021).

Tax implication in the space of DeFI style services are ambiguous between jurisdictions; however, some form is evolving with the following categorizations: lending rate taxes, wrapping (meaning putting the coin inside another that is traded on the desired exchange where the coin wrapped is not accepted at the desired exchange), interest expense and earnings, governance token appreciations, Ethereum Gas Fees deductions, and liquidation taxes on the fees earned to name a few.

The idea of smart contracts is the heart of the concept of decentralized finance. Smart contract technology is the backbone behind all decentralized autonomous organizations, such as those which operate without a centralized board. In essence, smart contracts allow users to set up a smart contract where the creator of the contract receives payments automatically from any sales of tokens that have been purchased.

Decentralized Finance (DeFI) (Swaps)

- DeFI performs much like traditional finance but without without the need of a central authority;
- Uses smart contracts between strangers anywhere in the world;
- DeFi platforms allow people to lend or borrow funds from others;
- Speculate on price movements on a range of assets using derivatives;
- Trade cryptocurrencies;
- Earn interest in savings-like accounts;
- Minimum to no KYC/AML and credit checks;
- 100% user responsibility.

UNISWAP SushiSwap 0x

Balancer AAVE Bancor

Uniswap (UNI) $26.88 7.1%

Sushi (SUSHI) $15.31 17.8%

Bancor Network Token (BNT) $4.32 4.1%

Balancer (BAL) $29.41 6.4%

Aave (AAVE) $401.38 4.0%

0x (ZRX) $1.17 11.9%

A good example of a smart contract system is the decentralized asset exchange or DeFI dapp. The goal of a DeFI dapp is to allow any asset to be listed and swapped on a decentralized exchange, free from any restrictions imposed by a centralized exchange. This is accomplished through a series of linked transactions linked to the previous one and all leading to a final destination. The final destination can be an asset list maintained by the decentralized autonomous organization.

By linking a decentralized autonomous organization to a real-time execution environment using the distributed ledger technology of the Internet, smart contract systems offer true freedom from traditional asset exchanges and financial markets. Distributed Ledger Technology, or Distributed Ledger Application, is what makes Distributed Ledger Technology work so well. The underlying logic of smart contracts runs through a decentralized computer program. The nature of computers and the Internet make this concept workable, even in a highly decentralized setting. But by using smart contracts to transfer value, Distributed Ledger Technology offers true freedom from the constraints of the traditional financial markets.

7. Decentralized Exchange

7.1. The Market Snapshot

The Decentralized Exchange 24 Hour Trading Volume is $4B with over 103 Million Monthly

7.2. What is a Decentralized Exchange

Decentralized exchanges (DEXs) facilitate peer-to-peer swaps by using smart contracts that automate the execution of trades[47]. Not all DEXs use the same infrastructure. Some DEXs use traditional order book models while others use emergent liquidity protocols[48]. A Decentralized Exchange (DEX) allows users to *trade* cryptocurrency tokens amongst one another without the need for an intermediary in what is referred to as ‘swaps’ or ‘swapping pairs’. The DEX offers ‘theoretically’ complete anonymity as well as non-custodial transactions; meaning your private key to your crypto wallet is not shared. Hackers target exchanges to access the central database and extract users’ private keys and withdraw their funds. Since DEXs do not retain your private keys and do not require identity checks this disincentivizes most hackers; however, some hackers prefer to use decentralized exchanges because they lack KYC requirements[49].

Decentralized finance uses smart contract technology. The way it operates is simple. Smart contracts are programmed by users and allow them to freely specify how they would want their tokens to be used. Once these programs are deployed, users are able to sell their tokens for a fixed price. The buyers of these tokens will then be able to purchase them from the seller for a predetermined amount. This is the definition of liquidity in the crypto world and is essentially the lifeblood of DeFi. This allows token holders to enjoy both advantages of a centralized exchange (smart contract) and the benefits of decentralized exchange (anonymity and lower fees), all from the comfort of their own homes.

Successful implementation of a DeFi system will result in more sellers and buyers creating more liquidity to leverage. Although this will only happen during times when the exchange rate is low, traders will still find a lot of opportunities to gain profit from their investments. Through the use of

---

[47] (2021) Decentralized Exchange platforms in crypto trading, Gemini. Available from: https://www.gemini.com/cryptopedia/decentralized-exchange-crypto-dex (accessed 29 September 2021).
[48]https://www.gemini.com/cryptopedia/decentralized-exchange-dex-crypto#section-decentralized-exchanges-vs-centralized-exchanges
[49] Crypto hacks: Crypto Exchange Hacks & Cryptocurrency hackers (n.d.) Coinbase, Coinbase. Available from: https://cointelegraph.com/magazine/crypto-exchange-hacks/ (accessed 29 September 2021).

pooling methods, it is now easier for many traders to obtain their desired liquidity at the lowest possible rates.

7.2.1. Liquidity Funded Exchanges

Liquidity Funded Exchanges are the next generation of decentralized exchanges that do not use order books to facilitate trades or set prices. Instead, these platforms typically employ liquidity pool protocols to determine asset pricing. Peer-to-peer in nature, these exchanges instantly execute transactions between users' wallets — a process some refer to as a swap. The DEXs in this category are ranked in total value locked (TVL), or the value of assets held in the protocol's smart contracts.

7.2.2. Order Book Enhanced Exchanges

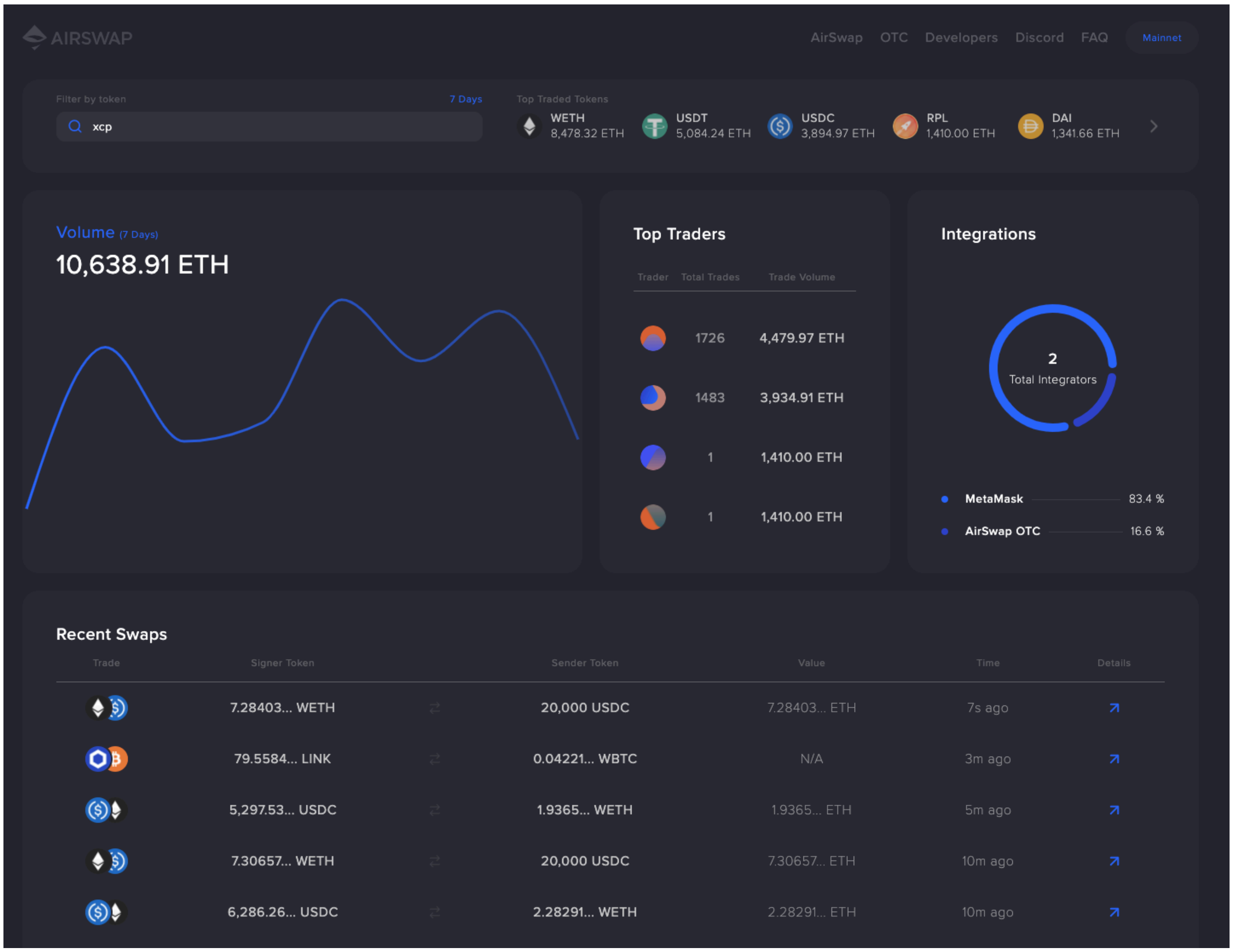
Order Book Exchanges manage order books and compile a record of all open buy and sell orders for a particular asset. The spread between these prices determines the depth of the order book and the prevailing market price. On DEXs with order books, this information is often held on-chain during trades, while your funds remain off-chain in your wallet. **Many of these types of DEXs specialize in a particular financial instrument that is executed in a decentralized manner.**

Top Decentralized Exchanges by Trading Volume Updated: 9/22/21

#	Exchange	24h Volume	Num Coins	Num Pairs	Visits	Most Traded Pair	% Market Share By Volume
1	Uniswap (v3)	$1,235,494,387	452	945	5,682,101.0	USDC/ETH $578,934,688	28.1%
2	PancakeSwap (v2)	$612,928,862	1982	11681	20,961,948.0	WBNB/BUSD $106,579,449	13.9%
3	Sushiswap	$448,982,147	343	685	249,486.0	ILV/0XC02A... $72,281,335	10.2%
4	Uniswap (v2)	$365,281,142	1694	3022	5,682,101.0	USDC/0XC02... $37,888,750	8.3%
5	Raydium	$301,626,433	90	177	2,319,812.0	SOL/USDC $102,093,946	6.8%
6	1inch	$275,440,817	297	307	352,479.0	USDC/ETH $99,755,157	6.3%
7	Trader Joe	$167,829,530	37	218	463,913.0	WAVAX/0XC7... $76,520,674	3.8%
8	Curve Finance	$131,884,054	23	63	742,525.0	USDC/USDT $37,299,557	3.0%
9	DODO	$82,523,841	9	17	417,392.0	USDT/USDC $54,691,862	1.9%
10	Spookyswap	$76,767,015	30	259	371,661.0	USDC/0X21B... $31,719,027	1.7%

7.3. Token Swaps

Token swaps are designed to lower overhead costs and reduce the time needed to exchange crypto assets for one another. This means that seamless crypto-to-crypto exchange services are used instead of the time-consuming and costly process of converting digital assets to fiat, then utilizing the fiat to purchase your desired coins.



Airswap (main dashboard in figure x) is an example of how counterparties can communicate prices and send orders to one another once they are connected.

Two types of swaps are common on the market: the Request for Quote and the Quote-Driven. The latter is replacing the human element.

Request for Quote (RFQ) market like Airswap, all orders are displayed from buyers and sellers from the apps website page with greater detail expanding into the Ethereum block explorer. Parties can communicate their intention to trade tokens to each other off-chain. [50] Provider's such as the NASH exchange improve on the use of conventional books, and ViteX using

[50] *https://www.reddit.com/r/defi/comments/lalnd2/uniswap_vs_airswap_explained/*

on-chain transparency visible to the public to prevent tampering, both still using order books. Parties may also request prices from an external third-party oracle in order to verify their accuracy. An Ethereum smart contract can be used to fill orders through atomic swaps[51] once the parties have reached an agreement.

Quote-Driven Market, such as Uniswap, uses an **Automated Market Maker (AMM)** to perform the swaps. This automatically posts the ask and bid prices that buyers and sellers are willing and able to accept. These smart contract-based services also use the Ethereum platform and allow extreme degrees of flexibility with liquidity pooling options. dApps such as SushiSwap, and Polkadot are all popular options for traders as well as Binance Smart Chain (BSC) that is the number one contender for Ethereum.

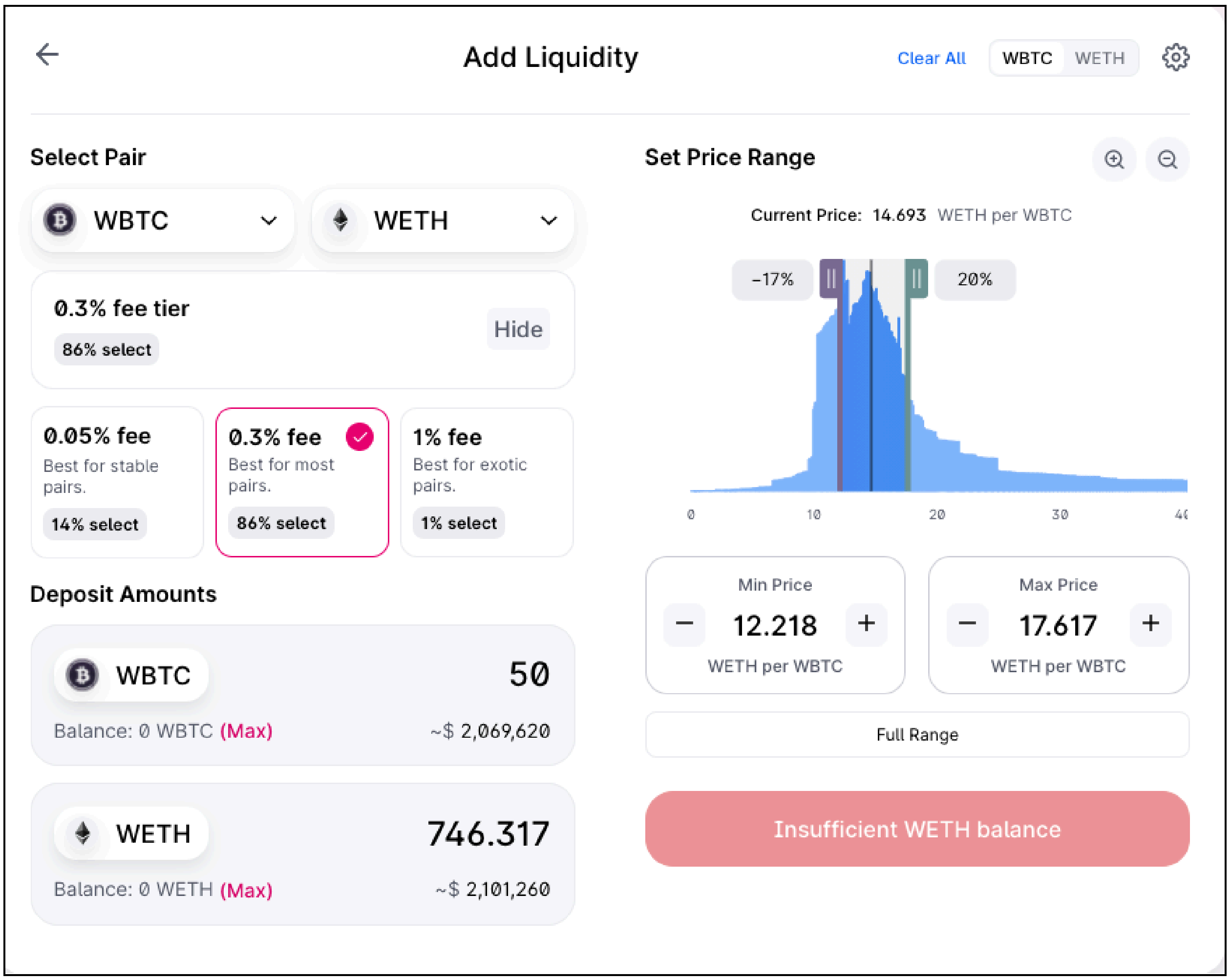


Figure (X) Uniswap adding liquidity to the pool with percent of fees breakdown.

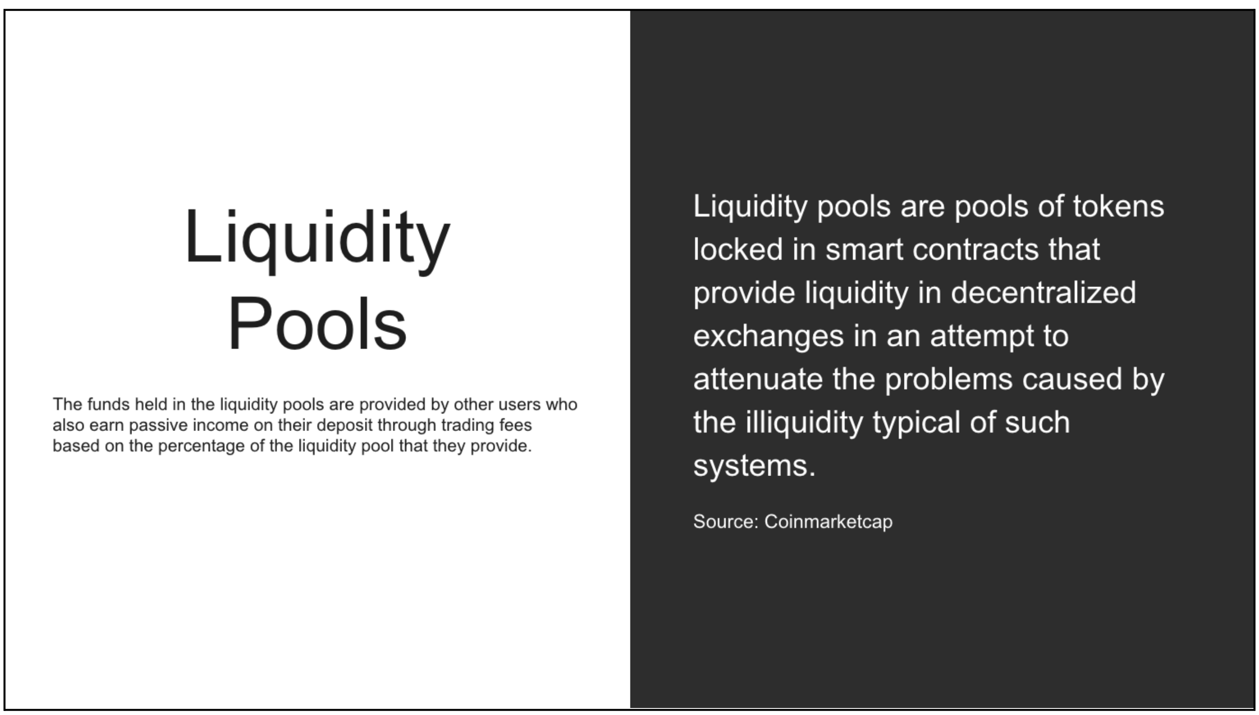
[51] *What are Atomic Swaps? Atomic Swaps allow two parties to trade tokens on two different blockchains. This type of trading mechanism is sometimes called atomic cross-chain trade. It completely eliminates the need to have third-party entities executing trades. This system allows crypto users to remain autonomous and facilitates trustless transactions that don't require them to know each other and are free from counterparty risk.*

Token swap services play a key role in the ongoing cryptocurrency renaissance. They remove the barriers that prevent people from entering the altcoin marketplace. These solutions can be used to provide an easier way to access cryptocurrency with low market capitalization. Token swaps are likely to continue to be a prominent part of the crypto industry due to their easy, quick, and simpler approach to digital asset trading.[52]

7.4. Liquidity Pools

The idea of a liquidity pool is simple enough.

To create a cryptocurrency market, liquidity providers (LPs) are users who add cryptocurrency to a liquidity pool. A liquidity pool is a group of funds deposited by anyone who holds cryptocurrency and wants to lend it to others. These Liquidity Providers (LP) rely on smart contracts to distribute fees and earnings. You don't need a counterparty as you do in traditional finance systems when you execute a trade on Automated Market Maker. You execute the trade against liquidity in the liquidity pool, which must be sufficient to cover the borrowers' loan. This measurement is called the Total Value Locked (TVL) and is a key measurement of a platform's liquidity in action.



[52] Uniswap WBTC/ETH Pool

7.5. Automated Market Maker (AMM)

This game has been changed by Automated Market Makers (AMM.) This is an important innovation that allows for on-chain trading without needing an order book. Trades can be executed without the need for a counterparty. This will enable traders to trade token pairs that are otherwise likely to be in a state of high illiquidity, not earning any interest.

AMMs make market-making easier because anyone can become a liquidity provider.

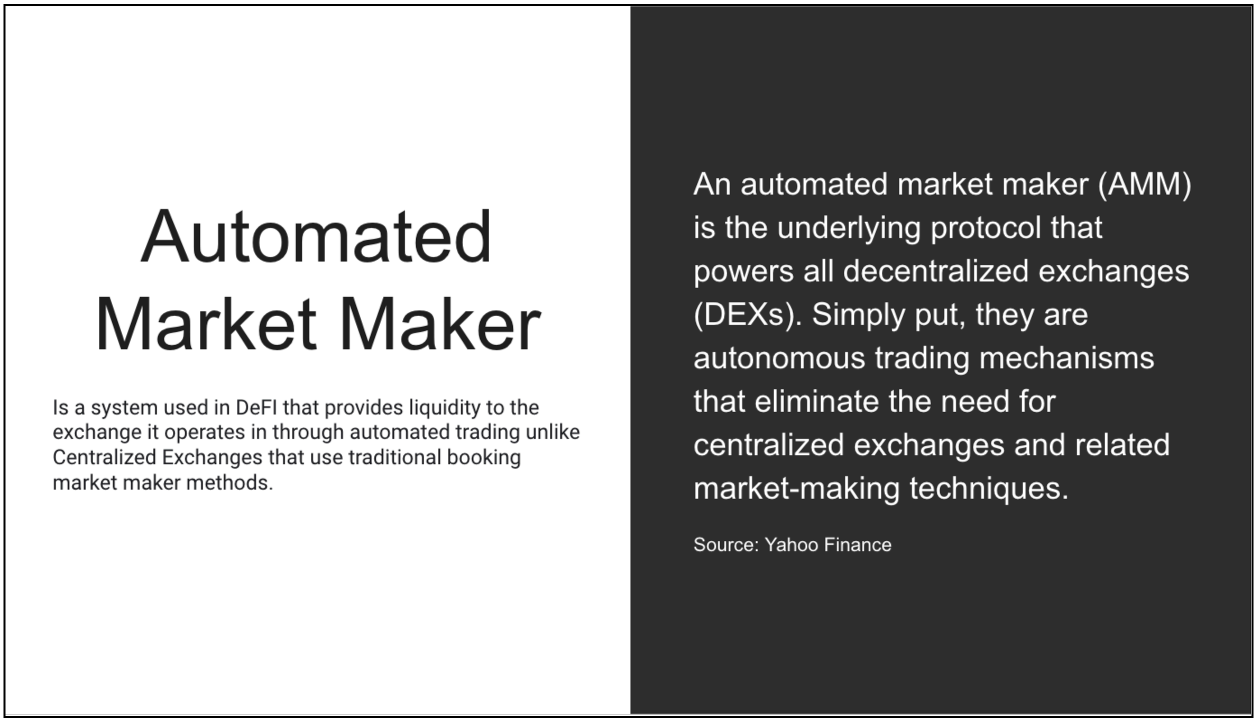
An automated market maker (AMM) is at the heart of all decentralized online exchanges (DEX). This service allows users from anywhere in the world to trade in the financial markets. Their popularity stems in part from their ability to take trading risks and leverage automation through mathematical algorithms. They allow traders to enter and exit trades without being tied to a computer or other external factors. This has made them an attractive option for many new traders and investors. However, as with any other technological innovation, there are some inherent problems with these machines. One of the most common complaints revolves around using a so-called “Duke System” algorithm that’s supposed to be used by all AMMs.

The “Duke System” is named after the creators of the algorithm, who are two men named Albert Perrie and John Grace. Their original intention was to create a mathematical formula by which a trader could specify the asset to buy and the timeframe by which the purchase should be made to gain optimal profits. However, this mathematical equation was supposed to apply equally to all assets regardless of their volatility. Because of this, many traders have claimed that the new system does not correctly represent the asset characteristics of the currencies it trades.

In particular, some claim that while the new system can correctly identify fundamental and technical indicators such as volatility, the process of using these indicators leads it to inappropriately attribute future prices to past prices and the need to buy or buy-sell orders during trading hours. **This process is said to result in an artificially high risk/reward ratio.**

Others argue that this is a problem caused by the fact that all or most traders will only focus on buying and selling during normal business hours to maximize their profit. They further claim that in a simultaneous sell order from another trader, the automated market maker will incorrectly attribute the order as being originated by a different person. This will result in the loss of money due to the difference in stop loss and take profit triggers that automatically cause the robot to exit a trade early to cover its risk of not making a profit.

Another issue revolves around the use of Uniswap, a unique index meant to measure the trends in the market’s trading volume. Some critics have claimed Uniswap is prone to inaccurate measurement, particularly because there are many possible interpretations of the trends in the trading volume. For example, one explanation for the excessive trading volume by automated market makers is that many people are looking for a quick way to generate money. At the same time, it is also true that large amounts of trading volume may indicate that the market is experiencing unusual activity, such as the sudden rise of one currency against another or changes in the rate of exchange between various currencies. To prevent losses caused by these factors, the Uniswap algorithm is supposed to be coupled with measures that will reduce the risk of losses in the automated trading system.



The automated market maker also employs what is called an emulator to adjust its risk level parameters. The emulator is a mathematical formula that takes the closing prices of the major financial markets and calculates the maximum amount of money made by selling certain assets in the exchange. Since this formula considers the closing prices of the markets, it is less susceptible to extreme changes in the values of assets that can affect the calculations of the ammulator. However, it is still vulnerable to the same factors that can affect the profitability of investments.

Decentralized Finance, also called DeFi, uses Cryptocurrency and decentralized ledger technology to handle global financial transactions. Cryptocurrency is money derived from a digital source and normally uses peer-to-peer technology. This means that instead of dealing with a single organization or bank, you are dealing with dozens of peers, including any number of computer networks, including the Internet itself. The main advantage of using Cryptocurrency is that it is highly secure and transaction-free, and completely fungible, meaning that one coin can be used for any transaction. However, some disadvantages of using Cryptocurrency in place of conventional financing sources have been raised.

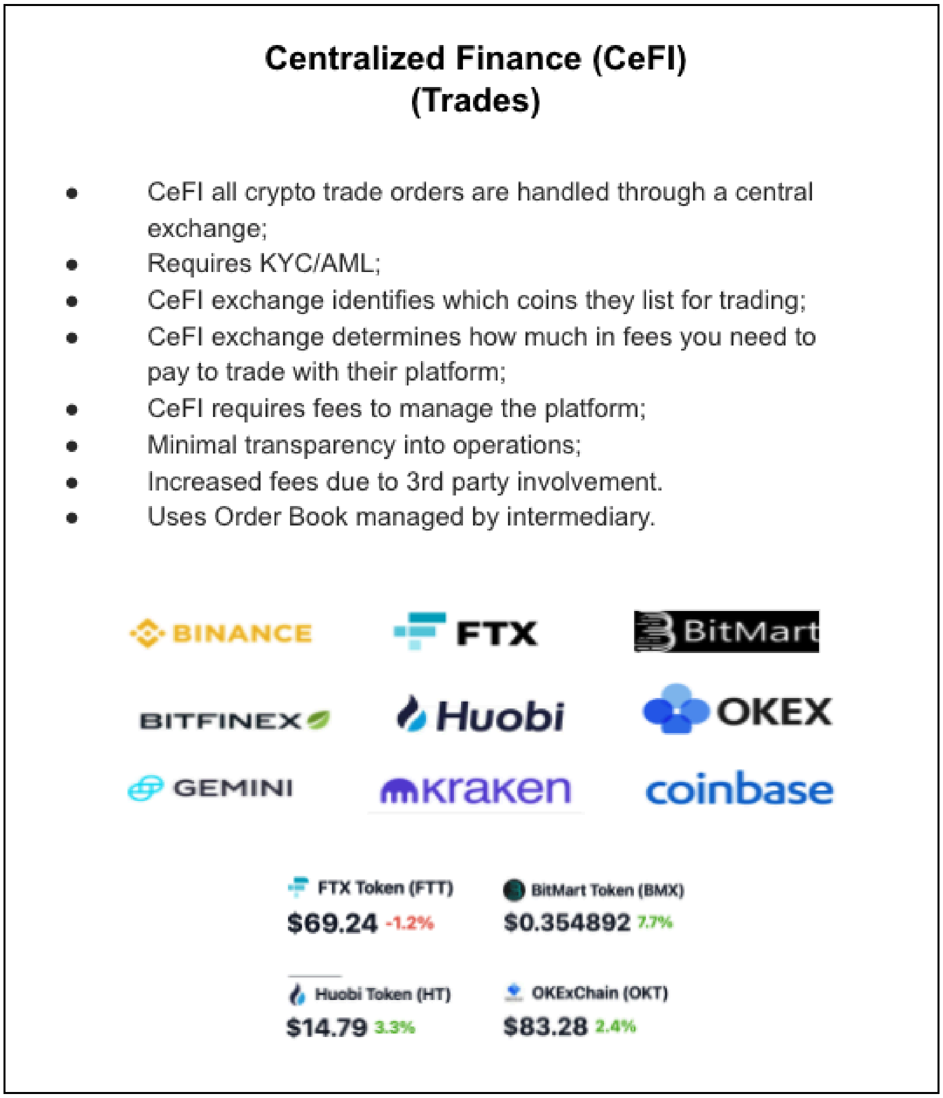
According to some experts, decentralized Finance may pave the way towards further innovations such as a global currency market or a World Wide Web such as W3. Besides, decentralization provides users with an unprecedented degree of control over their financial lives. They can specify precisely how much they want to spend when they want to spend, and what they want to invest in. Besides, decentralization also allows users to avoid the notorious pitfalls of fiat currencies, such as inflation, political instability, and financial crisis.

In conclusion, it is important to understand how the pricing mechanism within the automated trading system works in the DeFI ecosystem. Market makers do influence the way the prices of the assets are set on the decentralized exchange. This is why new traders need to learn how the system works before committing their money to the exchange and why GSHI cryptocurrency education is significant for attracting and retaining new customers.

8. Centralized Finance (CeFI)

8.1. What is a Centralized Finance (CeFI)

Centralized finance involves the most widespread and powerful form of private financial service in the world today: central banks. These organizations manage the money supply of a country or even the whole world. Centralized financing, in contrast to decentralized finance, is based on the assumption that some invisible or nearly invisible force governs the supply of money. Centralized banks are generally major financial institutions such as banks, mortgage companies, and lending companies. Governments around the world widely use centralized financing.



However, in contrast to decentralized finance, centralized finance assumes that money is created by a "free market" and regulated process. In other words, many intermediaries are needed, which decreases trust amongst all parties; this can be counterintuitive when using blockchain as a distributed ledger system[53]. Also, in contrast to the famous Mises-Wagner[54] [55] model, there is no need for a crypto lender to distinguish between secured and unsecured loans, as the collateral requirements are well above the principle, and will repay the lender regardless.

---

[53] Halverson, Sean 2018 - The Trust Machine
[54] https://mises.org/wire/ludwig-von-mises-and-nature-money
[55] In dealing with the nature of money, Mises relied heavily on the work of Carl Menger. The founder of the Austrian School had shown that money is not to be defined by the physical characteristics of whatever good is used as money; rather, money is characterized by the fact that the good under consideration is (1) a commodity that is (2) used in indirect exchanges, and (3) bought and sold primarily for the purpose of such indirect exchanges.

Centralized finance is the system of financial transactions that rely on the operation of a number of banking systems, instead of relying on only one central bank. The most prominent example of this type of finance is the "custodial banking system," which makes use of a bank deposit as a security for financial transactions, such as loans and trading.

Aside from Custodial Banking, there are a number of less famous systems of this nature:

- CDAC - Centralized Dealer Automated Clearing House is a system for clearing international interbank loans. It is a system for domestic banking that provides low-fee trading and clearing of most interbank loans. It was developed by the Central Banks of Australia, Canada, Germany, Ireland, Japan, Ireland, Portugal, South Africa, the United Kingdom, the European Central Bank, the United States, and Japan.

- CFTC - Commodity Futures Trading Commission is an agreement entered into in May 2021 between the Commodity futures Trading Commission and the futures trading industry. The CFTC is a United States-led effort to reform the futures and options markets. It is a system for domestic electronic trading similar to the NYSE, although the CFTC operates through six regional offices instead of the usual five.

The popularity of the digital currency has encouraged people who provide its services to offer these services worldwide. A number of financial institutions have already launched their own versions of the digital currency wallet. Services offered by such companies range from secure storage of your private keys to complete access to the blockchain ledger. In addition, some companies offer additional features like an online wallet and cold storage of the private keys.

Although the main function of such services is to make financial transactions faster, safer and cheaper, these services also have another important role: to reduce the cost of financial intermediation. Bank’s note the importance of reducing costs, particularly transaction costs, as increasing bank charges reduces the value of the bank's holdings of the underlying assets. The same reduction can be achieved by using smart contracts and digital currencies. This makes the financial industry more efficient, making it less risky while increasing its value through new business ventures.

9. Centralized Exchange (CEX)

9.1. The CEX Market Snapshot

The Centralized Exchange 24 Hour Trading Volume is $336B (Updated 9/22/2021)

9.2. What is a Centralized Exchange (CEX)

Investors may be confused by the idea of a "centralized exchange,” as digital currencies are often described as "decentralized." Investors can conduct trades from fiat into cryptocurrency or vice versa, and you can also use them for trading between different cryptocurrencies in contrast to decentralized exchanges. In comparison, only the swapping between pairs is made either based on an agreed-upon price between each token or through an AMM.

CEX's act as trusted intermediaries (counterintuitive term) in trades and are often custodians, storing and protecting your money. The leading exchanges provide a seamless digital asset trading experience, from security to fair market pricing to consumer protection and regulatory compliance to access various digital assets. 95% of digital asset trades were executed via centralized exchanges as of September 2020. This amounts to approximately USD 228 billion per month[56].

CEX's offer digital wallets for cryptocurrency storage. You are putting your trust in an exchange to keep your private key safe and your funds secure which also make a CEX a target for hackers. You don't have to worry about losing your private key or wallet. You should still do your research to make sure that the exchange has adequate security measures to protect your funds. Many exchanges provide custody services to financial institutions and investment companies that invest in cryptocurrency. However, they do not have the ability to manage their private keys. These assets are usually kept in cold storage by a CEX.

Although regulations can vary widely, centralized exchanges must be subject to those of the jurisdictions where they are located. One financial regulator may have oversight over cryptocurrency for the whole country or region, whereas the United States has a more fragmented regulation system. Some issues, like licensing, are handled state-by-state, while federal law mandates other aspects, such as KYC data collection or AML issues. To remain compliant, exchanges need to work with many federal and state actors which leave the US as a laggard in the adoption of cryptocurrency.

[56] https://www.gemini.com/cryptopedia/centralized-exchanges-crypto#section-services-offered-by-centralized-exchanges

Top Centralized Exchanges (CEX) by Trading Volume Updated: 9/22/21

#	Exchange	24h Volume (Normalized)	24h Volume	Visits (SimilarWeb)	# Coins	# Pairs
1	Binance	$24,874,150,170	$24,874,150,170	165,696,594.0	331	1239
2	Huboi Global	$5,342,755,511	$5,342,755,511	5,854,617.0	352	933
3	Coinbase Exchange	$5,272,544,346	$5,272,544,346	14,962,048.0	102	304
4	Crypto.com Exchange	$4,167,635,888	$4,167,635,888	9,764,011.0	119	217
5	FTX	$2,758,210,412	$2,758,210,412	21,690,867.0	270	435
6	KuKoin	$1,990,865,750	$1,990,865,750	11,934,432.0	450	887
7	Kraken	$1,440,669,537	$1,440,669,537	13,195,314.0	87	389
8	Bitfinex	$1,133,477,503	$1,133,477,503	5,001,192.0	155	374
9	Gate.io	$1,131,944,987	$1,131,944,987	12,469,972.0	917	1895
10	Binance US	$878,626,575	$878,626,575	3,879,846.0	59	115

10. Cryptocurrency Types

10.1. Stablecoins

The Stablecoin Market Cap is $128 Billion with a trading volume of over $76 Billion.

Stablecoins, such as USDC, DAI, and Tether is a cryptocurrency, which is always equal to $1. These types of cryptocurrencies can maintain a stable value through USD reserves, arbitrage, and complex code backed by cryptocurrency. [57]

What is a stablecoin exactly? Stablecoins are cryptocurrency that have been designed to reduce volatility by pegging to more stable assets. The most common use case for stablecoins is to be converted into fiat currency digital assets. It usually tracks the most popular national currencies, such as the US Dollar, Euro and British Pound. This allows you to benefit from blockchain technology and peer to peer value transfer, while not being subject to high volatility like bitcoin, ethereum or other cryptocurrencies.

Stablecoins, a new type of technology, have different implementations, liquidity, risk, acceptance, and acceptability.

Some Stablecoins are just about to hit the market and will be linked to the Consumer Price Index of certain countries. Stablecoins can be fixed for almost any item in theory. There are Stablecoins that can be fixed to fiat coins as well as precious items such as gold and silver.

Tether and some central Stablecoins require that a custodian regulate the currency and reserve a certain amount as collateral. To maintain order, Tether must keep the US Dollar in a bank account. The amount must equal the amount they issue. This prevents price fluctuations and is available for audit by the public through their transparency page.

Fiat-collateralized refers to the simplest version. It includes every Stablecoin currency that has been produced in that currency. The coin's issuer is responsible for production and liquidation. This is similar to the old practice of keeping a certain amount of gold in reserve at the central bank for each dollar printed.

The Crypto-collateralized version uses other cryptocurrency (e.g. ETH) to secure Stablecoins. These Stablecoins must use a series of protocols because other crypto values aren't stable. This ensures that $1 remains the price of the stablecoin - these are decentralized stablecoins such as Dai, ChainLink, and Uniswap all using different protocols in the ETH ecosystem.

[57] https://www.benzinga.com/money/best-crypto-loans-and-lending-platforms/

Non-Collateralized Stablecoins design is very different from other stablecoins because it is not backed with any collateral. It works in the same way as fiat currencies, except that it is controlled by a sovereign entity like a country's Central Bank.

Economists see stablecoins as having lower-cost, safer, more real-time, and more competitive payments than what consumers and businesses currently experience. They could make it easier for businesses to accept payments, and for governments to run conditional cash transfers programs (including the sending of stimulus money) faster. They could be used to connect the unbanked and underbanked sections of the population to the financial systems.

However, critics argue that without strong economic and legal frameworks, stablecoins could be unstable. They could crash like a weak currency board, "break down the buck" as money market funds did in 2008, and spiral into worthlessness. They could even recreate the chaos of the "wildcat banks" of the 19th-century[58].

[58] https://hbr.org/2021/08/stablecoins-and-the-future-of-money

10.1.1. Top 10 Stablecoins (based on Market Cap)

Rank	Coin	Market Cap (Updated 9/16)	Type	Summary
1	Tether (USDT	$69,367,510,370	CeFI	Pegged to the US Dollar.
2	USD Coin (USDC)	$29,237,517,241	CeFI	Bridge between dollars and trading on cryptocurrency exchanges.
3	Binance USD (BUSD)	$12,735,077,903	CeDeFI	Stable coin pegged to USD that has received approval from the New York State Department of Financial Services (NYDFS).
4	DAi (DAI)	$6,282,491,110	DeFI	Native stablecoin for the Maker protocol -The collateralized assets backing Dai are other cryptocurrency instead of fiat and are held within smart contracts rather than in institutions.
5	TerraUSD (UST)	$2,591,482,540	DeFI	Algorithmic stablecoin that is pegged to the US Dollar - isn't backed by US dollars in a bank account. Instead, in order to mint 1 TerraUSD, US$1.00 worth of TerraUSD's reserve asset (LUNA) must be burned.
6	TrueUSD (TUSD)	$1,401,336,197	CeDeFI	Provides its token holders regular attestations of escrowed balances, full collateral, and also the legal protection against misappropriation of underlying USD.
7	Pax Dollar (USDP)	$1,022,892,271	CeDeFI	Is the first digital asset to be issued by a financial institution and to be fully secured by the U.S. dollar.
8	Liquity USD (LUSD)	$604,986,692	DeFI	Is a decentralized borrowing protocol that allows you to draw 0% interest loans against Ether used as collateral.
9	Neutrino USD (USDN)	$527,112,609	DeFI	Decentralized dollar-pegged stablecoin running on the Waves blockchain, is ported into the Ethereum network. USDN yields its holders a sustainable revenue of up to 15% APY via the underlying LPoS consensus algorithm.
10	HUSD (HUSD)	$476,671,301	DeFI	HUSD Token is issued by Stable Universal, with a 1:1 USD peg.

Crypto traders often use Tether to purchase cryptocurrency instead of the US dollar. It allows them to have safety in a stable asset in times of volatility in crypto markets. Tether has the US Dollar in a bank account, and that amount must equal the amount of Tether they issue to maintain the system's order.

tether

Transparency

Current Balances

USD₮	
Total Assets	**$69,683,984,298.56**
Liabilities (USD₮ in Circulation on Omni)	
Total Authorized	$1,335,000,000.00
Less: Authorized but not issued	- $434,260,601.03
Less: Quarantined USD₮	- $32,303,805.00
Liabilities (USD₮ in Circulation on Eth)	
Total Authorized	$33,861,598,607.96
Less: Authorized but not issued	- $135,331,976.26
Liabilities (USD₮ in Circulation on Tron)	
Total Authorized	$35,993,266,846.70
Less: Authorized but not issued	- $2,064,174,763.55
Liabilities (USD₮ in Circulation on EOS)	
Total Authorized	$85,251,000.50
Less: Authorized but not issued	- $473.56
Liabilities (USD₮ in Circulation on Liquid)	
Total Authorized	$36,561,000.00
Less: Authorized but not issued	- $0.00
Liabilities (USD₮ in Circulation on Algorand)	
Total Authorized	$24,000,019.79
Less: Authorized but not issued	- $0.00
Liabilities (USD₮ in Circulation on SLP)	
Total Authorized	$6,000,899.85

EUR₮	
Total Assets	**€75,998,802.00**
Liabilities (EUR₮ in Circulation on Omni)	
Total Authorized	€1,610.54
Less: Authorized but not issued	- €166.55
Less: Quarantined EUR₮	- €0.00
Liabilities (EUR₮ in Circulation on Eth)	
Total Authorized	€150,000,050.00
Less: Authorized but not issued	- €74,003,062.98
Liabilities (EUR₮ in Circulation on Tron)	
Total Authorized	€0.00
Less: Authorized but not issued	- €0.00
Liabilities (EUR₮ in Circulation on EOS)	
Total Authorized	€0.00
Less: Authorized but not issued	- €0.00
Liabilities (EUR₮ in Circulation on Liquid)	
Total Authorized	€0.00
Less: Authorized but not issued	- €0.00
Liabilities (EUR₮ in Circulation on Algorand)	
Total Authorized	€0.00
Less: Authorized but not issued	- €0.00
Liabilities (EUR₮ in Circulation on SLP)	
Total Authorized	€0.00

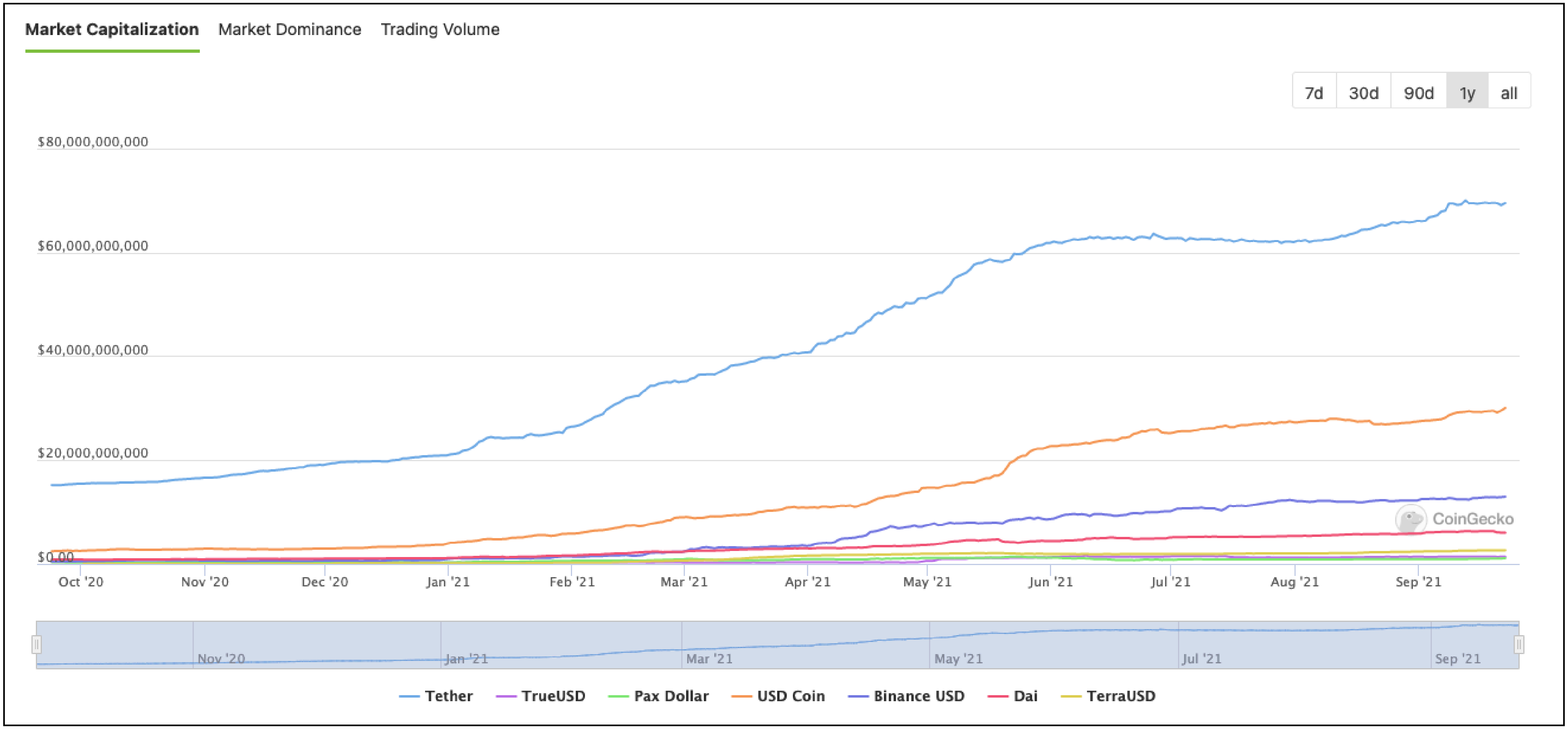
Source: Tether

Tether has become the most prominent player in crypto trading due to its US dollar collateral.

True stablecoins are non-interest-bearing coins designed to have stable value against a reference currency — say USD 1. Stability is achieved through two commitments. First, the issuer agrees to mint and buy back coins at par. Second, the issuer holds assets to back its obligation to redeem the outstanding stablecoins. This “reserve” provides comfort that the issuer can buy back all outstanding coins, on-demand. Reserve assets should be denominated in the currency of the reference asset, remain highly liquid during a crisis, and incur extremely small losses in a run or stressed market conditions.[59]

[59] https://flexiple.com/blockchain/what-are-stablecoins-and-how-do-they-work/

Stablecoins by Market Capitalization (9/22/21) (Source: https://www.coingecko.com/en/stab



10.2. Altcoins

Altcoins were launched in 2011, and now there are many thousands. Altcoins were initially created to improve aspects of Bitcoin, such as transaction speed and energy efficiency. Today, developers' goals can lead to altcoins serving a wide variety of purposes where there are thousands of altcoins available in the market.

Altcoins can be any cryptocurrency other than Bitcoin. This definition was common in the early days of cryptocurrency. Bitcoin was the most popular and well-known currency. As can be seen in the Summary column below - these Top 10 Altcoins serve specific purposes. They are an indicator of how the crypto community is responding to these protocols and platforms.

10.2.1. Top 10 Altcoins (based on Market Cap)

Rank	Coin	Market Cap (Updated 9/16)	Type	Summary
1	Ethereum (ETH)	$423,704,574,695	DeFI	Ethereum is a global, open-source platform for decentralized applications.
2	Cardano	$79,335,765,602	CeDeFI	The Cardano project is different from other blockchain projects as it openly addresses the need for regulatory oversight whilst maintaining consumer privacy and protections through an innovative software architecture.
3	Binance Coin (BNB)	$65,895,709,767	CeDeFI	Binance states that BNB was designed to be used to pay discounted fees on the Binance platform and also function as the native token powering the Binance Chain.
4	XRP	$51,894,950,073	DeFI	Ripple was made to connect banks, payment providers and digital asset exchanges, enabling real-time settlement expeditions and lower transaction fees.
5	DOGE	$32,073,468,120	DeFI	Dogecoin is a Litecoin fork. Dogecoin quickly developed its own online community and reached a capitalization of US$60 million in January 2014
6	Solana	$47,183,680,268	DeFI	Based on Proof of History (PoH); a proof for verifying order and passage of time between events. The purpose of PoH is used to encode the trustless passage of time into a ledger.
7	Polkadot	$36,630,003,369	DeFI	Platform that allows diverse blockchains to transfer messages, including value, in a trust-free fashion; sharing their unique

				features while pooling their security. In brief, Polkadot is a scalable heterogeneous multi-chain technology.
8	Uniswap	$14,023,526,621	DeFI	UNI is the governance token for Uniswap.
9	ChainLink	$14,073,236,415	DeFI	Aims to ensure that the external information (pricing, weather data, event outcomes, etc.) and off-chain computations (randomness, transaction automation, fair ordering, etc.) fed to on-chain smart contracts are reliable and tamper-proof.
10	Axie Infinity	$3,958,878,299	DeFI	AXS is the governance token for the Axie Infinity game. Token holders will be able to shape and vote for the direction of the game universe. This is unlike traditional games where all decisions are made by the game developers.

10.3. DeFI Cryptocurrency

DeFi or Decentralized Finance refers to financial services built on top of distributed networks with no central intermediaries. This is another categorization with degrees of variance in the industry as to what constitutes Top Alt Coins compared to Top DeFi Coins. In some instances, there will be an overlap between both of these categories.

DeFi crypto market cap for 9/22/2021 is $117,733,192,249.39 with a total trading volume of $9,2B in the last 24 hours at CoinGecko and $15.9B at TradingView. The classification and taxonomy create variances in how these industries are reported.

10.3.1. Top 10 DeFI Cryptocurrency (based on Market Cap)

Rank	Coin	Market Cap (Updated 9/16)	Price	Summary
1	Terra	$13,381,956,763	$33.42	Terra is a decentralized financial payment network that rebuilds the traditional payment stack on the blockchain. It utilizes a basket of fiat-pegged stablecoins, algorithmically stabilized by its reserve currency Luna.
2	Uniswap	$11,156,076,987	$21.43	UNI is the governance token for Uniswap.
3	Chainlink	$10,977,488,786	$24.17l	Chainlink is a framework for building Decentralized Oracle Networks (DONs) that bring real-world data onto blockchain networks, enabling the creation of hybrid smart contracts.
4	Dai	$6,040,559,635	$1.00	Dai is the native stablecoin for the Maker

				protocol. It is the world's first crypto-collateralized and decentralized stablecoin, whose value is soft pegged to the US Dollar.
5	cETH	$6,002,543,734	$60.09	Compound protocol balance token.
6	Pancake Swap	$4,559,145,356	$20.29	CAKE is the governance token for PancakeSwap.
7	cUSDC	$4,524,399,794	$0.02	Compound is an open-source, autonomous protocol built for developers, to unlock a universe of new financial applications. Interest and borrowing, for the open financial system.
8	cDAI	$4,048,254,073	$0.02	Compound protocol balance token.
9	Lido Staked Ether	$4,034,551,869	$3,093.96	Lido Staked Ether (stETH) is a token that represents your staked ether in Lido, combining the value of initial deposit and staking rewards. stETH tokens are minted upon deposit and burned when redeemed. stETH token balances are pegged 1:1 to the ethers that are staked by Lido and the token's balances are updated daily to reflect earnings and rewards. stETH tokens can be used as one would use ether, allowing you to earn ETH 2.0 staking rewards whilst benefiting from e.g. yields across decentralised finance products.
10	AAVE	$4,013,509,814	$303.89l	Aave is a decentralized money market protocol where users can lend and borrow cryptocurrency across 20 different assets as collateral.

10.3.2. Top 10 DEX (based on Market Cap) Updated: 9/22/2021

Rank	Swaps	Market Cap (Updated 9/16)	Num Pairs	Coins Offered	TVL*	Type
1	Uniswap	$11,208,728,484	943	451	$4,383,034,884	Governance
2	PancakeSwap(v2)	$4,568,957,483	11685	1987	$5,032,722,913	Governance
3	SushiSwap	$2,064,246,013	686	343	$4,479,121,325	Governance
4	Uniswap(v2)	$11,268,490,871	3024	1697	$4,406,404,228	Governance
5	1inch	$479,474,685	309	299	$34,945,674	Governance

6	Raydium	$767,658,575	177	90	$1,436,796,463	Foundation
7	Trader Joe	$380,686,689	220	37	$1,007,172,756	Governance
8	Curve Finance	$1,013,737,167	23	63	$13,567,599,462	Governance
9	DODO	$197,408,664	9	17	$252,751,085	Governance
10	Spookyswap	$84,873,854	260	30	$285,545,271	Governance

**Total Value Locked (TVL) - Capital deposited into the liquidity pool in the form of loan collateral.*

10.4. CeFI Cryptocurrency

CeFI or Centralized Finance refers to financial services that include central intermediaries who act on your behalf to conduct the financial transaction. The CeFI type tokens offer a variety of member benefits ranging from multipliers and returns across multiple fees used by the platform.

The CeFi crypto market cap for 9/22/2021 is $83,660,717,950 with a total trading volume of $2,883,439,856 in the last 24 hours at CoinGecko. The classification and taxonomy create variances in how these industries are reported.

10.4.1. Top 10 CEX Cryptocurrency (based on Market Cap)

Rank	Coin	Market Cap (Updated 9/16)	Price	Summary
1	Binance Coin	$58,460,794,880	$378.56	BNB was designed to be used to pay discounted fees on the Binance platform
2	FTX Token	$6,796,634,839	$55.92	Holders get a fraction of exchange fees, a fraction of the liquidation insurance fund, and can use the token as collateral and to get tighter OTC spreads on FTX.
3	OKB	$4,540,793,151	$17.09l	OKEx describes OKB is a global utility token issued by the OK Blockchain Foundation.
4	Crypto.com Coin	$4,233,688,133	$0.16	Loyalty token for discounted fees, priority token allocations, payment rewards, and member only offerings.
5	LEO Token	$2,830,984,559	$3.00	Utility token intended for the use on the Bitfinex exchange and other trading platforms managed by its parent company iFinex.
6	Huobi Token	$2,026,717,829	$12.45	Utility token - ERC-20 Compliant
7	KuCoin Token	$824,192,208	$10.63	A profit-sharing token that allows traders to

				draw value from the exchange.
8	Gate Token	$777,234,176	$4.83	Used to pay for transaction fees on GateChain's network.
9	LINK	$692,295,552	$115.58	Used to pay node operators for retrieving data for smart contracts and also for deposits placed by node operators as required by contract creators.
10	AscendEx Token	$$319,349,065	$0.43l	is a non-refundable functional utility token which will be used as the unit of exchange between participants on BitMax.

10.4.2. Top 10 CEX Trading Volume

Rank	Tokens	Market Cap (Updated 9/16)	Coins	Pairs
1	Binance	$17,125,150,367	332	1231
2	Huobi Global	$3,643,050,666	353	934
3	Coinbase Exchange	$3,158,730,315	102	307
4	Crypto.com Exchange	$2,914,930,220	121	220
5	FTX	$1,997,324,328	277	442
6	KuCoin	$1,376,146,447	467	915
7	Gate.io	$1,253,424,329	929	1944
8	Kraken	$905,641,493	89	393
9	Bitfinex	$693,356,408	156	376
10	Binance	$424,933,413	60	117

11. Algorithmic Trading

11.1. What Is Algorithmic Trading?

Algorithmic Trading is an automated procedure used in a trading system. These robots are programmed continuously to begin executing trade transactions any time of the day, provided that the set conditions for predetermined factors such as time, price, or level of the transaction are met. These robots usually employ special statistical philosophies and formulas to make financial moves in the exchange. In this case, the currency market uses algorithms to predict which currencies will rise and fall in value and then trade these based on mathematical computations and formulas. The concept is very similar to how a stock trader would use mathematical calculations and predictions to gain an edge over other stock traders.

The software also employs certain protocols and rules to ensure accuracy. However, the programs that accomplish this are not human, so it becomes important to know what these algorithms are, how they work, and why they make their decisions. These robots are not always accurate, so they can lose money by investing in one of these types.

The biggest advantages of these types of trading robots are the amount of money that can be made. Like a human broker, the traders do not control them, so they don't have to worry about losing money or having their investment stolen. They are completely automated, which means they follow a set of predetermined rules or protocols to make all of their investments. This allows all traders to focus on making their trades rather than worrying about the results of other people's transactions.

The most popular types of algorithmic trading robots are:

- **Portfolio Indexing** allows investors to trade and maintain crypto indexes. Rebalancing a portfolio is the act of trading coins so that each asset's value equals the original percentages. These bots allow traders to rebalance portfolios. This involves realigning portfolio assets to achieve the desired allocation split. Portfolio rebalancing will enable traders to set trading intervals and thresholds to help them maintain their target allocation. This gives them more control over their long-term risk levels.

- **The social or Copy Trading** concept is following already in-profit bots, signals, indexes, or Expert traders. You just need to connect your exchange account to a social copy trading platform, check their marketplace, and follow your preferred strategy. Then the platform will automatically perform or copy trades to your exchange account.

- **Manual Trading** is a platform that allows traders to automate tasks that are too complicated, time-consuming, or laborious to perform manually. These tasks can be automated in a variety of ways. You can create a trading bot for a single pair of trading pairs or design a more complex bot that trades on multiple pairs or exchanges. Many platforms provide ready-made bots that can be adjusted to suit your needs.

- **Automated Trading** is where a platform provides a trader with the opportunity to automate actions that may seem too complex, time-consuming, or laborious to execute manually. Automating these actions tasks ranges from designing a simple trading bot on a single trading pair or designing an advanced bot that trades on many pairs or even many exchanges. Also, most platforms offer ready-made bots where you just need to adjust a simple setting, and you are ready to go.

The research available to investors and scholars is growing year on year in finding the best algorithms for trading cryptocurrency. Much research has been done to date, and it has proven to be very successful in multiple trading analyses.

Kamrat (2018) [60] performed backtesting and comparing the trading systems (Original Turtle, Extended Turtle) on eight prominent cryptocurrencies. The Original Turtle Trading System had an average net profit margin of 18.59% (percentage net profit over total revenues) and 35.94% overall profitability (percentage winning trades over the total number of transactions) through the experiment. This was achieved in nearly 87 transactions over one year. Extended Turtle Trading System had a 52.75% average profitability and 114.41% average profit margin in 41 trades over the same period. The research revealed that the Extended Turtle Trading System could outperform the Original Turtle Trading System when trading cryptocurrency.

11.2. The Categorization Of Algorithms

The use of machine learning technology creates computer algorithms that automatically improve by finding patterns in data without any explicit instructions. Machine learning has seen a rapid advancement in recent years, which has allowed it to be used in cryptocurrency trading, particularly in predicting cryptocurrency returns.

Many methods can be used to identify profitable trades. However, most of them fall under these five categories[61].

- **Classification Algorithms.** Machine Learning Classification Algorithms. In cryptocurrency trading, Naive Bayes (NB), Support Vector Machine (SVM), KNearest Neighbours (KNN), Decision Tree [DT], Random Forest [RF], and Gradient Boosting [GB] algorithms are used.

- **Clustering Algorithms.** Clustering is a machine-learning technique that groups data points so that each group has some regularity. K-Means, a vector quantization that is used in clustering analysis in data mining, is called K-Means.

---

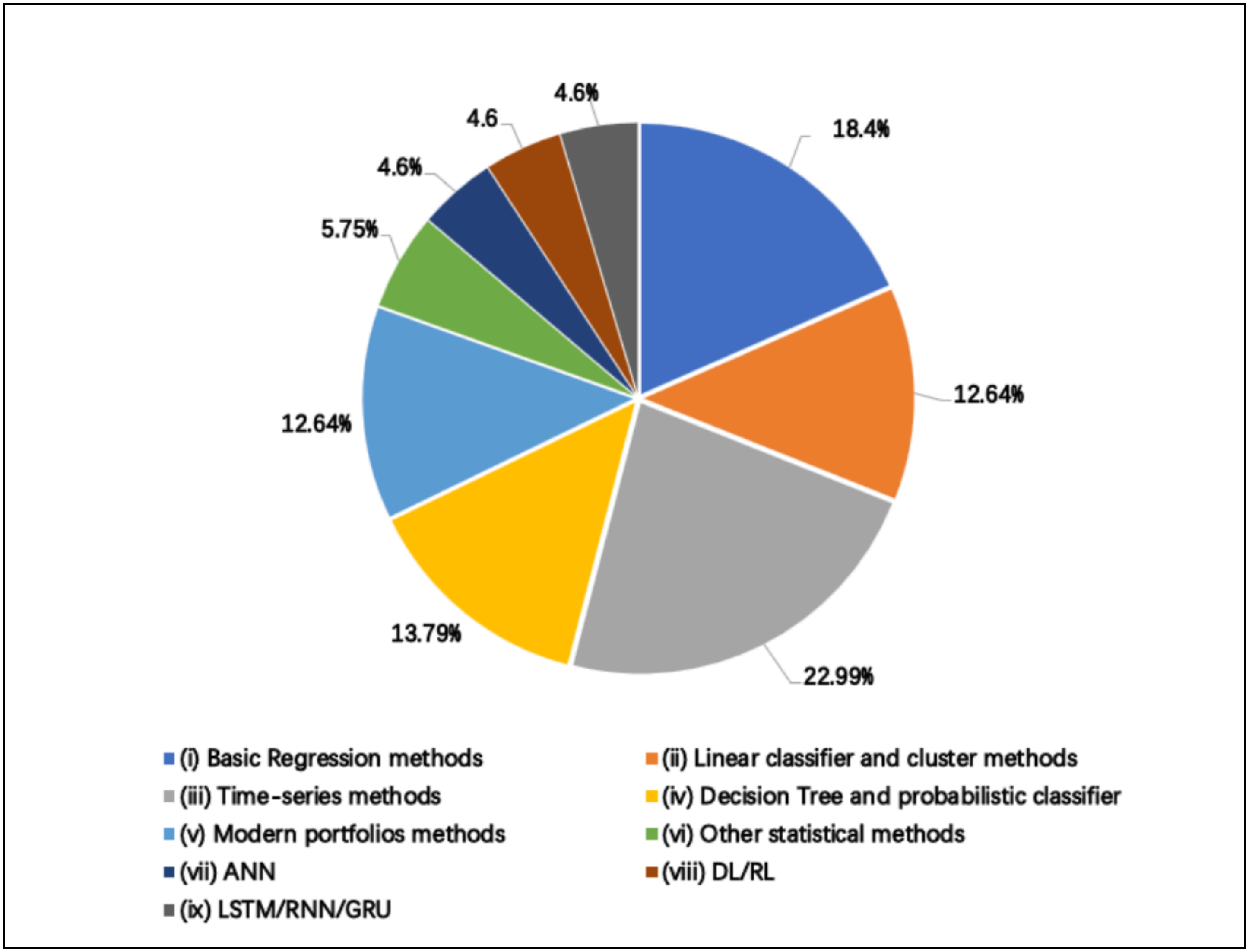
[60] Kamrat, S., Suesangiamsakul, N., Marukatat, R., 2018. Technical analysis for cryptocurrency trading on mobile phones, in: 2018 3rd Technology Innovation Management and Engineering Science International Conference (TIMES-iCON), IEEE. pp. 1–4.
[61] Fang F, Ventre C, Basios M, et al. (2021) Cryptocurrency trading: A comprehensive survey. arXiv.org. Available from: https://arxiv.org/abs/2003.11352 (accessed 30 September 2021).

- **Regression algorithms.** Regression is any statistical technique that attempts to estimate a continuous value. Linear Regression (LR), and Scatter-plot Smoothing are two common methods used to solve regression problems in cryptocurrency trading.

- **Deep Learning Algorithms.** Deep learning is a modern version of artificial neural networks (ANNs). This was possible thanks to the advancements in computing power. An ANN is a computer program that draws inspiration from the natural neural networks of an animal's brain. The system "learns", to perform tasks such as prediction using ex-samples. High computational complexity is what gives deep learning its superior accuracy. Many modern applications of artificial intelligence are based on deep learning algorithms. The most popular deep learning technologies for cryptocurrency trading are convolutional neural networks, recurrent neural networks, and gated recurrent units.

- **Reinforcement Learning Algorithms.** Reinforcement learning (RL) is an area of machine learning leveraging the idea that software agents act in the environment to maximize a cumulative reward. Deep Q-Learning (DQN) and Deep Boltzmann Machine (DBM) are common technologies used in cryptocurrency trading using RL. Deep Q learning uses neural networks to approximate Q-value functions.

Current research distribution using various algorithms in trading cryptocurrencies. Source: Cryptocurrency Trading - A Comprehensive Survey.



11.3. The use of Social Sentiment for cryptocurrency trading

Trading cryptocurrency is currently done using methods that are similar to those used to trade commodities or stocks. However, these algorithms are not always well-suited to predict cryptocurrency prices. Contrary to stock exchanges that can shut down for hours or even days, cryptocurrency trading and prediction seem more predictable and consistent than those of the stock market[62].

Research on the subject of identifying emotional buying, and selling triggers to predict cryptocurrency movements and develop profitable trading strategies based upon those predictions is increasing at a rapid rate. However, it is important to check the suitability of these methods used in research sentiment analysis in the cryptocurrency market.

In recent years social platforms have enjoyed a lot of popularity. This is mainly due to the human desire to interact and the ease with which virtual connections can be made and maintained. Social networks offer an enormous amount of information and opinions on all topics. This data has become a priority in many research projects and language processing activities and is in the best interest for GSHI to pursue.

Technically, sentiment analysis is part of Natural Language Processing (NLP.) It involves training a classifier to recognize and establish attributes/features that can be used later in the processing of new input data or evaluating its nature[63].

Artificial intelligence and machine-learning techniques have been used to predict stock markets over the past decade. The most popular techniques are Neural Networks (NNs), Support Vector Machine (SVMs), and Random Forests. Deep learning methods are derived from NNs. They have been used for forecasting the price of Bitcoin, Digital Cash, and Ripple. Recurrent Neural Networks (RNN) were used to predict market direction in the case of the NASDAQ composite index with promising results[64]. Various institutional and educational research using social analysis have produced cryptocurrency yields above 100% depending on the algorithm used.

In the below example, we get an idea of how a Natural Language Processing program may identify keywords that are attributed to a predetermined Impact (Additive) for scoring. Words such as “ban” are associated with negative sentiments (-1.0) and the word “moon” is associated with a positive input (+0.3). Identifying the keywords and constant tuning of the machine learning model will only increase the accuracy and confidence level of the prediction.

[62] Valencia F (2019) Price Movement Prediction of Cryptocurrencies Using Sentiment Analysis and Machine Learning. San Pablo, Querétaro, Mexico: Entropy.
[63] Dulau T-M (2019) CRYPTOCURRENCY – SENTIMENT ANALYSIS IN SOCIAL MEDIA. Sciendo, dissertation.
[64] McCoy M and Rahimi S (2020) Prediction of highly volatile cryptocurrency prices using social media. International Journal of Computational Intelligence and Applications. Available from: https://www.worldscientific.com/doi/10.1142/S146902682050025X (accessed 9 July 2021).

Figure (X) Keyword scoring shows positive and negative scores for certain words.

Word	Impact (Additive)
ban	−1.0
banned	−1.0
bans	−1.0
banning	−1.0
crackdown	−0.7
crack	−0.7
cracks	−0.7
cracking	−0.7
suspend	−0.7
suspended	−0.7
suspends	−0.7
suspending	−0.7
halt	−0.7
halts	−0.7
halted	−0.7
hacked	−0.5
heist	−0.5
hacks	−0.5
hack	−0.5
regulations	−0.8
moon	+0.3

Source: McCoy M and Rahimi S (2020) Prediction of highly volatile cryptocurrency prices using social media. International Journal of Computational Intelligence and Applications.

The breadth and scope for Sentiment Analysis is beginning to surface in layman's terms; however, there is still a significant amount of research to be conducted using various Artificial Intelligence and Machine Learning Techniques which still makes this landscape ambiguous with varying performance results.

Figure (X) Example results of research using MLP, SVM, and RF modeling showing how these different algorithms are producing varying results.

Table 3. Results of applying multi-layer perceptron (MLP), support vector machine (SVM) and random forest (RF) using Twitter data, market data or both for predicting daily market movements for Bitcoin.

Model	Accuracy (95% CI)	Precision	Recall	F_1 Score
MLP Twitter	0.39 (±0.02)	0.38	0.39	0.38
MLP Market	0.72 (±0.03)	0.74	0.72	0.71
MLP Twitter and Market	0.72 (±0.06)	0.76	0.72	0.72
SVM Twitter	0.50 (±0.03)	0.29	0.50	0.37
SVM Market	0.55 (±0.03)	0.53	0.56	0.47
SVM Twitter and Market	0.55 (±0.03)	0.31	0.56	0.40
RF Twitter	0.44 (±0.04)	0.50	0.80	0.62
RF Market	0.61 (±0.04)	0.67	0.25	0.36
RF Twitter and Market	0.44 (±0.04)	0.28	0.44	0.34
Random	0.50 (±0.28)	0.49	0.50	0.50
Majority	0.55 (±0.0)	0.31	0.56	0.40

Source: McCoy M and Rahimi S (2020) Prediction of highly volatile cryptocurrency prices using social media. International Journal of Computational Intelligence and Application

11.4. Common Trading Strategies Encountered In The Cryptocurrency Market

There are numerous different platforms out there for algorithmic trading, and many people use multiple ones. But there are some advantages of algorithmic trading and some disadvantages for new traders who are just learning about this method. New traders often fall into the trap of jumping in too fast and becoming frustrated because the results of their trades are not what they expected. For experienced traders, the biggest advantages of these types of trading platforms are that the investor has a chance to try different options. The biggest disadvantages are that most of them involve a bit of risk, and no one should ever get into the market without doing their research first. Once a trader understands how the platform works and the results, they will be able to trade successfully.

The most common type of cryptocurrency trading strategies are:

- **Momentum-based Strategies or Trend Following Algorithmic Trading Strategies** seek to profit from the continuance of the existing trend by taking advantage of market swings and used when speed is important.

    1. **Value investing:** Value investing generally relies on long-term reversion of mean, while momentum investing is based upon the time gap before the mean reversion occurs.

    2. **Momentum Investing:** Momentum means chasing performance but systematically taking advantage of other performance chasers making emotional decisions.

    3. **Earnings Momentum Strategies:** An earnings momentum strategy may profit from the under-reaction to information related to short-term earnings.

    4. **Price Momentum Strategies:** A price momentum strategy may profit from the market's slow response to a broader set of information, including longer-term profitability.

- **Arbitrage Algorithmic Trading Strategies** identify trigger events that seek profit from statistical modeling that searches for price discrepancies across exchanges. A currency can, very briefly, be available at different prices on various exchanges simultaneously. A crypto arbitrage algorithm will use its exceptional speed to buy the coin on the exchange where it is cheapest and then sell it on the exchange where the price is highest to make a profit before the temporary price anomaly resolves itself. For instance, if BTC price falls under (x), then Dogecoin will fall by (y), but Dogecoin has not fallen, so sell Dogecoin to make a profit (common with pairs trading.)

Example Arbitrage Coding for Dai. This piece of code is looking for the best trading opportunities with other coins across different exchanges.

```
// Compare if last price is overvalued and too high away from the average - in this case, want to sell aka go short
if (last > movingAverage + PositionBand && position === 'out') {
  const quoteBalance = await ocean.wallet.getTokenBalance({
    etherAddress: process.env.BOT_ADDRESS,
    tokenAddress: myPair.quoteToken.address
  })
```

- **Statistical Arbitrage Algorithmic Trading Strategies** Is a group of trading strategies that utilize mean reversion analyses to invest in diverse portfolios of up to thousands of securities for a very short period, often only a few seconds but up to multiple days.

- **Automated Market Making Algorithmic Trading Strategies** use an automated market maker (AMM) as the underlying protocol that powers all decentralized exchanges (DEXs). Simply put, they are autonomous trading mechanisms that eliminate the need for centralized exchanges and related market-making techniques. DEXs promote autonomy such that users can initiate trades directly from non-custodial wallets (wallets where the individual controls the private key.).

Algorithmic traders also find some great benefits in using backtesting. Backtesting is simply the process of trading against historical market trends to test the accuracy of your trading bot. The idea is that while the backtesting takes place, the traders will not be spending any money. By running the system back and forth as if it were performing in real-time, cryptocurrency traders can see what effect is occurring in the market.

12. Loans

12.1. How Does Crypto Lending Work

You have a savings account with your bank, which means that you are likely earning an interest rate of around 0.5% annually. What if you could do the same thing with a crypto loan platform where you lend your crypto to borrowers and earn more than 10%? This crypto loan market is rapidly expanding and is more accessible to people who don't have a financial history, underbanked consumers who don't have a bank account, and self-employed workers who struggle to access loans - BTC loans can be practically instant.

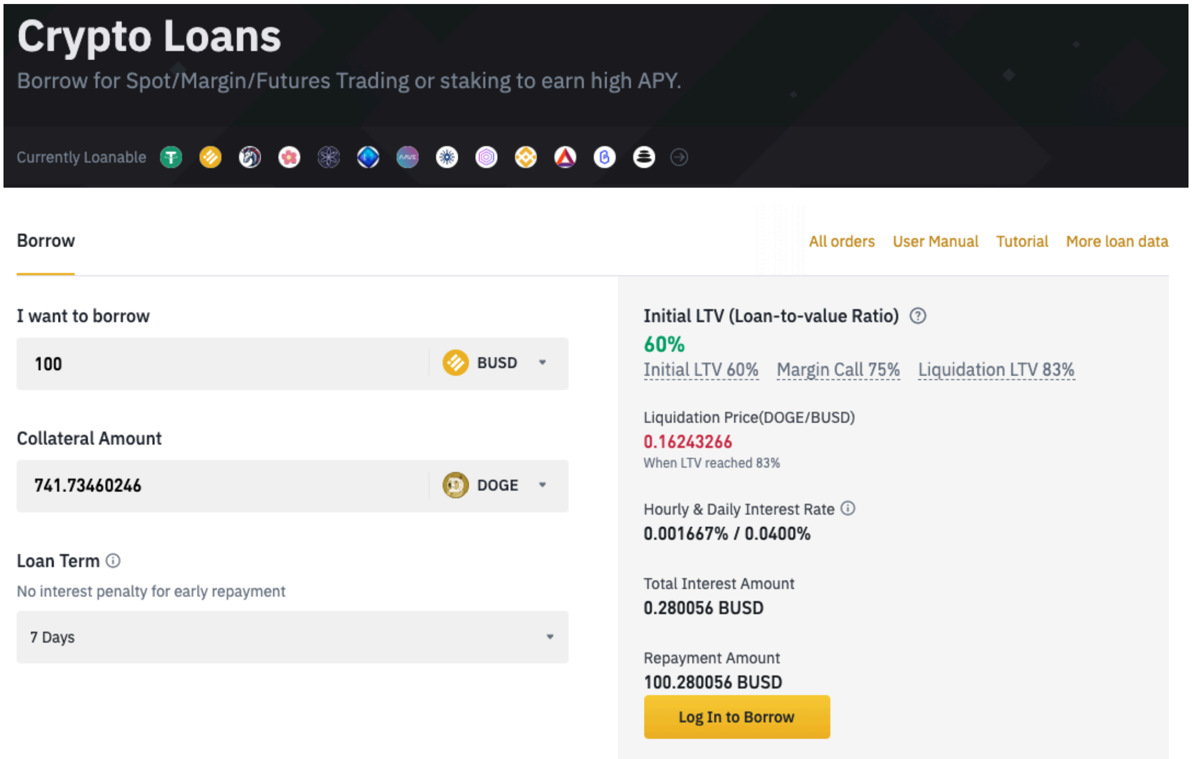
You'll also be able to make your assets liquid without triggering a taxable event — and you can adjust the loan to suit your needs. Users can also switch between crypto assets, so you could deposit Ether and borrow Tether, all on the same platform. This can potentially change the competitive dynamics of the capital relationship between the firm and small businesses.

Cryptocurrency Lending

Cryptoholders searched for new ways to make crypto assets work for their benefit while in a Hold On For Dear Life (HODL) position during the 2020 pandemic.

As a result, crypto loans quickly gained popularity. You can lend your cryptocurrency to someone looking for a loan against your crypto asset, which will be given to the borrower as a stablecoin or, in some instances, even fiat currency. This allows the borrower to keep their crypto asset and not sell it, often due to the appreciation nature of the coin minus the volatility.

The question immediately asked is what protects the lender if the crypto asset falls below the current price? This is where the Loan to Value (LTV) ratio comes into play. Meaning that if the borrower's cryptocurrency falls below a certain threshold which would increase the LTV, then liquidation of the borrower's collateral initiates, and the sale of the borrower's crypto assets is used to pay back the lender.



Borrowing BUSD against DOGE for 7-Days

Both the DeFi (non-custodial) and CeFi (custodial) protocols offer unique features. CeFi acts as a broker in the execution and repayment of loans. This means that both the lenders and borrowers must give up any collateral or assets until the amount borrowed is paid in full. Each party will have no control over its assets. These platforms also have a KYC process, which means anonymity is not possible.

DeFi protocols use smart contracts to execute loans. This process is automated, and the agreement will be executed when all requirements are met. One does not have to go through the KYC/AML process.

Also, DeFi protocols are expanding rapidly in this space, with new concepts such as Flash Loans constantly emerging as new opportunities.

12.2. Flash Loans

Flash loans are made using smart contracts and are a rapidly emerging trend for DeFi protocols because they are immediate and do not require collateral. This means that these protocols will not allow funds to change hands unless certain conditions are met without human or centralized intermediaries. The rule for a flash loan is that the borrower must repay the loan before the transaction's end. Otherwise, the smart contract will reverse the entire transaction to protect the liquidity pool.

A Flash Loan is useful for traders who want to profit quickly from arbitrage opportunities in two different markets that are pricing cryptocurrency differently.

Traders can make money by looking for price discrepancies across several different exchanges. Say two markets are pricing ABC-coin differently. It's priced at $1 on Exchange A and $2 on Exchange B. A user can use a flash loan and call a separate smart contract to buy 100 ABC coins for $100 at Exchange A, then sell them for $200 at Exchange B. The borrower then repays the loan and pockets the difference. Other uses include increasing collateral backing a loan for another type of collateral.[65]

<u>AAVE is a leading emerger in the Flash Loan space. This logic tree is an example of a recommended approach for constructing a flash loan opportunity on the AAVE platform.</u>

For developers, a helpful mental model to consider when developing your solution[66]:

1. Your contract is called the LendingPool contract, requesting a Flash Loan of a certain amount of reserves using flashLoan().
2. After some sanity checks, the LendingPool transfers the requested amounts of the reserves to your contract, then calls executeOperation() on your contract (or another contract that you specify as the _receiver).
3. Your contract, now holding the flash loaned amounts, executes any arbitrary operation in its code.
    a. If you are performing a **'traditional' flash loan**, then when your code has finished, you transfer the flash loaned amounts of reserves back to the LendingPool.
        i. The LendingPool contract updates the relevant details of the reserves and **pulls** the flash loaned amount + fee.
            1. This is different from v1 flash loans, where the flash loaned amount needed to be pushed back to the LendingPool contract.
        ii. If the amount owing is not available (due to a lack of balance or approval), then the transaction is reverted.
    b. If you are performing a **flash loan to incur debt** (see the mode parameter in the flashLoan() function), then a debt will be incurred.

---

[65] https://medium.com/@bneiluj/flash-boys-arbitrage-dao-c0b96d094f93
[66] https://docs.aave.com/developers/guides/flash-loans

i. All of the above happens in 1 transaction (hence in a single ethereum block).

Because flash loans are still in their infancy, there are risks associated with them. Flash loans are still developing protocols. There has been a long list of attacks with different characteristics. Two of the most popular attacks were conducted on Ethereum trading and the lending protocol bZX. The borrower tricked the lender into believing that they had repaid the loans in full. But the borrower wasn't. The borrower manipulated the stablecoin price temporarily to make the loan repayment more expensive. Another recent event saw one entity use a flash loan to obtain additional votes in a MakerDAO election that significantly impacted the entire community.

Flash Loans can be useful in certain instances, for traders looking to quickly profit from arbitrage opportunities when two markets are pricing a cryptocurrency differently.

There are numerous benefits to the rapidly growing crypto loans market, and why these types of loans are more attractive than traditional secured loans[67].

To recap the reasons why crypto loans will play a significant role in the cryptosphere:

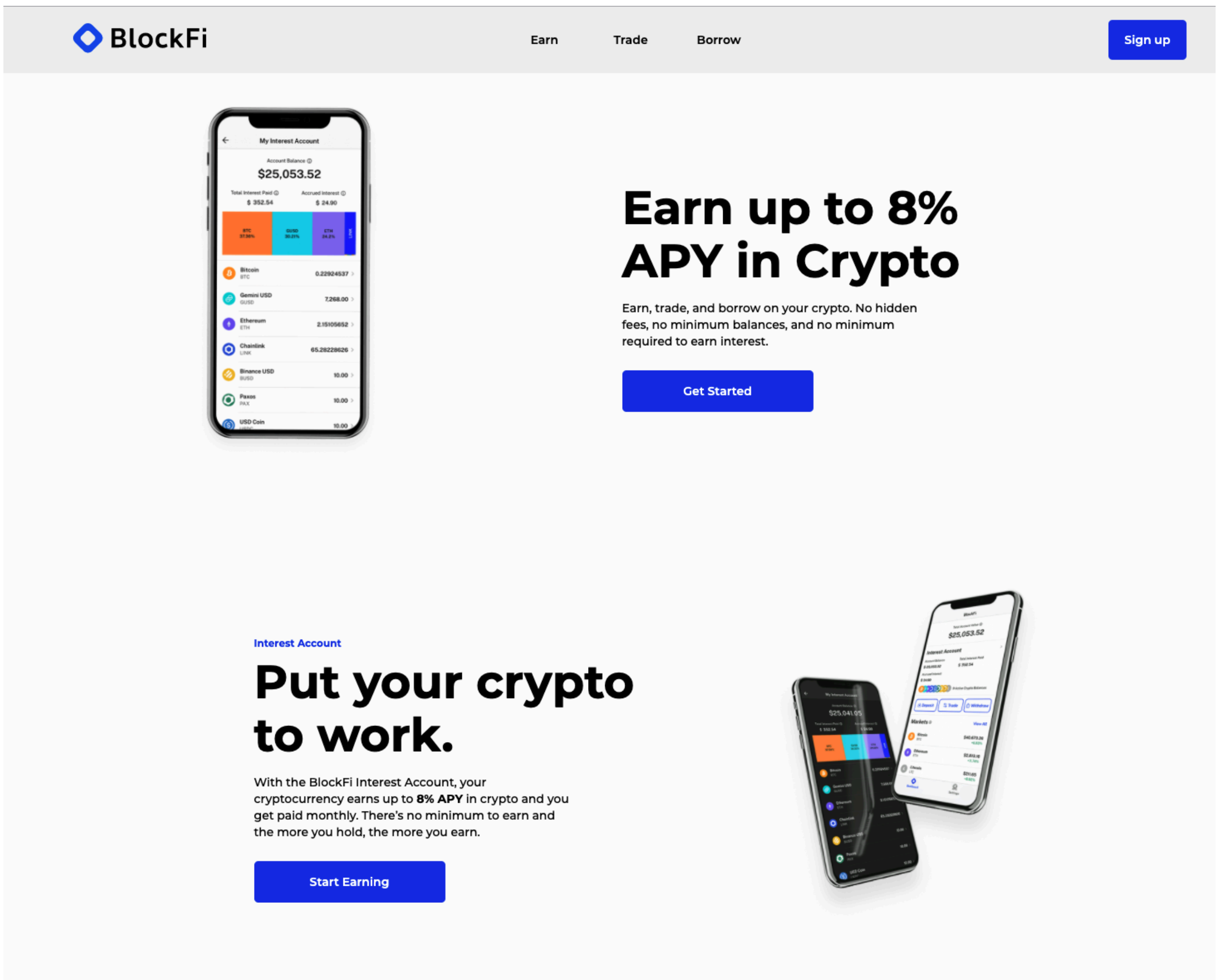
- Smart contract guarantees: because of the overcollateralization mechanism, crypto loans are paid back, which eliminates the costs associated with vetting the borrower's credit against risk.
- Crypto-lending protocols repay lenders interest rates far exceeding traditional finance rates averaging .05% in favor of rates exceeding 10%.
- Your asset value determines the amount of the loan you are able to obtain based on the protocols Loan to Value Ratio (LTV.)
- You can choose a loan currency. This will depend on the platform you
- use and the amount that you need. You can choose from a selection of cryptocurrency or fiat.
- As a borrower, I don’t have to sell my crypto, but only lock them into a smart contract, and if the LTV threshold falls below a certain event, the liquidation process is triggered.
- No credit checks: Crypto-lending platforms and exchanges don't typically run credit checks on applicants. This makes it attractive for people with poor credit and no credit history, or with no access to banking or have no banking history.
- Fast funding: After you have been approved, your loan money will be available within hours if not minutes.
- Anyone can borrow crypto: There are many cryptocurrency exchanges that offer "interest" accounts, where you can lend your digital assets to others and receive a high return (sometimes as high as 10 percent).

[67] https://saltlending.com/loan-to-value-ltv-explained-9ff7182d446f/

12.3. Top CEFI Lending Platforms

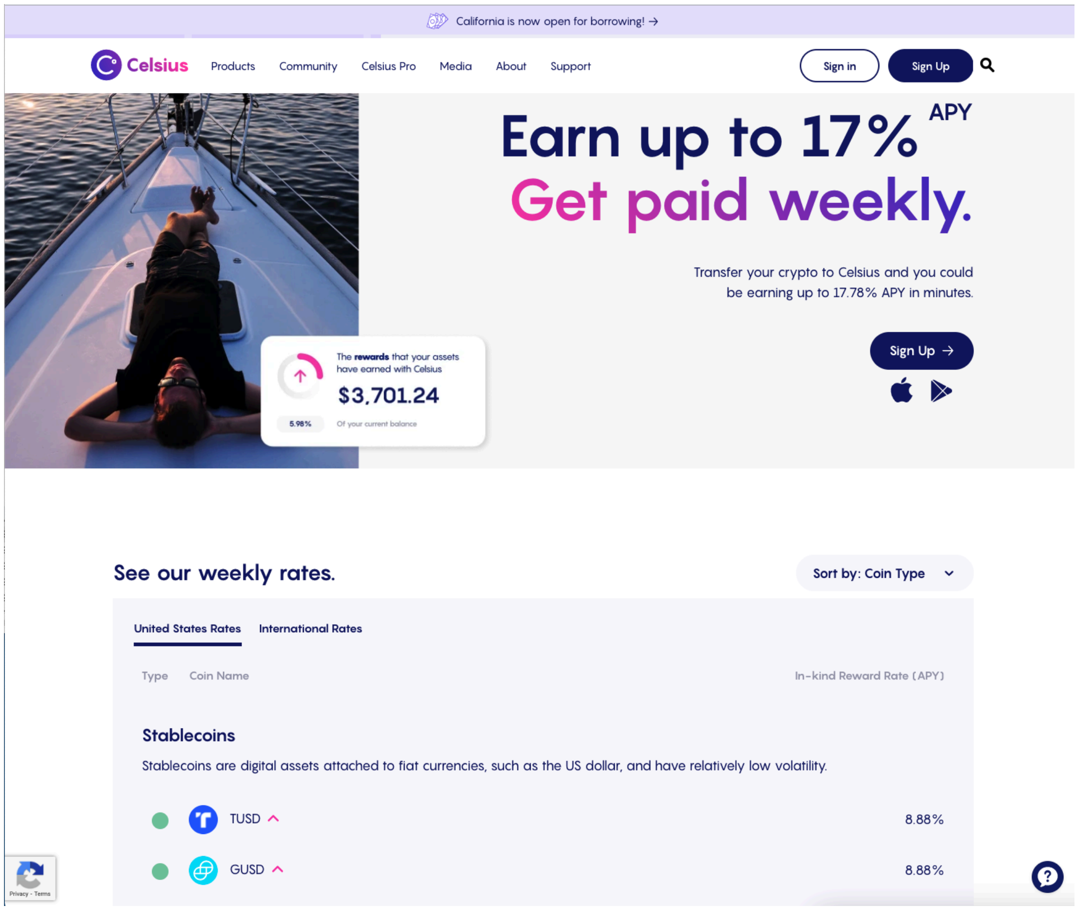
BlockFi allows customers to borrow USD against their cryptocurrency. Flori Marquez, Zac Prince, are the founders of BlockFi. BlockFi was established in August 2017. It is located in Jersey City, New Jersey. BlockFi has raised $350 Million Series D and has a $3B valuation.

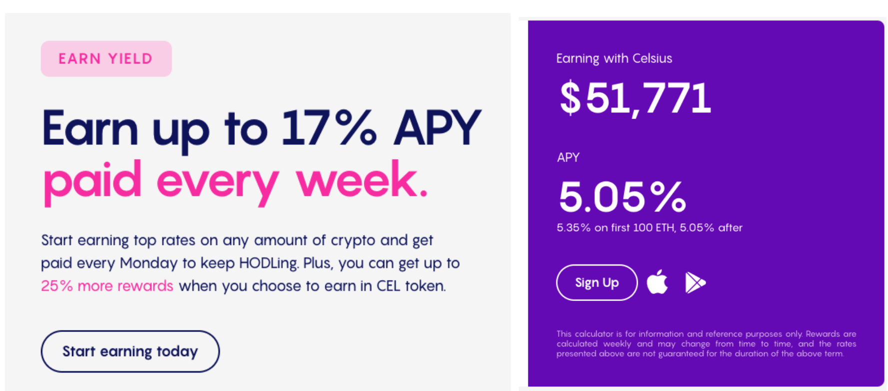
BlockFi allows you to earn as much as 8% per year on your cryptocurrency. BlockFi also enables you to borrow against your coins. You can borrow against your coins instead of having to sell them when you have the funds. This allows you to avoid selling into a down market. We'll show you how BlockFi can offer loan services and will enable you to earn interest by depositing.[68]



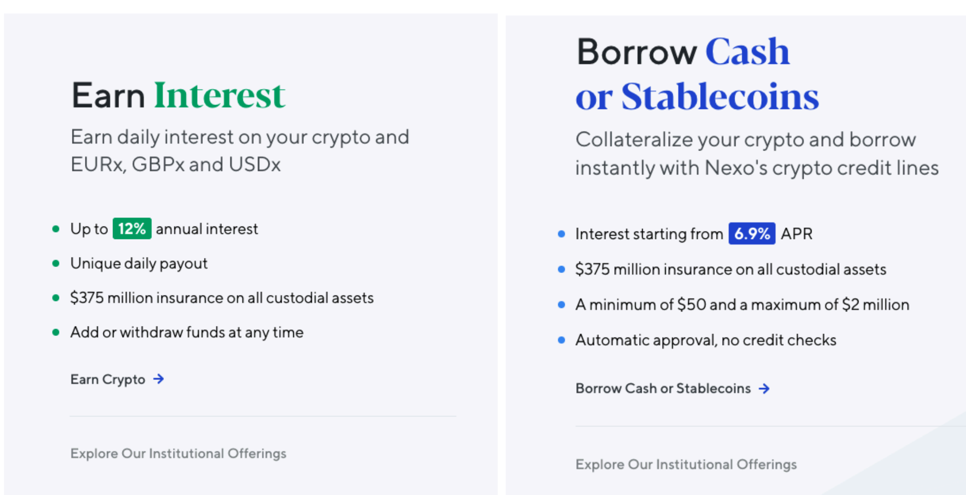
[68] https://thecollegeinvestor.com/34536/blockfi-review/

Celsius Network is an extremely popular loan platform with more than $10B in assets and 485,000 users. Its extremely low borrowing rates (for now), which start at 1%, are much lower than the average CeFi borrowing rate at the time of writing at 4%. Celsius can support 25 coins and offers flexible, but not capped, LTV rates.



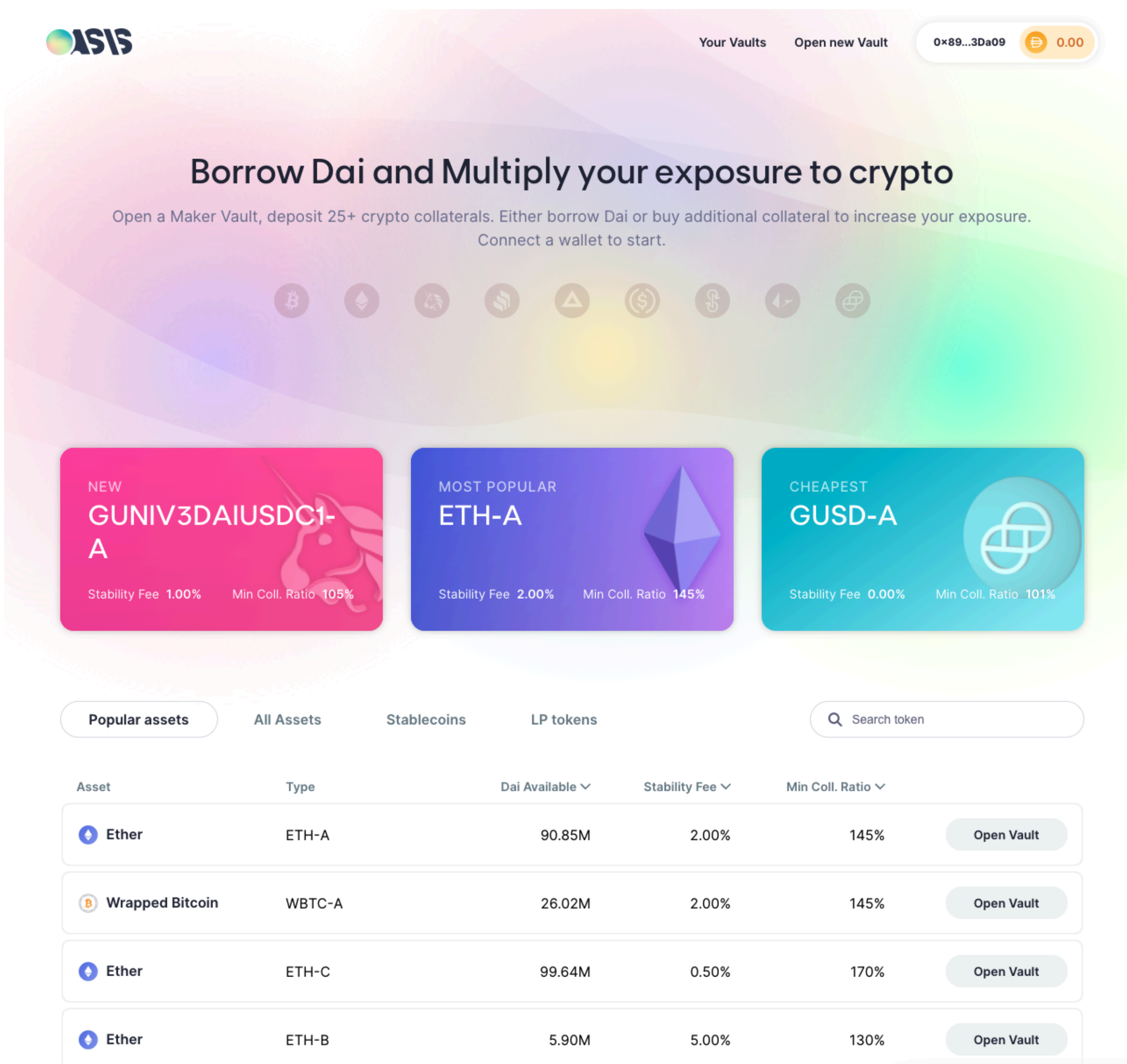


Nexo, another popular CeFi loan platform, is distinguished by its $375 million insurance policy on all custodial assets. It has more than 1.5 million users and $12B in assets. Nexo usually has slightly higher LTV and slightly lower borrowing rates. It supports 20 currencies.

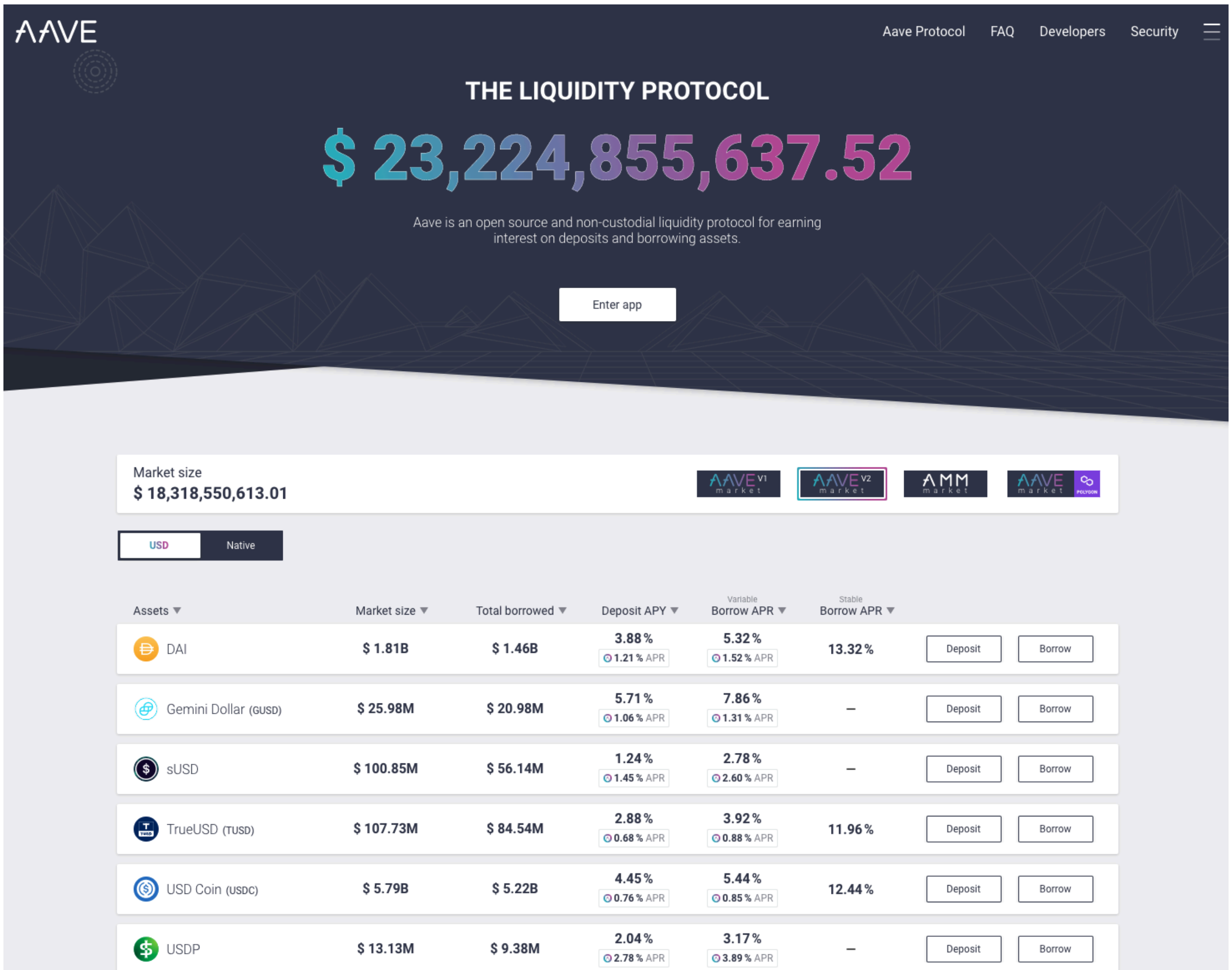


Oasis Borrow is the hub for crypto loans in the Maker ecosystem, which supports DAI stablecoin. Because they can be used to finance yourself from yourself, which makes Maker loans attractive. Oasis Borrow allows you to put up an asset and agree to make a certain amount of DAI or repay your loan before unlocking your asset. You are borrowing crypto to generate the collateral asset.

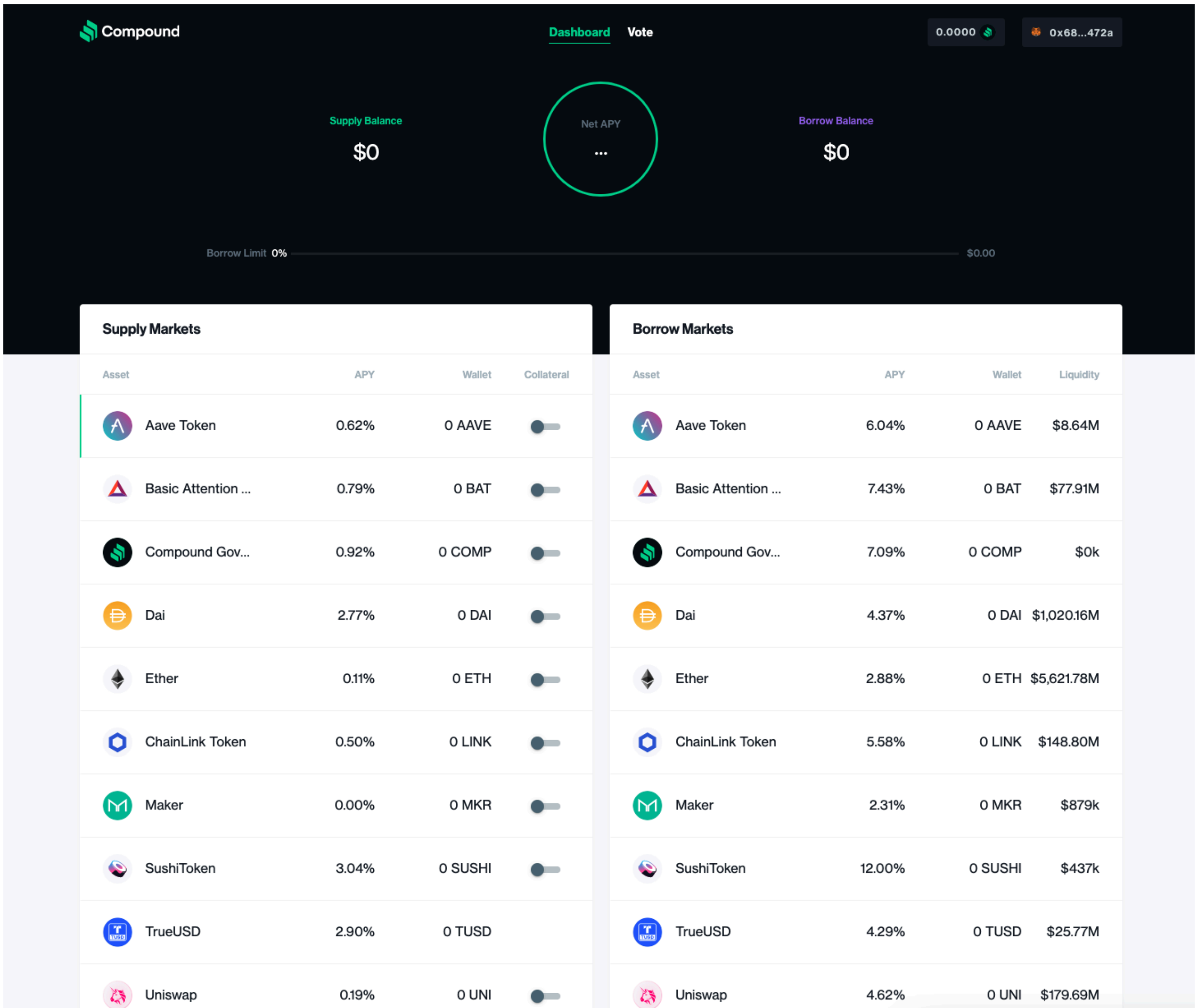
Maker offers a 50% or 75% LTV rate on crypto loans. The higher rate also receives higher stability fees (which can be variable). Smart contracts on Maker automatically liquidate accounts with lower LTVs than the agreed rate. This comes with a 13% liquidation penalty. Maker currently holds the title of largest DeFi lending project with more than $7.3B in assets and more loans than $2.7B.



AAVE also offers DeFi loans and is the pioneer of Flash Loans. This is a first in its field. It has a lower LTV rate than large competitors and very low borrowing rates. It accepts 24 coins for collateral, as opposed to Maker's 18 and Compound's 9.



Compound requires its users to support the protocol by depositing cryptocurrency, which earns interest. The amount that they have contributed is represented in cTokens. Their value increases over time. This cToken supply balance may be used as collateral to secure a loan. The LTV rates for Compound are usually slightly higher than MakerDAO. Compound does not charge borrowing fees, and the liquidation threshold is lower. It will liquidate only 50% of an under-collateralized loan with an 8% liquidation penalty.



13. Non-Fungible Token (NFT)

13.1. What is a Non-Fungible Token?

The trade volume for non-fungible tokens (NFTs) surpassed $250m in 2020. This is a fourfold increase over 2019, and the market capitalization of this nascent tokens industry, which includes artwork, game collectibles, and tokens representing a variety of items, reached $338m. Fast forward 6-months, and the record-breaking sales volume for non-fungible tokens rose to $2.5 billion during the first half of 2021.

Unlike cryptocurrencies, where each token is equal, non-fungible tokens are unique and limited in quantity. More or less, "Non-fungible" means that the item is unique and cannot be replaced by another. A bitcoin, for example, is fungible. You can trade it for another bitcoin and get the same thing. However, a one-of-a-kind trading card is not a fungible[69]. You would have something entirely different if you traded it for another card. These tokens can be digital assets, or tokenized versions of real-world assets. Because NFTs can't be interchangeable, they could serve as proof of ownership and authenticity within the digital realm.

NFTs can be any digital content (such as music, drawings), but the main excitement right now is about using the tech for digital art sales[70]. Imagine if digital assets could be created similar to Bitcoin, but each unit would have a unique identifier. This would make them unique from the rest (i.e. non-fungible). This is the essence of an NFT.

How can these NFT's be valuable? Like any valuable item, its value is not inherent to it. The value is assigned by those who believe it is valuable. Value is simply a shared belief. It doesn't really matter what it is; fiat currency, precious metals, classic cars all are assigned value because people believe they do.

NFTs are an important building block of a blockchain-powered digital economy. Many projects have tried NFTs in various use cases including gaming and digital identities, licensing certificates, fine art, and certificates. NFTs allow fractional ownership of high-value items. **NFTs are available for trading in open markets, such as Treasureland, BakerySwap and Juggerworld (BSC), and OpenSea (Ethernet).** These markets allow buyers and sellers to connect, and each token's value is unique. NFTs can be subject to price fluctuations due to market supply or demand.

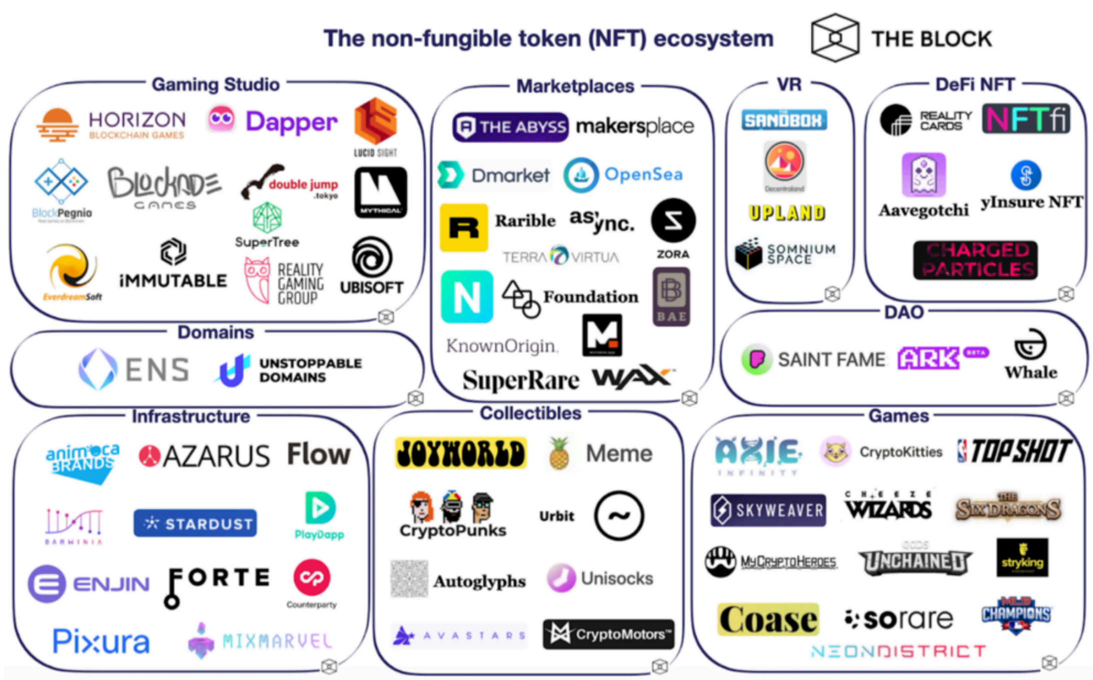
[69] DeNicola L (2021) What to know about non-fungible tokens (nfts) - unique digital assets built on Blockchain Technology. Business Insider, Business Insider. Available from: https://www.businessinsider.com/nft-meaning (accessed 23 September 2021).
[70] Clark M (2021) NFTs, explained. The Verge, The Verge. Available from: https://www.theverge.com/22310188/nft-explainer-what-is-blockchain-crypto-art-faq (accessed 22 September 2021).

A university could issue an NFT to students who have completed a degree. This would allow employers to easily verify the applicant's education[71]. A venue could also use NFTs for selling and tracking event tickets. NFTs may have many applications beyond the art world, as the technology and concept improves.

Selling NFTs has been a lucrative business in the art world. Here are a few examples you may have heard about[72]:

- Digital artist Beeple sold "Everydays — the First 5000 Days" for $69.3 million through a Christie's auction.
- A 20-second video clip of LeBron James "Cosmic Dunk #29" was sold for $208,000.
- A CryptoPunk NFT sold for $1.8 million at Sotheby's first curated NFT sale.
- Twitter CEO Jack Dorsey auctions an NFT of his first tweet, which sells for $2.9 million.

Blockchain technology opens up new possibilities for digital collectibles that go beyond traditional financial applications. NFTs, which can represent physical assets in digital form and have the potential to solve numerous copyright and ownership issues.



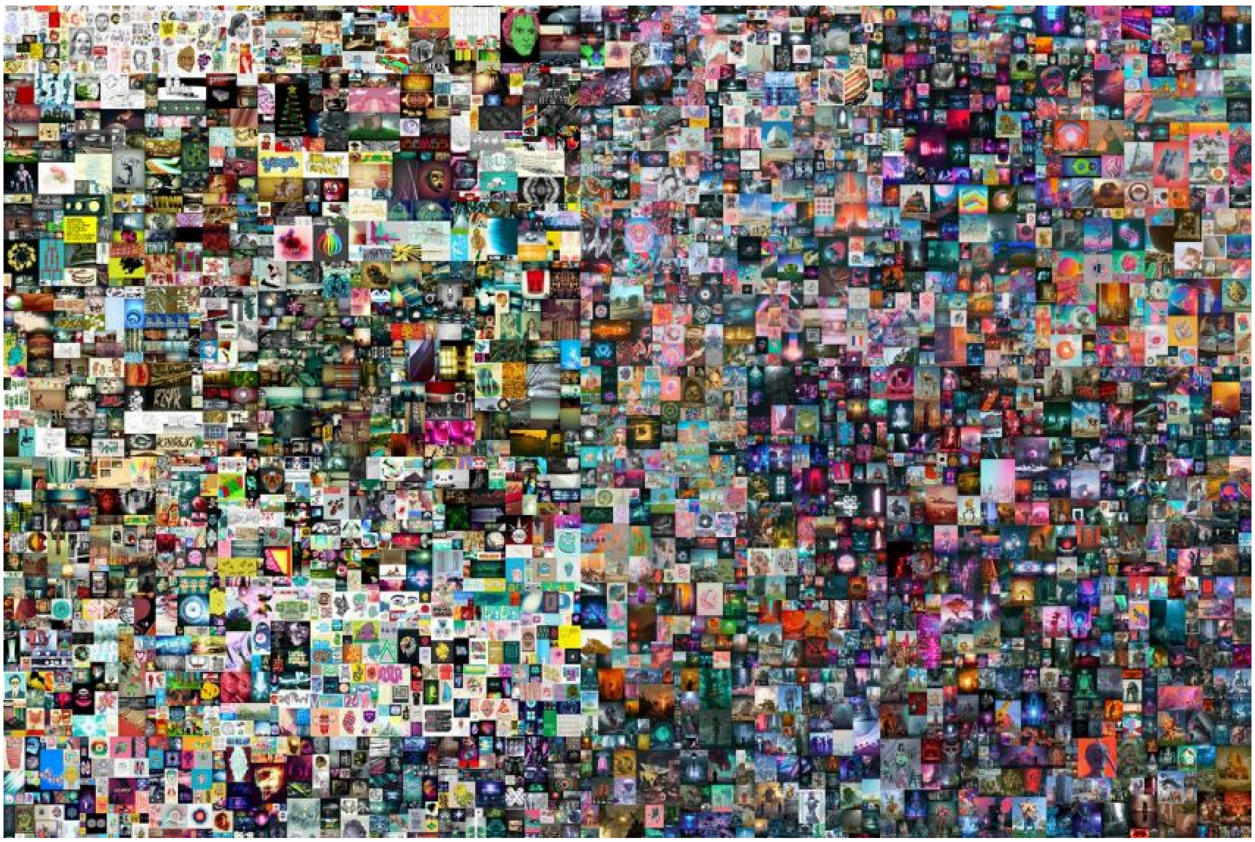
Source: The Block Non-Fungible Token (NFT) Ecosystem

[71] Binance Academy (2021) A guide to crypto collectibles and non-fungible tokens (nfts). Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/a-guide-to-crypto-collectibles-and-non-fungible-tokens-nfts (accessed 21 September 2021).
[72] DeNicola L (2021) What to know about non-fungible tokens (nfts) - unique digital assets built on Blockchain Technology. Business Insider, Business Insider. Available from: https://www.businessinsider.com/nft-meaning (accessed 23 September 2021).

13.2. Notable NFT Sales to Date

Everydays: the First 5000 Days ($69.3 million)

Mike Winkelmann, nicknamed Beeple, has sold his piece "Everydays — The First 5000 Days" for $69.3 million, breaking the record for digital artwork sales and becoming one of Christie's Auction House's most valuable sales.



“Everydays — The First 5000 Days” is a JPG file that contains a collage of images created by Beeple, a single picture is usually sold at auction, but here you’ll find almost everything the artist has made since he began displaying his work online in 2007.

CryptoPunk #3100 ($7.58 Million)

CryptoPunks is a completely new form of NFT. It was sold at a price of $7.58 million.

CryptoPunk is a one-of-a-kind creature that belongs to a group of nine aliens. CryptoPunks, you see, only has 10,000 punks in stock. Furthermore, just 9 of them are aliens.

Crossroads ($6.6 Million)

Another of our most expensive NFTs is Crossroads. This is also another Beeple work; this one sold just days before the massive selling of Everydays. The artist was also able to sell this work on Nifty Gateway. This is a single piece of artwork rather than a collection, unlike Everydays. As a result, the appraised value of this artwork has increased.

The artwork is based on the presidential election in 2020. The artist created two versions of this, one for if Trump wins and the other for if Trump loses. And the video would vary depending on the election results.

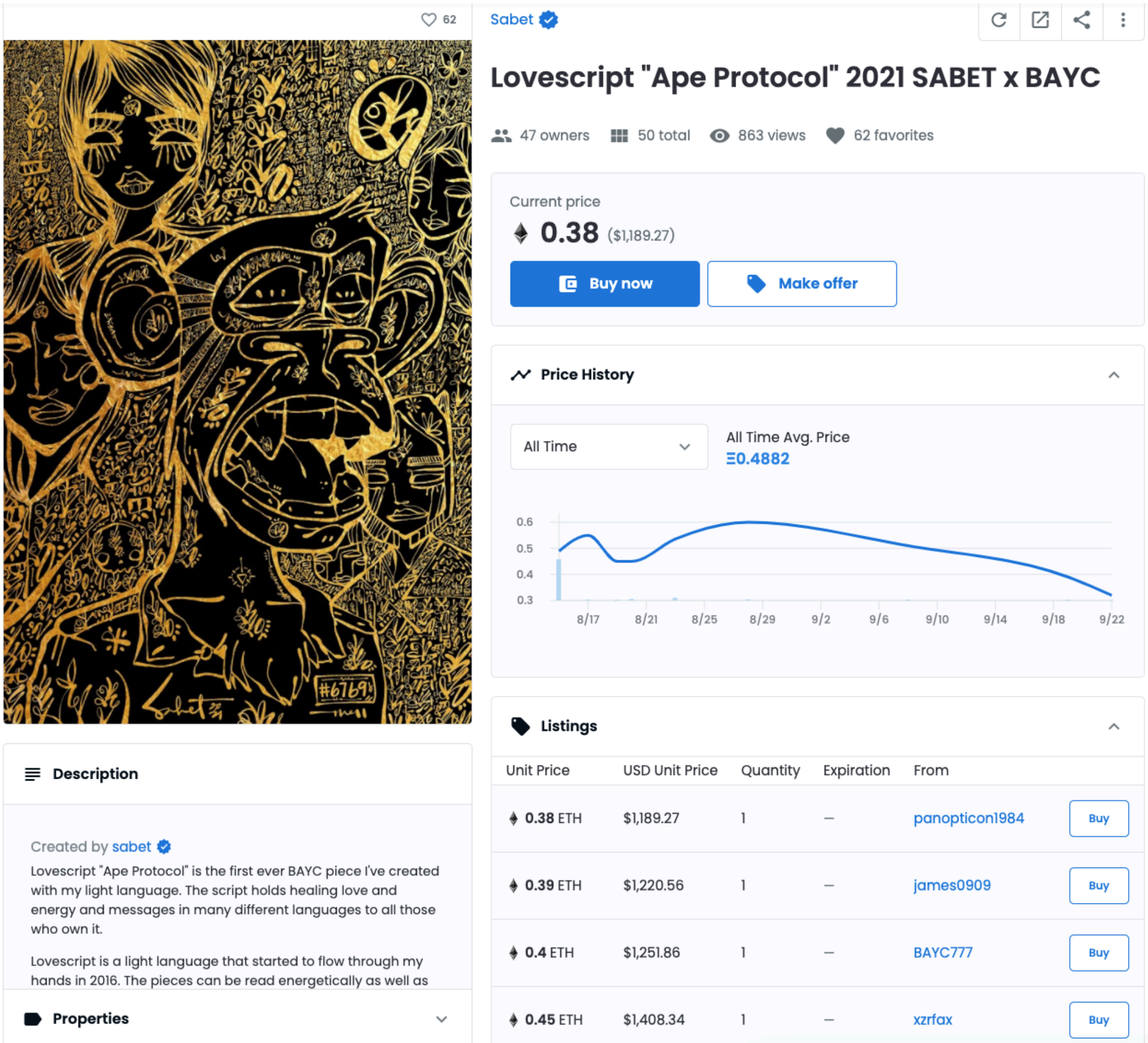
The first Tweet ($2.9 Million)

The first tweet sent by Twitter creator Jack Dorsey has been sold for $2.9 million (£2.1 million) to a Malaysian businessman. Mr. Dorsey auctioned off the tweet, which stated "just putting up my twttr," for charity on March 21, 2006. Sina Estavi, a Malaysian customer, compared the transaction to purchasing a Mona Lisa artwork.

The 2.9 million dollars earned were donated to GiveDirectly, a nonprofit specializing in giving financial resources to the underprivileged, and its COVID-19 assistance initiatives in Africa.

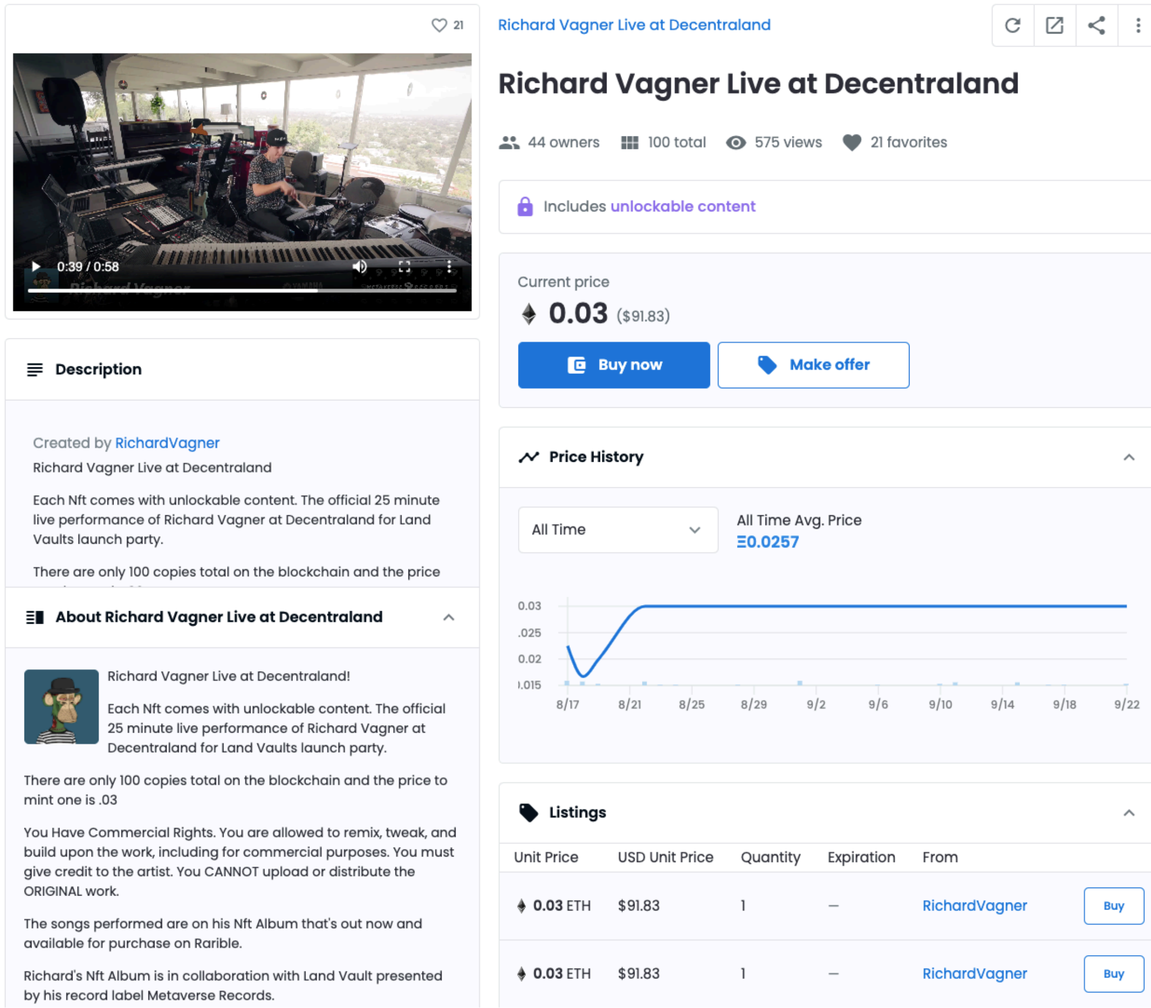
13.3. Types of NFT's to date[73]

13.3.1. Art

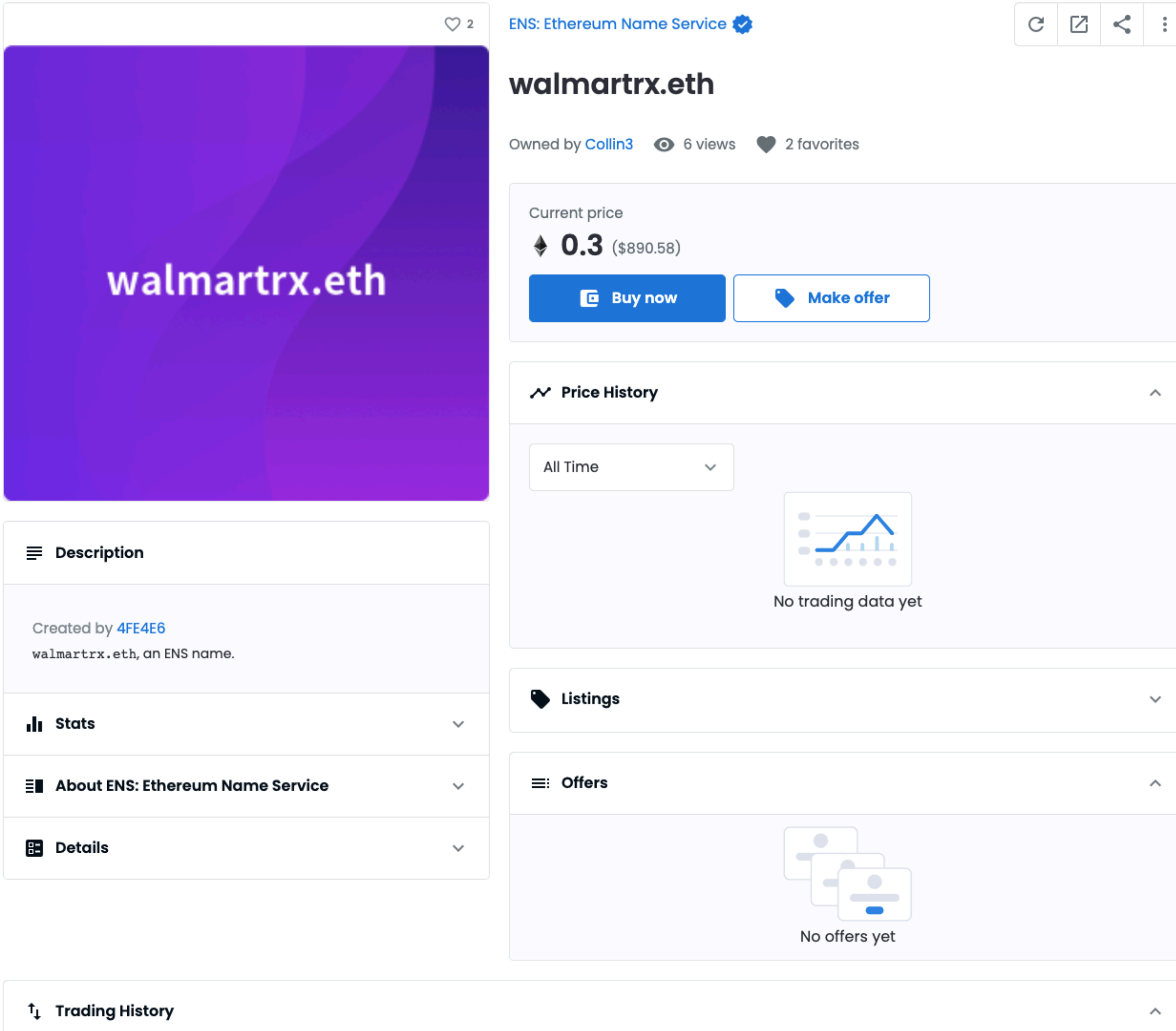


73 New categories and types are constantly evolving.

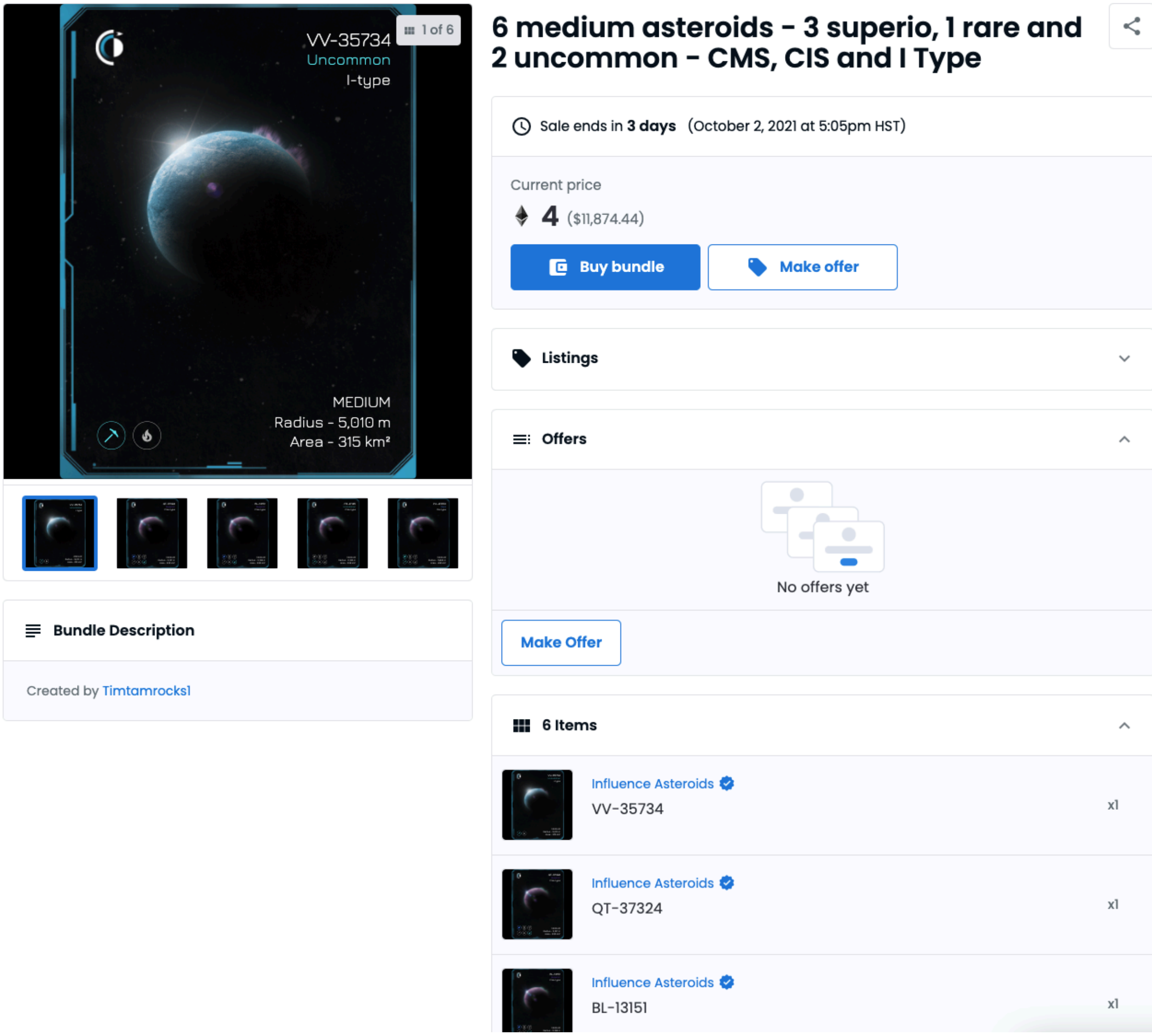
13.3.2. Music


21
0:39 / 0:58
Richard Vagner Live at Decentraland
Richard Vagner Live at Decentraland
44 owners
100 total
575 views
21 favorites
Includes unlockable content
Current price
0.03 ($91.83)
Buy now
Make offer
Description
Created by RichardVagner
Richard Vagner Live at Decentraland
Each Nft comes with unlockable content. The official 25 minute live performance of Richard Vagner at Decentraland for Land Vaults launch party.
There are only 100 copies total on the blockchain and the price
About Richard Vagner Live at Decentraland
Richard Vagner Live at Decentraland!
Each Nft comes with unlockable content. The official 25 minute live performance of Richard Vagner at Decentraland for Land Vaults launch party.
There are only 100 copies total on the blockchain and the price to mint one is .03
You Have Commercial Rights. You are allowed to remix, tweak, and build upon the work, including for commercial purposes. You must give credit to the artist. You CANNOT upload or distribute the ORIGINAL work.
The songs performed are on his Nft Album that's out now and available for purchase on Rarible.
Richard's Nft Album is in collaboration with Land Vault presented by his record label Metaverse Records.
Price History
All Time
All Time Avg. Price
Ξ0.0257
0.03
.025
0.02
0.015
8/17
8/21
8/25
8/29
9/2
9/6
9/10
9/14
9/18
9/22
Listings
Unit Price
USD Unit Price
Quantity
Expiration
From
0.03 ETH
$91.83
1
—
RichardVagner
Buy
0.03 ETH
$91.83
1
—
RichardVagner
Buy

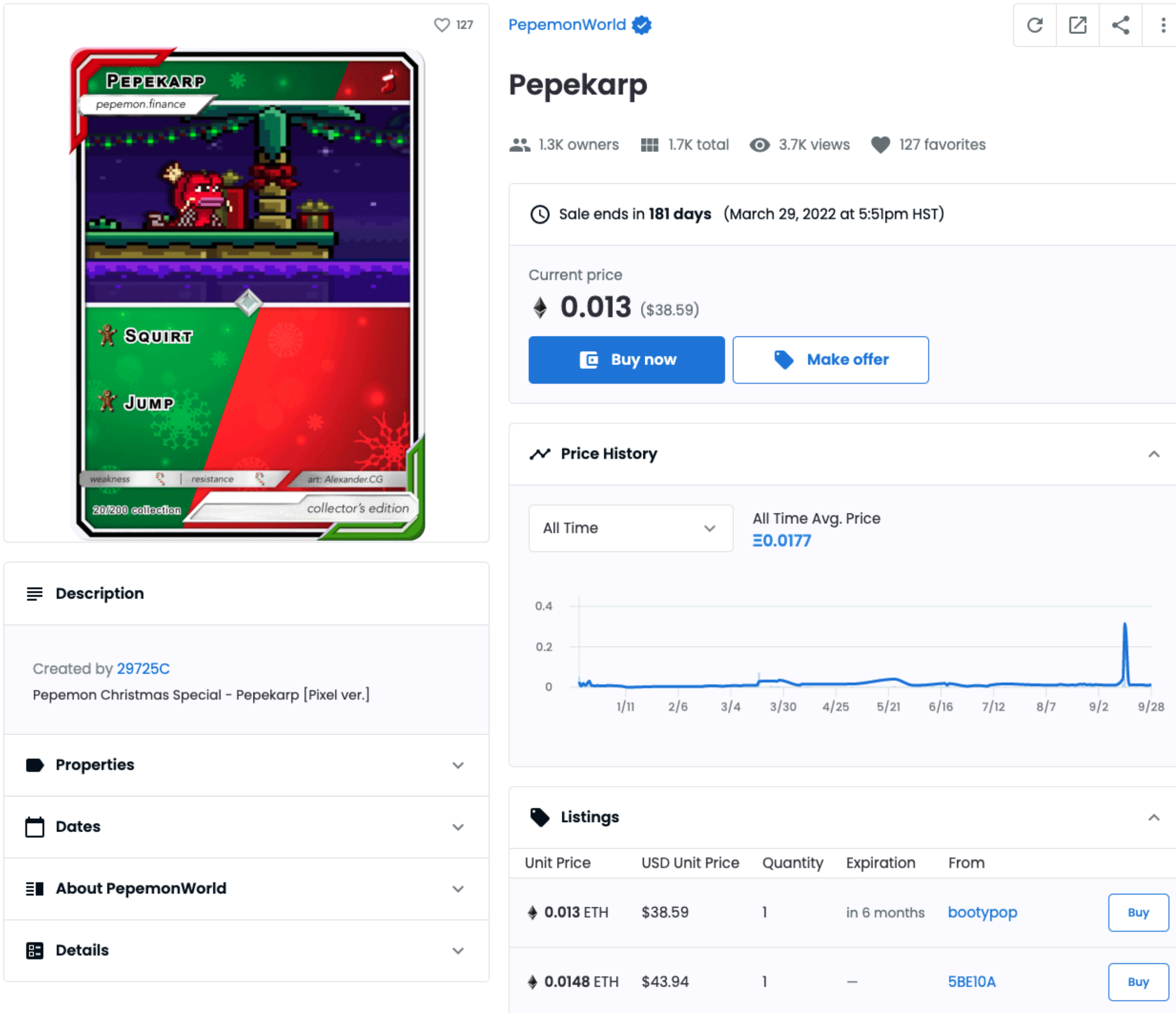
13.4. Ethereum Based Domain Names



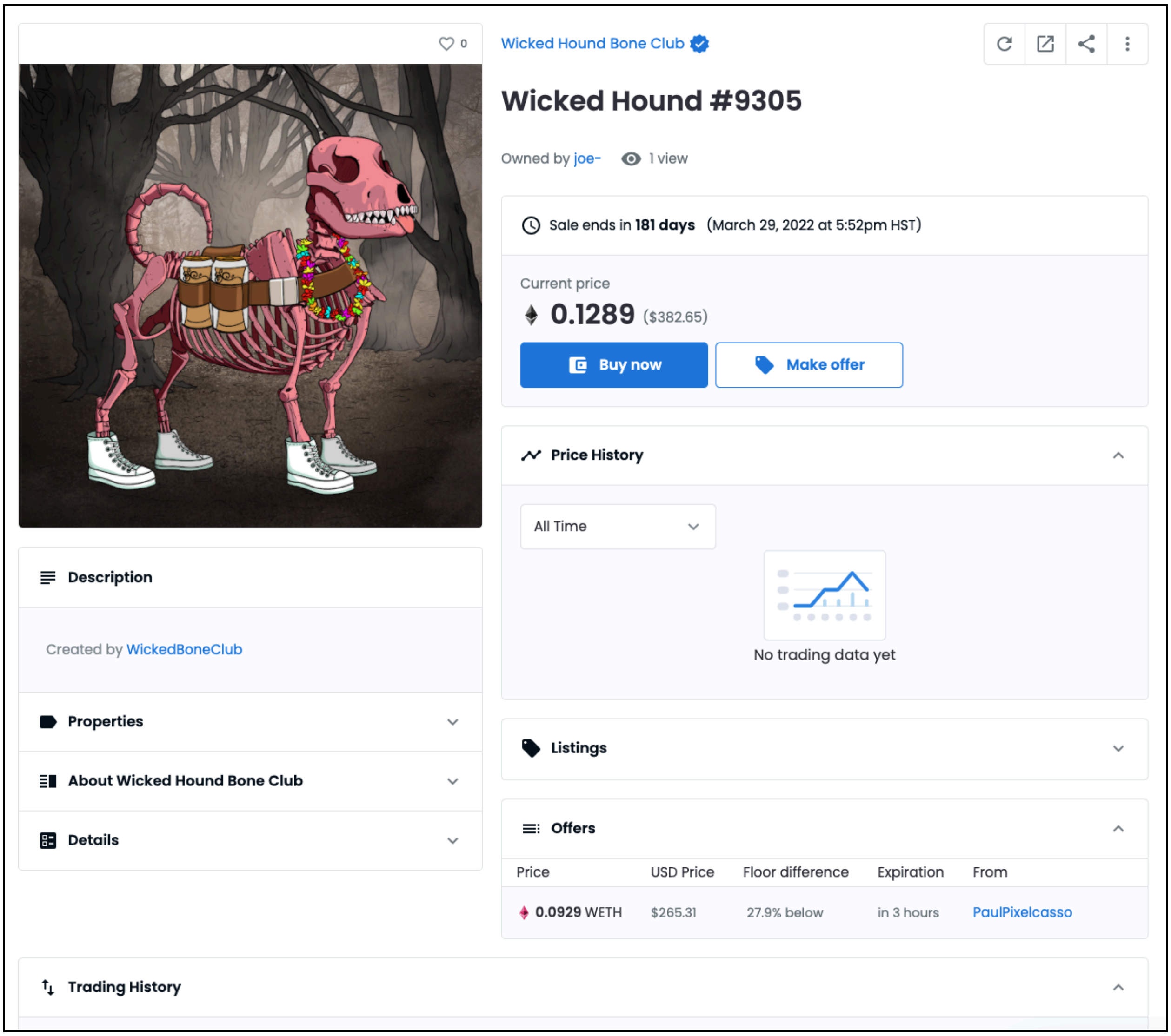
13.5. Virtual Worlds



13.6. Trading Cards



13.7. Collectibles



14. Yield Farming & Liquidity Mining Explained

14.1. What is Yield Farming?

Yield farming in crypto is a relatively new concept gaining significant popularity using Permissionless Liquidity Protocols (PLP). At the simplest level, yield farmers via yield farming strategies move assets around within several different DeFi protocols that are constantly working to find the pool offering the best yield over some time [74]. Several decentralized

[74] Binance Academy (2021) What is yield farming in Decentralized Finance (DEFI)? Binance Academy, Binance Academy. Available from:

exchanges exist, such as Aave, Uniswap, Compound, and others. The return on liquidity provision varies depending on demand, so yield farmers are constantly looking to move to where they can get the best rate of return.

This year, Yearn Finance aimed to help liquidity providers find the best yield at any given time, giving rise to a class of specialty tools aimed at helping those with assets to stake find the best deals in exchange for providing their tokens to liquidity pools. Yield farming is closely linked to an automated market maker (AMM) model. It usually involves liquidity providers (LPs), and liquidity pools[75]. Let's find out how it works.

Yield Farming

Is the fastest growing DeFI driver with a market cap growth of $500 million to $10 billion in 2020.

Yield farming protocols incentivize liquidity providers (LP) to stake or lock up their crypto assets in a smart contract-based liquidity pool. These incentives can be a percentage of transaction fees, interest from lenders or a governance token. These returns are expressed as an annual percentage yield (APY). As more investors add funds to the related liquidity pool, the value of the issued returns decrease accordingly.

Source: Coinmarketcap

14.2. How Yield Farming Works.

Liquidity providers deposit funds in a liquidity pool. This pool powers a marketplace that allows users to borrow or exchange tokens[76]. These platforms can be used at a cost. Liquidity providers are paid according to their share in the liquidity pool. This is how AMM works. Distribution rules will depend on how the protocol is implemented. In the end, liquidity providers will receive a return on liquidity provided to the pool. The most common metrics are the Annual Percentage Rat (APR), and Annual Percentage Yields (APY). APR does not take into

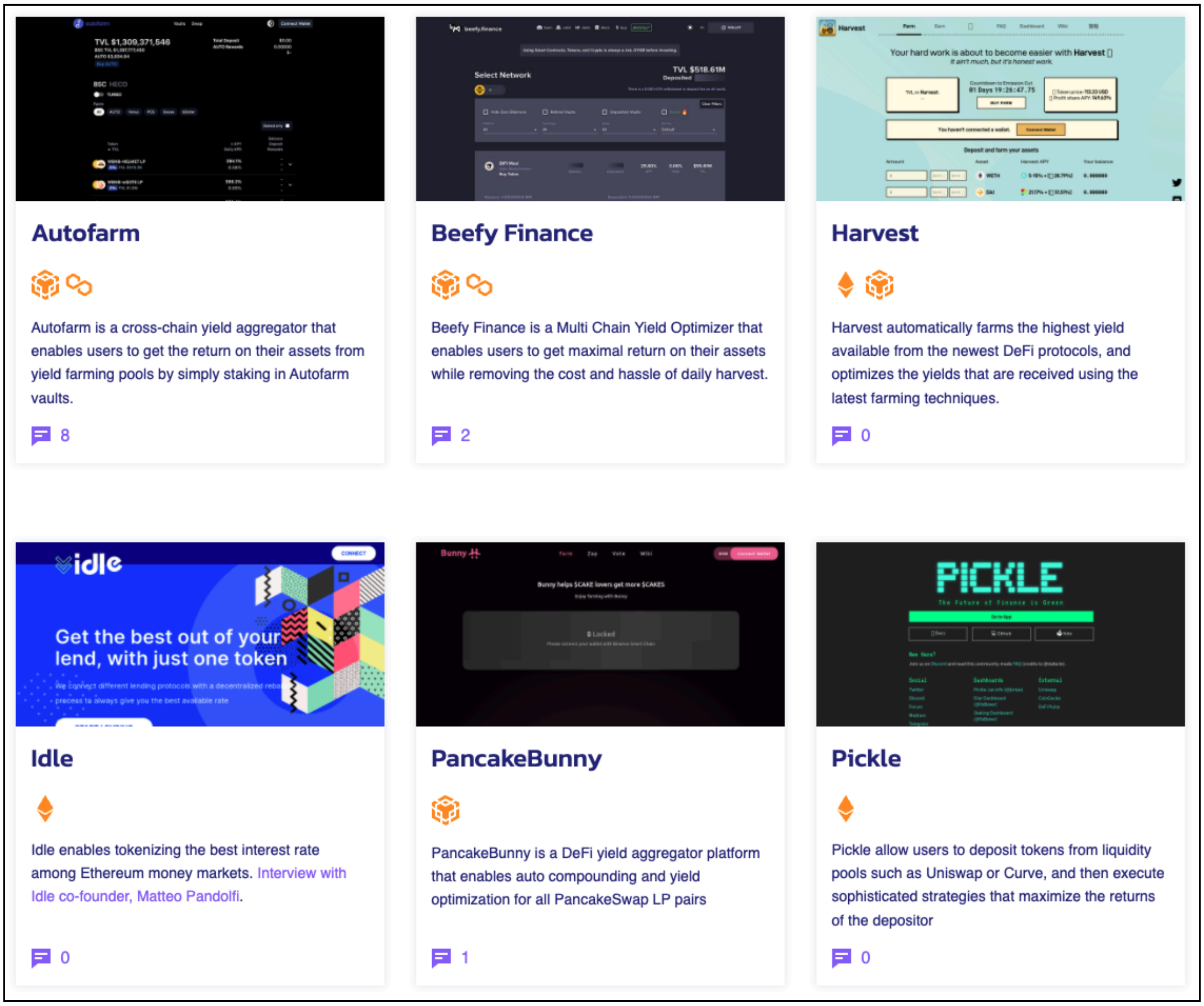
---

https://academy.binance.com/en/articles/what-is-yield-farming-in-decentralized-finance-defi (accessed 23 September 2021).

[75] CoinMarketCap (2020) What is yield farming?: CoinMarketCap. CoinMarketCap Alexandria, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/article/what-is-yield-farming (accessed 16 September 2021).

[76] Giove B (2021) A guide to multi-chain yield farming. Bankless, Bankless. Available from: https://newsletter.banklesshq.com/p/a-guide-to-multi-chain-yield-farming (accessed 28 September 2021).

consideration the effects of compounding. APY does. In this instance, compounding refers to reinvesting profits in order to generate higher returns. Be aware, however, that APR and APY can be interchangeably used.



Yield Aggregators - Source: https://defiprime.com/yield-aggregators#bsc

Since DeFi was first introduced to the cryptocurrency market on the Ethereum blockchain, it has since spread to other blockchains such as Binance Smart Chain (BSC). Binance Exchange created BSC to compete with Ethereum in April 2019. Due to the increase in users and the complexity of DeFi transactions, high transaction fees have plagued Ethereum. BSC is more centralized, which speeds up transaction processing and significantly lowers transaction fees than its competitor. These fees are charged by the Ethereum network in Ether (ETH), while BSC charges them with Binance Coin (BNB).

Ethereum is home to the highest-value DeFi platforms (Aave and Curve, Uniswap, etc.). However, BSC has sufficiently large projects such as PancakeSwap or Venus Protocol to compete with the Ethereum network. Ethereum-based platforms cannot use Ether or other tokens made on Ethereum. Most of these tokens are known as ERC-20 tokens. BSC's native token, BNB, is used by its platforms as well. Other tokens on the network are also available (mostly called BEP-20 tokens). Although BSC transactions are still cheaper than DeFi, Ethereum gas fees have fallen since then. You can now find the best yield farms on both chains.

There are multiple types of yield farming projects offering different financial services, mostly to earn astonishingly high interest. Large banks might earn you 0.01% to 0.25% a year, but these sub-percent yields can't compete with the 20% to 250% earnings some DeFi platforms.

Uniswap: ~20% to 50% APR

Uniswap, which has more than $5.5 billion in total value, is second to Curve Finance as the largest decentralized exchange (DEX). Swaps can be made with Ethereum, thousands of ERC-20 tokens and the platform allows for staking in liquidity pool to facilitate swaps. The liquidity providers receive a portion of the trading fees for each swap. If they have a sufficient principal deposit, they can earn substantial interest. The market and pool fluctuations affect the interest rates for Uniswap as well as all other DEXes.

Aave: ~0.01% to 15% APR

According to DeFiPulse.com, Aave is the DeFi king when it comes to total value locked. It has a staggering value exceeding $10 billion. Aave, a decentralized platform on Ethereum and the Polygon sidechain, offers low-interest cryptocurrency lending and borrowing. Aave's borrow APRs are among the highest on the market because so many investors have put crypto into Aave in order to earn interest. Variable borrowing of stablecoins, such as DAI, USDC or USDT, can range from 3% to 7% at time of writing.

PancakeSwap: ~8% to 250% APR

PancakeSwap is a decentralized exchange (DEX) similar to Uniswap. It works in the same way as Uniswap, but it is on Binance Smart Chain (BSC), which has a lot more features that focus on gamification. It has the largest total value of all BSC DeFi projects, at $7 billion. PancakeSwap provides BSC token swaps and interest-earning stake pools. It also offers a gambling game that lets users guess the future price for BNB, as well as non-fungible tokens (NFT).

Curve Finance: ~2.5% to 25% APR

Curve Finance, the largest DEX, has a total value of $7.9 Billion at the time this article was written. Curve Finance also makes the most of its locked funds with its market-making algorithm. This is a benefit to users who perform swaps and liquidity providers.

14.3. The Liquidity Mining Market

Liquidity mining is said to have played a role in the 2020 DeFi boom. It also contributed to the monthly volume growth of decentralized exchanges — from $39.5 million in January 2019 to $45.2 billion in January 2021. As of September 2021, its total value locked is estimated at $169 billion[77].

14.4. What is Liquidity Mining?

In 2020, liquidity mining was popularized by new decentralized exchanges like Uniswap and Compound. This had a profound effect on DeFi's sector and changed it forever.

A liquidity pool is nothing more than an Automated Market Maker that provides liquidity to avoid large price swings for an asset. An AMM is a smart contract that regulates trading. Since smart contracts are decentralized, users do not have to trade the order book of an exchange. Instead, they effectively trade with other users[78]. Only traded pairs are allowed, which implies that the pool is always filled with two distinct cryptocurrencies.

Many platforms also offer governance tokens as an incentive, to date this appears to be the distinguishing factor between Liquidity Mining and Yield Farming. Backers get additional incentives from investing their liquidity in the platform. These tokens can then be traded freely on the market, but when retained they provide several benefits, including voting rights for further development. Potential price increases for tokens can generate additional profits for the platform. This is evident in the AMM with SushiSwap and UniSwap.

14.5. How Liquidity Mining works

To facilitate trades on decentralized platforms, liquidity is provided by people. For a portion of the fees, users can place their cryptocurrency in the DEX to earn passive income.

You can also earn rewards for depositing crypto assets to the exchange. Liquidity providers get a percentage of every trade. Most platforms offer passive income through liquidity mining, with additional incentives and rewards. Liquidity providers (LPs) get tokens called LP tokens when they supply liquidity to a pool. These tokens are proportional to how much liquidity the provider has provided to the pool[79]. A trade is made when the pool facilitates it where the rate is divided equally among all LP token owners.

---

[77] Sun Z (2021) What is total value locked? Nasdaq. Available from: https://www.nasdaq.com/articles/what-is-total-value-locked-2021-09-29 (accessed 3 October 2021).
[78] What is liquidity mining? (2021) Shrimpy Academy. Available from: https://academy.shrimpy.io/post/what-is-liquidity-mining (accessed 24 September 2021).
[79] Team PP (2021) Liquidity mining in defi: What is it & how does it work? PixelPlex, PixelPlex. Available from: https://pixelplex.io/blog/what-is-liquidity-mining/ (accessed 3 October 2021).

For example, let's take Ether (ETH), and USD Coin (USDC). To make things easier, the price of Ethereum can be equal to 1,000 USDC. The pool is made up of equal amounts of USDC and ETH by liquidity providers. Someone depositing 1 ETH would need to match itto with 1000 USDC.

Because the pool has liquidity, anyone can trade ETH for USDC based on funds they have deposited. They don't need to wait for a counterparty who will match their trade. Providers of liquidity are incentivized to contribute with rewards. They receive a pool token, which is a token that represents their stake, each time they deposit. The pool token in this example would be USDC/ETH[80].

Users who use the pool to trade tokens are entitled to a proportionate share of trading fees. If the USDC/ETH pool trading fees are 0.3%, and a liquidity provider contributed 10% to the pool, then they will be entitled 10% of 0.3% total trade value.

If a user wishes to withdraw their stake from the liquidity pool they can burn their pool tokens.

When a platform is lacking liquidity due to low volume or interest, not only will it become difficult for traders to buy and sell their assets, but these illiquid tokens can cause unpredictable price swings when there are only a few large individual transactions. Therefore, tokens with high price volatility and inefficient conversions are unlikely to be adopted. A DEX must provide liquidity in order to remain competitive and increase firm revenue by collecting trading fees on their platform.

14.6. Popular Yield Farming and Liquidity Pools

Asset	Total Value Locked (TVL)	Market Cap / TVL Ratio*	Estimated Returns
Compound Finance (COMP)	$10,457,024,327	.19	1yr: 175% 1h: .1%
Maker (MKR)	$13,095,114,443	.18	1yr: 367% 1h: -0.7%
Synthetic Network Token (SNX)	$2,043,786,818	.94	1yr: 148% 1h: 1.1%
Aave (AAVE)	$14,437,628,494	.29	1yr: 464% 1h: -0.5%
Uniswap (UNI)	$4,679,969,182	2.93	1yr: 576% 1h: -0.7%
Curve DAO Token (DAO)	$14,665,973,558	.08	1yr: 278%

[80] Kuznetsov N (2021) DEFI liquidity pools, explained. Cointelegraph, Cointelegraph. Available from: https://cointelegraph.com/explained/defi-liquidity-pools-explained (accessed 25 September 2021).

			1h: -2.1%
Balancer (BAL)	$3,248,905,897	.08	1yr: 45.3% 1h: -1.0%
Yearn.Finance (YFI)	$5,061,907,527	.23	1yr: 51% 1h: -0.7%

*Ratio of market capitalization over total value locked of this asset. A ratio of more than 1.0 refers to its market cap being greater than the total value locked. MC/TVL is used to approximate a protocol's market value vs. the amount in assets it has stacked/locked. Source: Coingecko

15. Tokenizing Future Yields

DeFi makes it possible for smart contract developers and financial creators to engineer new protocols that create new investment classes of traditional financial models by building on top of existing protocols. Tokenizing Future Yields is a new protocol recently introduced to cryptocurrency. It functions much the same way as a traditional finance model where you trade on the future interest rates within a Bond market. This market is estimated to be $119 trillion in value, according to SIFMA. Future Yield tokenization can be a useful tool for capital efficiency and risk hedging; as investors seek to hedge against the volatility of yield rates, the main beneficiaries are retail and corporate yield farmers.

We have seen more efforts to tokenize future yield from Aave and Compound interest-bearing tokens in the past few months. Protocols that allow users to access prospective yield instantly are emerging. These protocols tokenize and separate the interest and principal portions of future yield and give them instant access[81].

It is exciting to be able to tokenize the yield that billions of dollars are collateralizing. This opens of up new avenues for users to deploy innovative strategies.

Splitting interest-bearing assets into Future Yield Tokens and Principal Tokens allows you to tokenize the yield from these tokens.

We are witnessing the creation of a financial instrument that will allow you to trade future interest rates in DeFi. Farmers will also be able to hedge their risk and grow their own yield. To allow yield-generating asset-holders the ability to generate additional yield and give traders the opportunity to trade future yield streams with no underlying collateral.

[81] What is Pendle Crypto? how to buy it? (n.d.) Stock Market News and Research. Available from: https://kalkinemedia.com/news/cryptocurrency/what-is-pendle-crypto-how-to-buy-it (accessed 1 September 2021).

APWine is an example of only a handful of platforms that allows users to deposit yield-bearing assets and exchange them for tokens that represent their principal and yield[82]. Instead of depositing regular assets like DAI or USDC as most DeFi protocols do, users can only deposit interest-bearing tokens like aDAI (Aave's interest-bearing DAI token), yyUSD vault tokens from Yanel), iFARM (Harvest Finance's interest-bearing FARM token) onto the APWine platform. Two tokens are available for those who do so:

1. Interest-bearing tokens or IBT, which represent users' claim to their deposits. By depositing 1,000 aDAI coins, you can get IBT that will allow you to claim these 1,000 tokens.

2. Future yield tokens (or FYT) are future yield tokens that represent the future yield generated by an asset during a specific time period (30 days of the beta test). The 1,000 aDAI that are deposited will produce 1,000 FYT. Each token represents the yield of 1 aDAI for a 30-day period.

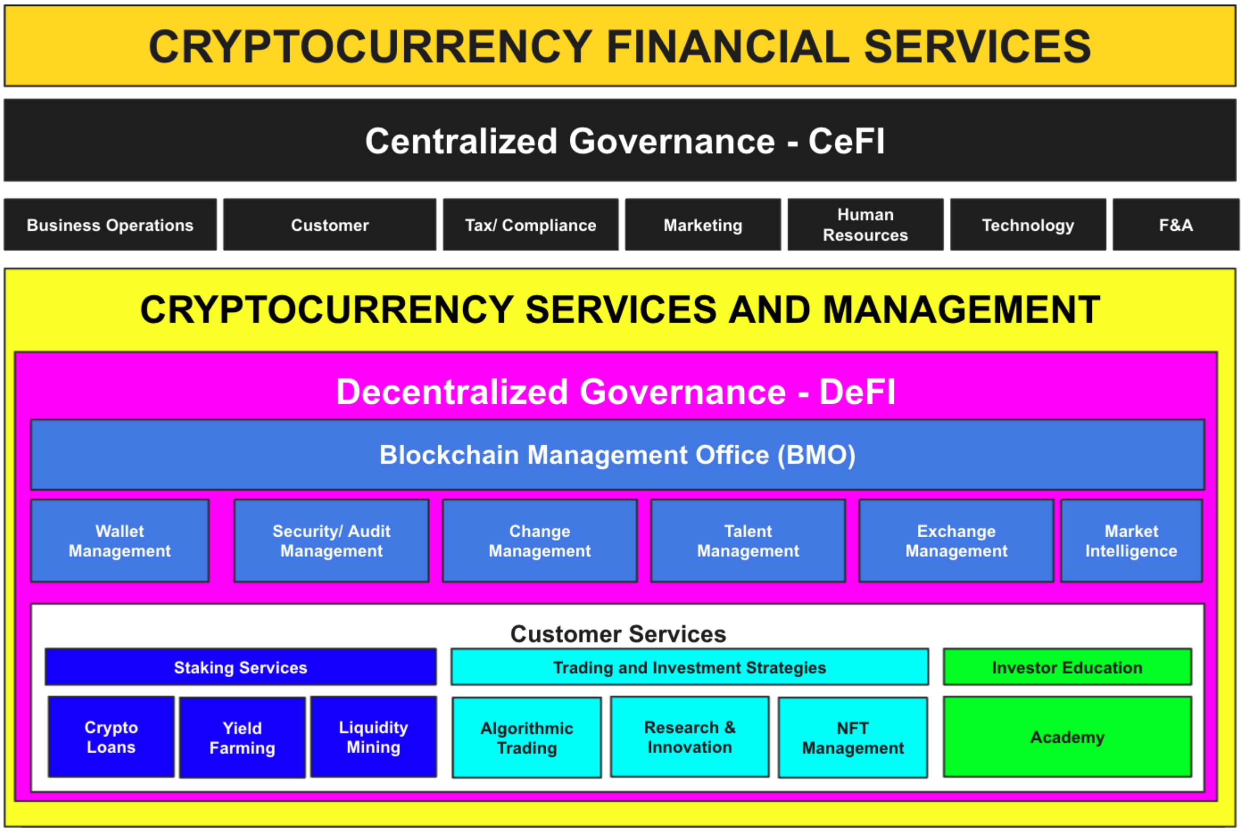
Let's say that Aave's DAI variable interest rate is currently 12%. To lock in the current rate you can sell future yield tokens at a discount to get a maximum return up to $10. Your earnings are locked in regardless of future yield rate movements. It is possible to get it, but if yield rates fall, you will have effectively reduced them. However, if rates were ever to rise, future yield tokens would provide a higher return if you hadn't sold them at the beginning[83].

Tokenizing future yields is another example of the opportunities in the cryptocurrency space that are evolving rapidly. The jury is still out on if hedging against the yields of tokens will be viable or profitable. Regardless, it is another trend emerging in this space where it is recommended GSHI pay particular attention too.

---

[82] Sawinyh N (2021) Apwine - The Protocol for future yield tokenisation. DeFi, Defiprime. Available from: https://defiprime.com/apwine (accessed 18 September 2021).
[83] Win Win K (2021) Tokenizing Future Yield. CoinGecko.

16. GSHI - The Future Bank



16.1. GSHI Strategy For Cryptocurrency Services

GSHI's transformation into delivering cryptocurrency services will require constant monitoring of the market to identify evolving market opportunities - some from traditional finance and others from the cryptocurrency market where traditional finance does not provide services for today.

Building the operating capabilities for GSHI going forward will consist of a decentralized and centralized governance model; the CeDeFI model featuring a decentralized stablecoin offering a utility and or governance token.

The Decentralized Model

This is where the future of corporate governance and smart contract technology intersect. Since no one will be in charge of the governance process, it becomes imperative that the corporate governance experts (the people who study, analyze, design, develop, deploy and oversee) become members of the digital community. By being a part of the digital community, these experts will influence the future development of cryptocurrency technologies. This way, the future of corporate governance and smart contract technologies will be based on open source technology, which can integrate into any system and extend its functionality beyond the current system.

The decentralization aspect of the operating model will provide standard DeFI services through utility tokens and smart contracts that allow those contributing to the staking services through voting rights.

Decentralization is managed by a Blockchain Management Office (BMO) that provides support services based on popular demand through voting. The BMO functions in similar ways to the traditional Vendor/Supplier Management Office and follows Policies and Procedures that dictate the behavior of the DeFI Governance Model. The DeFI model would also benefit from periodic smart contract audits, as previously discussed.

For example, CipherTrace delivers cryptocurrency AML compliance solutions for some of the world's largest banks, exchanges, and other financial institutions because of its best-in-class data attribution, analytics, proprietary clustering algorithms, and coverage of 2,000+ cryptocurrency entities, more than any other blockchain analytics company. This protects from money laundering risks, illicit money service businesses, and virtual currency payment risks[84].

In order to be approved by the regulators, a diverse collection of metrics must be calculated in order for a particular cryptocurrency to be listed on a cryptocurrency exchange.

These measurements can be anything from supply and demand indicators, to profit margins, to metrics that compare different currencies against one another. The use of these metrics, coupled with the mathematical algorithm that is used to identify which cryptocurrency should be included in the marketplace, gives rise to a market that will be free of scammers, rip-offs, and frauds and instill investor confidence in GSHI products and services.

Two primary methods of monitoring may be used by GSHI: manual tracking and automation:

- **Manual** monitoring involves receiving reports on a weekly or monthly basis and analyzing the information.

- **Automation** is more hands-on; however, the software must generate an adequate report daily. Both methods are necessary for cryptocurrency compliance, as losses may not be properly investigated if a company does not monitor activity closely enough. GSHI will benefit greatly from including the correct monitoring tools, e.g., keyword triggers for compliance notifications events.

---

[84] https://sourceforge.net/software/product/CipherTrace/

GSHI will greatly benefit from developing a standardized interface that allows investors to easily access the information they need to make informed decisions regarding which cryptocurrency to purchase. By creating standardized interfaces for the different cryptocurrencies, investors will get the same information and applications as they would if they were visiting a traditional finance platform. This will allow easier comparisons between the various currencies traded on online CEX and DEX similar to those found for trading with major financial institutions such as NYSE, NASDAQ, and AMEX.

16.2. GSHI Short-Term Objectives

The GSHI's short-term objectives are to develop and operationalize the market mechanisms needed to generate liquidity for the decentralized component of the business model. The following short-term goals will create an immediate competitive advantage:

- Identify trends through indicators on movement in DeFI that will increase the GSHI Total Value Locked (TVL) through smart contract swaps and arbitrage opportunities.

- Empower liquidity providers to access liquidity pools or yield farms at any time for idle assets, giving rise to a class of specialty tools aimed at helping those with assets to stake to find the best deals in exchange for providing their tokens to liquidity pools.

- Provide a bug bounty program for developer recognition and compensation for reporting bugs, especially those about security exploits and vulnerabilities. This is a way to detect software and configuration errors that can slip past developers and security teams and later lead to big problems.

- Prepare for nefarious means, e.g., there would be no way to stop an extremely large fund from buying up a significant position in a governance token, holding it in several different wallets, proposing a sweetheart deal to a DAO, and then using its holdings to vote it through[85].

- Use market trading algorithms through mining/social/copy and or replicate through advancements in findings e.g.; SVM, K-Means Clustering, and RNN's as algorithmic examples across trading pairs and while also identifying ripple effects across multiple cryptocurrencies.

16.3. GSHI Long-Term Objectives

- Creating transparency for regulators to view as a source to measure the effectiveness of implemented regulations.

---

[85] Daos prepare for the next crypto winter with Treasury diversification - coindesk (2021) CoinDesk Latest Headlines RSS. Available from: https://www.coindesk.com/business/2021/06/03/daos-prepare-for-the-next-crypto-winter-with-treasury-diversification/ (accessed 5 September 2021).

- Develop annual Yield-Farming strategies each year to pay for developers, designers and business development without touching the principal.

- Increase NFT management presence that can introduce well-needed liquidity to leverage across other services areas.

- Provide more opportunities for liquidity providers to earn passive income from liquidity mining by offering additional rewards and incentive programs.

- Identify multi-chain partners and develop smart-yield-engines and liquidity aggregators where any financial hub can integrate into (e.g., cryptocurrency payment gateway and wallets.)

- Identify trends through indicators on movement in DeFI.

- To provide capital flexibility through terms that provide high liquidity options - requiring engineering to identify market opportunities - a mechanism that allows the principle to also generate in an AMM while mitigating impairment loss.

17. References

Alper T (n.d.) Pos Coins, Lightning, defi & dexes in danger as US Bill Chaos intensifiesTim. Crypto News. Available from: https://cryptonews.com/news/proof-of-stake-coins-in-danger-as-us-infrastructure-bill-cha-11337.htm (accessed 4 September 2021).

Aneesh A (n.d.) Technologically Coded Authority: The Post-Industrial Decline in Bureaucratic Hierarchies.
Anwar Sheluchin PhD Student (2021) A national digital currency has serious privacy implications. The Conversation. Available from: https://theconversation.com/a-national-digital-currency-has-serious-privacy-implications-130520 (accessed 4 September 2021).

Austin R (2021) Altcoins are the alternative digital currencies to bitcoin - here's what they are and how they work. Business Insider, Business Insider. Available from: https://www.businessinsider.com/what-is-altcoin#:~:text=Since%20the%20emergence%20of%20Bitcoin,of%20more%20than%209%2C000%20altcoins. (accessed 2 September 2021).

authors A, Kosti N and Additional informationFundingThis research was supported by the following Israeli Science Foundation Grants: ISF 324/2018 (2020) Legislation and regulation: Three analytical distinctions. Taylor & Francis. Available from: https://www.tandfonline.com/doi/full/10.1080/20508840.2019.1736369 (accessed 28 September 2021).

Becker S (2021) What to know about staking - the process of locking up crypto holdings to earn rewards and interest. Business Insider, Business Insider. Available from: https://www.businessinsider.com/staking-crypto (accessed 1 October 2021).

Binance Academy (2021) A guide to crypto collectibles and non-fungible tokens (nfts). Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/a-guide-to-crypto-collectibles-and-non-fungible-tokens-nfts (accessed 21 September 2021).

Binance Academy (2021) What is staking? Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/what-is-staking (accessed 1 October 2021).

Binance Academy (2021) What is staking? Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/what-is-staking (accessed 1 October 2021).

Binance Academy (2021) What is yield farming in Decentralized Finance (DEFI)? Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/what-is-yield-farming-in-decentralized-finance-defi (accessed 14 September 2021).

Binance Academy (2021) What is yield farming in Decentralized Finance (DEFI)? Binance Academy, Binance Academy. Available from: https://academy.binance.com/en/articles/what-is-yield-farming-in-decentralized-finance-defi (accessed 23 September 2021).

Bushell-Embling D (n.d.) NFT market will see "massive increase" in volume. Available from: https://www.technologydecisions.com.au/content/it-management/news/nft-market-will-see-massive-increase-in-volume-32437496 (accessed 3 October 2021).
Cengiz F (2021) What the EU's new mica regulation could mean for cryptocurrencies. EUROPP. Available from: https://blogs.lse.ac.uk/europpblog/2021/07/05/what-the-eus-new-mica-regulation-could-mean-for-cryptocurrencies/ (accessed 3 September 2021).

Clark M (2021) NFTs, explained. The Verge, The Verge. Available from: https://www.theverge.com/22310188/nft-explainer-what-is-blockchain-crypto-art-faq (accessed 22 September 2021).

CoinMarketCap (2020) What is yield farming?: CoinMarketCap. CoinMarketCap Alexandria, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/article/what-is-yield-farming (accessed 16 September 2021).

CoinMarketCap (2021) Smart contract audit: CoinMarketCap. CoinMarketCap Alexandria, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/glossary/smart-contract-audit (accessed 4 September 2021).

CoinMarketCap (2021) What is crypto lending?: CoinMarketCap. CoinMarketCap Alexandria, CoinMarketCap. Available from: https://coinmarketcap.com/alexandria/article/what-is-crypto-lending (accessed 15 October 2021).

Crowell B (2020) Crypto lending, explained. Cointelegraph, Cointelegraph. Available from: https://cointelegraph.com/explained/crypto-lending-explained (accessed 27 September 2021).
Crypto hacks: Crypto Exchange Hacks & Cryptocurrency hackers (n.d.) Coinbase, Coinbase. Available from: https://cointelegraph.com/magazine/crypto-exchange-hacks/ (accessed 29 September 2021).

Dao Overview - a list of Ethereum's top Daos and dao structures (2021) DeFi Rate. Available from: https://defirate.com/daos/ (accessed 4 September 2021).

Dao Overview - a list of Ethereum's top Daos and dao structures (2021) DeFi Rate. Available from: https://defirate.com/daos/ (accessed 6 September 2021).

Daos prepare for the next crypto winter with Treasury diversification - coindesk (2021) CoinDesk Latest Headlines RSS. Available from: https://www.coindesk.com/business/2021/06/03/daos-prepare-for-the-next-crypto-winter-with-treasury-diversification/ (accessed 5 September 2021).

Decentralized finance (2021) Wikipedia, Wikimedia Foundation. Available from: https://en.wikipedia.org/wiki/Decentralized_finance (accessed 4 September 2021).

DeNicola L (2021) What to know about non-fungible tokens (nfts) - unique digital assets built on Blockchain Technology. Business Insider, Business Insider. Available from: https://www.businessinsider.com/nft-meaning (accessed 23 September 2021).

Dulau T-M (2019) Cryptocurrency – Sentiment Analysis In Social Media. Sciendo, dissertation.
Eliot Couvat E (2021) Liquidity mining (staking) for social tokens. GCR - Home -. Available from: https://globalcoinresearch.com/2021/09/14/liquidity-mining-staking-for-social-tokens/ (accessed 1 October 2021).

Elliptic (2019) Elliptic discovery enables banks to assess their exposure to cryptoassets. Blockchain Analytics & Crypto Compliance Solutions. Available from: https://www.elliptic.co/media-center/elliptic-discovery-enabling-banks-identify-crypto-exposure (accessed 5 September 2021).
European Parliament P and Council P (2020) Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on Markets in Crypto-assets, and amending Directive (EU) 2019/1937. Brussels: European Commission.
Fang F, Ventre C, Basios M, et al. (2021) Cryptocurrency trading: A comprehensive survey. arXiv.org. Available from: https://arxiv.org/abs/2003.11352 (accessed 30 September 2021).

Gaubys J (n.d.) How many people have smartphones? [Jul 2021 update]. Oberlo, Oberlo. Available from: https://www.oberlo.com/statistics/how-many-people-have-smartphones (accessed 21 September 2021).

Gemini (2021) Decentralized Exchange platforms in crypto trading, Gemini. Available from: https://www.gemini.com/cryptopedia/decentralized-exchange-crypto-dex (accessed 29 September 2021).

Giove B (2021) A guide to multi-chain yield farming. Bankless, Bankless. Available from: https://newsletter.banklesshq.com/p/a-guide-to-multi-chain-yield-farming (accessed 28 September 2021).

Government by algorithm (2021) Wikipedia, Wikimedia Foundation. Available from: https://en.wikipedia.org/wiki/Government_by_algorithm (accessed 3 September 2021).

Hansen P (2021) New crypto rules in the European Union – Gateway for mass adoption, or excessive regulation? New Crypto Rules in the European Union – Gateway for Mass Adoption, or Excessive Regulation? Available from: https://law.stanford.edu/2021/01/12/new-crypto-rules-in-the-eu-gateway-for-mass-adoption-or-excessive-regulation/ (accessed 4 September 2021).

Hoogendoorn R (2021) NFT market cap bigger than bitcoin cash or litecoin. DappRadar Blog RSS. Available from: https://dappradar.com/blog/nft-market-cap-bigger-than-bitcoin-cash-or-litecoin (accessed 24 September 2021).

Kuznetsov N (2021) DEFI liquidity pools, explained. Cointelegraph, Cointelegraph. Available from: https://cointelegraph.com/explained/defi-liquidity-pools-explained (accessed 25 September 2021).

Liebkind J (2021) Daos, blockchain, and the potential of ownerless business. Investopedia, Investopedia. Available from: https://www.investopedia.com/news/daos-and-potential-ownerless-business/ (accessed 4 September 2021).

Liebkind J (2021) Daos, blockchain, and the potential of ownerless business. Investopedia, Investopedia. Available from: https://www.investopedia.com/news/daos-and-potential-ownerless-business/ (accessed 4 September 2021).

Matt Hussey AH (2021) What is a Decentralized Autonomous Organization (DAO)? Decrypt, Decrypt. Available from: https://decrypt.co/resources/decentralized-autonomous-organization-dao (accessed 3 September 2021).

McBride L (2021) New EU proposal looks to tighten regulations for sending cryptocurrency, Cointelegraph. Available from: https://cointelegraph.com/news/new-eu-proposal-looks-to-tighten-regulations-for-sending-cryptocurrency (accessed 4 September 2021).

McBride L (2021) New EU proposal looks to tighten regulations for sending cryptocurrency. Cointelegraph, Cointelegraph. Available from: https://cointelegraph.com/news/new-eu-proposal-looks-to-tighten-regulations-for-sending-cryptocurrency (accessed 3 September 2021).

McCoy M and Rahimi S (2020) Prediction of highly volatile cryptocurrency prices using social media. International Journal of Computational Intelligence and Applications. Available from: https://www.worldscientific.com/doi/10.1142/S146902682050025X (accessed 9 July 2021).

Pandey I (2021) Liquidity mining will be the future of Capital Markets. Hacker Noon. Available from: https://hackernoon.com/liquidity-mining-will-be-the-future-of-capital-markets-ds1s32bg (accessed 3 October 2021).

Peaster WM (2021) How to join a dao. Bankless, Bankless. Available from: https://newsletter.banklesshq.com/p/how-to-join-a-dao (accessed 5 September 2021).

Person and Samuel Shen AG (2021) Cryptocurrency exchanges scramble to drop Chinese users after Beijing's ban. Reuters, Thomson Reuters. Available from: https://www.reuters.com/technology/cryptocurrency-exchanges-rush-cut-ties-with-chinese-users-after-fresh-crackdown-2021-09-27/ (accessed 28 September 2021).

Reiff N (2020) The largest cryptocurrency hacks so far. Investopedia, Investopedia. Available from: https://www.investopedia.com/news/largest-cryptocurrency-hacks-so-far-year/ (accessed 2 September 2021).

Sawinyh N (2021) Apwine - The Protocol for future yield tokenisation. DeFi, Defiprime. Available from: https://defiprime.com/apwine (accessed 18 September 2021).

Simonite T (2020) The "Autonomous Corporation" called the DAO is not a good way to spend $130 million. MIT Technology Review, MIT Technology Review. Available from: https://www.technologyreview.com/2016/05/17/160160/the-autonomous-corporation-called-the-dao-is-not-a-good-way-to-spend-130-million/ (accessed 5 September 2021).

Stater H (2021) Best defi yield farms. Benzinga. Available from: https://www.benzinga.com/money/best-yield-farms/ (accessed 25 September 2021).

Sun Z (2021) What is total value locked? Nasdaq. Available from: https://www.nasdaq.com/articles/what-is-total-value-locked-2021-09-29 (accessed 3 October 2021).

Team PP (2021) Liquidity mining in defi: What is it & how does it work? PixelPlex, PixelPlex. Available from: https://pixelplex.io/blog/what-is-liquidity-mining/ (accessed 3 October 2021).

TheJuanSC (2021) We need to talk about liquidity pools in defi, ASAP. Medium, Geek Culture. Available from: https://medium.com/geekculture/we-need-to-talk-about-liquidity-pools-in-defi-asap-e9a2b27769e (accessed 24 September 2021).

TheJuanSC (2021) We need to talk about liquidity pools in defi, ASAP. Medium, Geek Culture. Available from: https://medium.com/geekculture/we-need-to-talk-about-liquidity-pools-in-defi-asap-e9a2b27769e (accessed 3 October 2021).

Using algorithms to trade in crypto currencies (2020) Small Business Insider. Available from: https://smallbusinessinsider.co.uk/using-algorithms-to-trade-in-crypto-currencies/#_edn2 (accessed 16 September 2021).

Valencia F (2019) Price Movement Prediction of Cryptocurrencies Using Sentiment Analysis and Machine Learning. San Pablo, Querétaro, Mexico: Entropy.

Vermaak W (2021) What's the difference between staking, yield farming, and liquidity mining?: Certik Foundation blog. CertiK Foundation ⛓ Building Secure Blockchain Infrastructure. Available from: https://www.certik.org/blog/whats-the-difference-between-staking-yield-farming-and-liquidity-mining (accessed 17 September 2021).

Walsh D (2021) Europeans don't want the EU to regulate Cryptos, Euronews poll finds. euronews. Available from: https://www.euronews.com/next/2021/09/01/majority-of-europeans-want-their-countries-to-regulate-crypto-not-the-eu-exclusive-euronew (accessed 27 September 2021).

What is cryptocurrency algorithmic trading and how is it changing the way we trade? (2021) Arbismart. Available from: https://arbismart.com/blog/what-is-cryptocurrency-algorithmic-trading-and-how-is-it-changing-the-way-we-trade/ (accessed 17 September 2021).

What is defi in crypto? The Complete Defi & Yield Farming Crypto Tax Guide (2021) Cryptocurrency Tax Software, ZenLedger. Available from: https://www.zenledger.io/guides/the-complete-defi-crypto-tax-guide (accessed 14 September 2021).

What is liquidity mining? (2020) Bit2Me Academy. Available from: https://academy.bit2me.com/en/what-is-liquidity-mining/ (accessed 27 September 2021).

What is liquidity mining? (2021) Shrimpy Academy, The Shrimpy Team. Available from: https://academy.shrimpy.io/post/what-is-liquidity-mining (accessed 16 September 2021).

What is liquidity mining? (2021) Shrimpy Academy. Available from: https://academy.shrimpy.io/post/what-is-liquidity-mining (accessed 24 September 2021).

What is Pendle Crypto? how to buy it? (n.d.) Stock Market News and Research. Available from: https://kalkinemedia.com/news/cryptocurrency/what-is-pendle-crypto-how-to-buy-it (accessed 1 September 2021).

Wiesflecker L (2021) What is liquidity mining, and how does it work? Medium, Coinmonks. Available from: https://medium.com/coinmonks/what-is-liquidity-mining-and-how-does-it-work-d0ab491e607 (accessed 2021).

Wiesflecker L (2021) What is liquidity mining, and how does it work? Medium, Coinmonks. Available from: https://medium.com/coinmonks/what-is-liquidity-mining-and-how-does-it-work-d0ab491e607 (accessed 23 September 2021).

Will Warren (2018) Decentralized Exchange. Coin Center. Available from: https://www.coincenter.org/education/key-concepts/decentralized-exchange/ (accessed 10 September 2021).

Win Win K (2021) Tokenizing Future Yield. CoinGecko.

18. Attachments